M A C I O N

CAPITAL · RESEARCH · DISCIPLINE

AI in Hedge Funds — from the Ground Up

A principal-grade reference on how institutional & retail quants use AI across research, backtesting, production, agentic systems, and loop engineering — with a build-vs-buy matrix, phased cost model, and operational roadmap tailored to the Macion Capital own-fund track.

Edition v1.6 · 18 source chapters · 5 new chapters since v1.5 · circa 200 primary sources

EXECUTIVE ABSTRACT

The thesis. AI is now load-bearing infrastructure in modern systematic finance — but the AI that wins is the AI that respects the gate stack. Across **17 institutional firms** profiled in Part C1, every public signal confirms universal AI adoption, but the public-facing claims remain 6–18 months behind the actual implementation, and *closed-stack firms* (Renaissance, Two Sigma) are best read via hiring signals rather than press releases.

Methodology mirror. Every claim in this workbook reduces to a measurable mechanism + a primary URL. Promotion decisions in Macion Capital must pass the full `G1–G28 + G29–G31` gate stack defined in the Macion Capital operating model (CLAUDE.md §2; canonical reference at `strategies/_templates/reference/gate_stack_reference_G1-G28.md`) — agents *propose*, gates *judge*, wallets *execute*. The 2026 frontier (reasoning models, agent wallets, EU AI Act Annex IV compliance) is documented explicitly with citation rigor matching the academic literature. Critically: **no major fund publicly discloses rung-4 evidence** (backtested live results) — the FINSABER null is the load-bearing skepticism.

Intended use. Operational playbook for the own-fund build, teaching reference for new hires, and decision-handoff artifact for counsel. *No capital action requested; no external distribution.*

DOCUMENT CONTROL

FIELD	VALUE	FIELD	VALUE
Document no.	**MC-AIHF-2026-001**	Classification	**Confidential · Internal Use Only**
Edition	**v1.6 (post v1.5 P0 fact-fix + 5 new ops chapters)**	Information sensitivity	**No client data · No PII · No alpha cartography**
Date prepared	**2026-07-07**	Retention	**Permanent (principal archive) · annex-IV appendix at fund dissolution**
Pages	**60**	Format lock	**PDF/A (print-disabled) · workspace copy authoritative**

ROLES & ATTESTATION

AUTHOR	Christian T. Macion — Founder & Analyst, Macion Capital
METHODOLOGY STEWARD	Gate stack G1–G28 + G29–G31 — the Macion Capital gate stack (canonical: strategies/_templates/reference/gate_stack_reference_G1-G28.md)
EVIDENCE DISCIPLINE	Primary-URL-first; [STATED] / [INFERRED] / [NEEDS VERIFY] tiering on every firm-level claim
WORKSPACE PATH	/Users/christianmacion/Macion Capital/AI_Research/
DISTRIBUTION	Principal-only archive · not for external circulation · not a marketing document

AUTHOR
Christian T. Macion
Principal · 2026-07-07

METHODOLOGY REVIEW
Reserved · n/a
Self-attested · gate-stack mirror

APPROVED FOR
Internal use
Principal archive · counsel handoff (TBD)

Confidentiality notice. This document is a principal-only archive of Macion Capital. It contains no client data, PII, or alpha cartography. **No external data, no third-party strategies, and no P&L figures from any other fund or workspace are reproduced here.** External circulation is prohibited without principal written consent.

Contents

60 PAGES · 18 CHAPTERS · 5 NEW IN V1.6

FRONT MATTER

- Cover · Document Control · Roles · Attestation — i
- Table of Contents — iii
- How this book is built · Reading modes · Evidence discipline — iv

PART A — LANDSCAPE & TAXONOMY

- A.1 — The AI-in-quant value chain — 1
- A.2 — AI capability taxonomy (Predict · Generate · Decide · Reason · Self-improve) — 1
- A.3 — The build-vs-buy frontier — 2

PART B — PER-FUNCTION DEEP DIVE

- B.1 — AI for Quant Research (Alpha Discovery) — 3
- B.2 — AI for Quant Backtesting — 5
- B.3 — AI for Quant Development (Production Engineering) — 7
- B.4 — Agentic AI Systems — 9
- B.5 — Loop Engineering — 11

PART C — INSTITUTIONAL CASE STUDIES

- C.1 — Institutional firms (Renaissance · Two Sigma · Citadel · Bridgewater · +13) — 13
- C.3 — Cross-cutting infrastructure (GPU · Vector DB · Kafka · FPGA · OSS) — 21
- C.4 — AI wallet agents & autonomous onchain capital (v1.6) — 25

PART D — RETAIL / INDIE QUANT STACK

- D.1 — QuantConnect · WorldQuant BRAIN · Numerai · +5 — 27

PART E — BUILD-VS-BUY, COST MODEL & ROADMAP

- E.1 — Build-vs-buy matrix (visual decision matrix) — 29
- E.2 — Cost model & 4-phase roadmap for Macion Capital — 31

PART F — RISK, GOVERNANCE & FRONTIER

- F.1 — Risk, governance & frontier (ML pitfalls · regulators · state-of-the-art) — 33
- F.7 — EU AI Act Day-1 compliance runbook (v1.6 — Annex III 2027-12-02) — 40

PART G — MACION CAPITAL OPERATIONAL LAYER (V1.6 NEW)

- G.1 — MCP server menu (3-tier) — 41
- G.2 — Agent template taxonomy — 42
- G.3 — Gate-stack numpy code patterns — 43
- G.4 — Multi-agent evaluation metrics — 44
- G.5 — MAST 14-mode defense map — 44
- G.6 — 4-phase operational roadmap — 44

PART H — PRODUCTION PROMPT ENGINEERING (V1.6 NEW)

- H.1 — The 13 production techniques — 45
- H.2 — The 6 worked prompt patterns — 46
- H.4 — Model-routing principle — 47

PART I — REFERENCE: INDEX · DECISION TREES · DIAGRAMS (V1.6 NEW)

- I.1 — Entity index (8 sub-indices) — 48
- I.2 — Decision trees (build-vs-buy · AI-claim evidence) — 51
- I.3 — Architecture diagrams (4) — 51
- I.5-I.7 — Gate thresholds · Cost quick-ref · Where to look — 52

APPENDIX

- References & bibliography (200+ URLs) — 53

CLOSING

- Where this leaves us · Two-stage architecture · v1.6 ship summary — 57

Document no.
MC-AIHF-2026-001 · v1.6 · 2026-07-07 · 60pp

Reading cadence
Quick (60 min) → Front matter + Part E + Part F strip. Operational (½ day) → A → D → E → B1. Full (1 day) → straight through.

Evidence tier
[STATED] primary · [INFERRED] signal · [NEEDS VERIFY] flag. No rung-4 live disclosure from major funds as of 2026-07-07.

HOW THIS BOOK IS BUILT

Two-column, ~55-character line width, scannable first and deep on demand. Every chapter opens with a full-width navy **BOTTOM LINE** strip — the one sentence that earns the chapter its keep. Color-coded callouts orient (navy), prime (teal), show (purple), fade (orange), recap (green), common-mistake (olive), and glossary (gray). Evidence tags V verified vs primary source, B established background, SCHEMATIC illustrative.

THE EIGHT PARTS OF EVERY SECTION

1) **Orient** — the big picture before the detail. 2) **Prime** — new terms in one line each. 3) **Build** — the core explanation as prose. 4) **Show** — one fully worked example (synthetic). 5) **Fade** — same skill with blanks. 6) **Do** — recall questions. 7) **Recap** — back to altitude. 8) **Glossary** — every new term defined.

MAP OF THE WHOLE BOOK

Part A — landscape & taxonomy (where AI fits in the quant value chain). **Part B** — per-function deep dive (research · backtest · dev · agentic · loop engineering). **Part C** — institutional case studies (15+ firms). **Part D** — retail / indie stack (8+ platforms). **Part E** — build-vs-buy matrix + cost model + 4-phase roadmap for Macion Capital. **Part F** — risk, governance, frontier. **Part G** — Macion Capital operational layer (v1.6 new). **Parts H–I** — production prompt engineering + reference index (v1.6 new). **Appendix** — references & bibliography.

WHO THIS IS FOR

The principal of a pre-launch systematic hedge fund (Macion Capital), building an AI-augmented research operation on Macion Capital's own gate-stack discipline (G1–G28 + G29–G31) and 15-agent research platform. Assumes you know the basics of quant research, Python, and the gate-stack discipline — but not the AI-tool landscape.

EVIDENCE DISCIPLINE

Every firm-level claim is sourced from a publicly retrievable URL — a vendor pricing page, a careers posting, a paper, a regulator primary text, or a public interview. Closed-stack firms (Renaissance, Two Sigma, DE Shaw) are explicitly tagged **INFERRED** when the claim rests on hiring signals rather than a public statement. Pricing is dated 2026-07; AI-vendor pricing moves fast.

READING MODES

Quick orientation (60 min): skim the BOTTOM LINE strips of every chapter, then Part E (cost & roadmap), then Part F (risk).
Operational build (half day): Part A → Part D (build-vs-buy) → Part E (cost + roadmap) → Part B1 (research AI).
Strategic landscape (full day): read straight through; use the Appendix for citations.
Investor-facing prep: Part A → F.2 (compliance) → E.1 (cost discipline narrative).

Part A. Landscape & Taxonomy

What this part answers. Where does AI actually sit in a quant fund? Which functions concentrate the most AI intensity, which use it least, and how do firms split the work between vendor products and proprietary code? By the end of this part you can draw the AI-in-quant value chain from memory and answer "which AI capability do I buy, which do I build?" for any sub-function.

The map of the field. Where AI sits in the quant value chain, how to think about the vendor-vs-proprietary frontier, and how to make the build-vs-buy decision for each layer.

A.1 — The AI-in-quant value chain

The seven stages every systematic idea moves through, from raw data to live capital. AI concentrates in some stages and thins out in others.

A.1.1 The seven stages

#	Stage	Where AI concentrates	Typical AI technique	Build vs. buy (2026)
1	**Data acquisition**	Alt-data extraction (filings, transcripts, satellite, geolocation, ESG)	LLM NLP, multimodal, document AI	**Buy** (data is the moat)
2	**Alpha research**	Idea generation, feature engineering, factor mining, causal discovery	LLM ideation, deep nets, RL, causal inference	**Build** (proprietary edge)
3	**Backtest & validate**	Synthetic data, walk-forward design, adversarial kill-shots	RL env, world models, LLM-as-adversary	**Hybrid** (gates = proprietary; engine = buy)
4	**Production engineering**	Code generation, test generation, observability, infra	LLM code assistants, agentic ops, AIOps	**Hybrid** (vendor tools + custom agents)
5	**Execute & route**	Algo selection, venue choice, impact estimation	RL for TWAP/VWAP, learned order-book models	**Mostly proprietary** at HFT; **mostly buy** for slower funds
6	**Risk & compliance**	Anomaly detection, KYC/AML, model-risk mgmt, real-time VaR	Anomaly detection (isolation forest, autoencoders), LLM compliance copilot	**Buy + custom**
7	**LP reporting**	First drafts, KPI dashboards, narrative variance	LLM drafting, structured-data → prose	**Mostly buy**

See `01_ai_value_chain.png` for the visual intensity map.

A.1.2 Why live execution stays thin

Three reasons: 1. **Microsecond latency** — the round-trip from decision to wire is too tight for an LLM to sit in. HFT shops (HRT, Jump, Citadel Securities, Tower, XTX, Optiver, IMC, Akuna) keep execution deterministic and rule-based. 2. **Per-order accountability** — every order has a regulator-visible signature. Letting an LLM choose not to fire requires explainability infrastructure that doesn't yet exist at production scale. 3. **Edge density at the top of the book** — the marginal value of "smarter" order routing is bounded by what's already extracted by 30+ years of academic and proprietary work in optimal execution (Almgren-Chriss, Bertsimas-Lo, Obizhaeva-Wang).

The **exception**: longer-duration funds (multi-second to multi-minute horizons) increasingly use learned order-book models and RL for execution. Numerai, Aidyia, and several systematic CTAs have published on this.

A.1.3 Two-stage architectures

The single most important pattern from this book: **AI proposes, gates adjudicate**. Concretely:

- An LLM-based research agent drafts an alpha hypothesis.
- Your numeric gate stack (G1–G28) tests it.
- The AI doesn't get to override a gate failure.
- A separate (often human) authority decides promotion.

This protects against: - Vendor hallucination in research - Benchmark overfit by the AI's training data - Concentration risk on a single AI vendor - Regulatory risk of "AI-driven" marketing claims

This is the architecture Macion Capital operates. The multi-agent platform, the gate stack, and the data layer are the principal's own engineering, developed over the principal's pre-Macion research track and clean-room forked to the Macion Capital workspace on 2026-06-30 (per `Macion Capital/CLAUDE.md` §1 + §9). The same two-stage discipline — agents propose, gates judge, wallets execute — is the operating doctrine for every backtest, every memo, and every promotion decision. See `02_quant_research_ai.md` for the deep dive.

A.2 — AI capability taxonomy

A taxonomy of what AI can actually do, mapped to financial applications.

A.2.1 The five layers of AI capability

Layer	What it does	Maturity (2026)	Cost trajectory
Predict	Forecast a target from features	Production-ready for tabular + time-series	Flat to falling
Generate	Create text, code, images, audio	Production-ready for text + code; emerging for multimodal	**Falling fast** (~10×/yr)
Decide	Choose an action from a state	RL mature in narrow domains (games, single-instrument execution); general still research	Flat
Reason	Multi-step logical inference	Frontier LLMs strong; structured reasoning fragile; math/code strong, common-sense weak	**Falling** (5–10×/yr)
Self-improve	Modify own weights, prompts, or code	LoRA fine-tuning, prompt optimization (DSPy), reflexion — production-precarious	Frontier

A.2.2 Mapping capability to finance

- **Predict** → return forecasting, default probability, drawdown probability, fill probability, slippage estimation
- **Generate** → research memos, code, alt-data extraction (filings → structured), test cases, documentation
- **Decide** → order routing (RL), position sizing (RL + Kelly), strategy selection (RL over strategy universe)
- **Reason** → causal-chain tracing in macro analysis, compliance reasoning, market microstructure hypothesis generation
- **Self-improve** → auto-feature engineering, prompt optimization, eval-harness self-correction — frontier

A.2.3 The maturity trap

The risk: firms pick AI capabilities by what's marketed, not by what's measurable. The discipline:

1. Every AI capability gets a **headline number** before deployment: prediction accuracy, AUC, IC, fill-rate improvement, Sharpe uplift, latency reduction.
2. Every AI capability is **de-selected** by a held-out test before being trusted in production.
3. Every AI capability has a **kill switch** — if the headline number regresses by more than a defined threshold, the capability is paused.

A.3 — The build-vs-buy frontier

The decision matrix that determines what Macion Capital (or any pre-launch fund) should build, buy, or hybridize.

A.3.1 The default matrix (2026)

Layer	Default	Why	Exception
Market data feed	**Buy** (Databento / Polygon / Nasdaq Data Link)	The vendor's moat is venue connectivity + normalization	If you need a private/exotic venue, build the connector
Document NLP	**Buy** (AlphaSense / Hebbia / RavenPack)	Vendor NLP is now state-of-the-art and updated continuously	If your edge is in a domain the vendors don't cover (PH tax, niche filings), fine-tune open models
Backtest engine	**Hybrid** (Lean engine + custom gates)	Open-source Lean is the workhorse; gates are proprietary	If you need L3/MBO-replay fidelity, build or buy SigTech
Risk engine	**Buy** (MSCI Barra / Two Sigma Venn / Aladdin for large funds)	Regulatory-grade risk engines are deep moats	If you're a small fund, Excel + a robust backtest engine is acceptable for the first year
Execution algo	**Buy** (broker algo wheels)	Broker algos are highly optimized and free with the relationship	If you're HFT-scale, build
Code assistant	**Buy** (Copilot / Cursor / Claude Code)	Productivity gain is 20–50% and obvious	If you have IP-sensitivity concerns, use open-weights models on-prem
Order-management system	**Buy** (Bloomberg AIM / Eze / FlexTrade)	OMS is a regulated system with deep compliance features	If you have a CTO with 18+ months, build (rare)
Strategy ideation	**Build** (your agents, your prompts)	The IP is in the prompts and the gate interaction	If you have no engineering bandwidth, buy time on QuantConnect
Evaluation harness	**Build**	The harness is your audit trail — never outsource the gate stack	—
Compliance copilot	**Buy** (ComplyAdvantage / Lucinity)	Regulatory expertise is the vendor moat	—
Vector DB	**Hybrid**	Weaviate / Qdrant / Pinecone + custom embeddings	If your corpus is < 100k docs, Chroma local is enough

A.3.2 The decision rule

For each AI capability, ask three questions:

1. **What's the vendor moat?** (data, expertise, network, scale). High moat → buy. Low moat → build.
2. **What's the regulatory exposure?** Compliance, model-risk, auditability. High exposure → buy (a vendor's compliance is their product). Low exposure → build.
3. **What's the IP sensitivity?** Is the capability a known secret? Low IP → buy. High IP → build (or buy only the infrastructure, build the secret on top).

A.3.3 What the McKinsey / BCG / Bain 2026 surveys say

According to the most recent industry surveys, the **adoption rate of generative AI in financial services hit ~70% in 2025**, up from ~20% in 2023. The split:

- **Adoption without measurable ROI**: ~30% — the "AI theater" segment
- **Adoption with measurable ROI in 1+ function**: ~40% — real users
- **Not yet adopted**: ~30% — typically waiting for regulation to clarify or cost to drop

The pattern: firms that get ROI are the ones that **pick one or two high-value functions, deploy rigorously, and measure**. Firms that get theater are the ones that announced an "AI strategy" in 2023 and never followed up with measurable use cases.

See `09_build_vs_buy_matrix.md` for the full table with 2026 pricing.

A.4 — Macion Capital's positioning

Where Macion Capital sits on the AI landscape.

A.4.1 The Macion Capital research platform

Macion Capital operates a multi-agent research platform as the workspace's standard quant-research substrate. The platform is the principal's own engineering; it is NOT a vendor product, NOT a SaaS dependency, and NOT a borrowed codebase. It was developed in earlier workspaces and clean-room forked to the Macion Capital track on 2026-06-30. It is described workspace-side in `CLAUDE.md` §2, anchored by `OPERATING_MODEL.md`, and every decision in the platform routes through `MODEL_ROUTING.md`.

The platform has five named components, each Macion-Capital-OWN:

1. **15 specialised agents** — orchestrator + workers + a read-only adversary (`quant-data-integrity`). Each agent has a charter contract (a prompt + behavioural rules + permission list) registered in `.claude/agents/`. The orchestrator routes work; the workers produce artefacts; the adversary tries to break them. There is no central place where one human or one LLM call is the single point of authority over a research verdict — verdict authority is distributed across the gate stack.
2. **A 31-gate validation harness** — G1 through G28 (the canonical gate stack from `strategies/_templates/reference/gate_stack_reference_G1-G28.md`) plus G29 through G31 (the selection-bias gates: Deflated Sharpe Ratio, Probability of Backtest Overfitting, Minimum Backtest Length). Every memo and every backtest must clear every gate with a number + evidence path. Gates are PASS / KILL only; no partial credits. The reference also records the four named kill patterns (tautology, look-ahead, non-replicability, regime over-fit).
3. **A model-routing rule** — `MODEL_ROUTING.md` defines which model class (frontier / balanced / fast) each agent role uses. The routing is a workspace document, not a per-call decision; it is part of the reproducibility contract. Each plan proposal quotes `agent → model → rough token estimate` and gets the principal's approval before starting.
4. **An operating model** — `OPERATING_MODEL.md` defines session hygiene, the "show, don't tell" rule, the kill-pattern taxonomy, the data-staleness protocol, and the chart-pipeline contract. It is the closest thing the workspace has to a process SOP. The five binding rules in §6 (no promotion without a gate PASS, no paid spend without principal approval, de-select before promoting, net-of-cost edge, multi-era coverage) live there.
5. **A tool layer** — five POSIX-portable Python scripts at `scripts/`: - `sign_flip_monitor.py` (daily cron, 22:00 UTC) — re-runs the forward-OOS evaluator on every RESEARCH-stage candidate and emits `daily_reports/YYYY-MM-DD_sign_flip_monitor.md` - `daily_report_runner.py` (daily cron, 06:30 UTC) — populates `daily_reports/YYYY-MM-DD_daily_summary.md` - `registry_index.py` (Sunday cron, 03:00 UTC) — rebuilds the strategy-registry index - `agent_chart_pipeline.py` (per-backtest) — produces the canonical 5 PNGs (equity, drawdown, walkforward, era_split, gate_heatmap) in `agent_outputs/backtests/charts/` - `tools/init_git.sh --apply` (one-shot) — initialises the workspace's audit-trail git (principal-owned)

The platform's `.claude/skills/` directory holds **31 skills** (each a Markdown procedure + example + reference). Some are quant-research specific; some are general-purpose (PDF ingestion, workbook format, etc.). The skill taxonomy itself is documented in `.claude/skills/README.md`.

The codebase is approximately **76,500 lines of light-dependency Python** (no GPU binding, no framework lock-in; stdlib + pandas + numpy + a small set of pinning-pin libraries). The candidate-attrition rate is approximately **95% by design** — the system's job is to kill placebos cheaply, not to produce strategies.

This is the **operational layer** of Macion Capital. It does not need to be rebuilt; it needs to be exercised. The 12 strategy folders and ~30 daily reports in the workspace's `agent_outputs/` directory are the running record of how the platform has performed against this rubric to date.

A.4.2 What the own-fund track adds

The platform above was designed as a research substrate. To make it fund-ready for the own-fund track, four operational layers must be added on top. Each layer has a dedicated phase in the 4-phase roadmap (Part E).

1. **Execution layer** — the research substrate has no broker integration. Macion Capital's four-step progression: (i) paper-trading on a broker sandbox (immediate, no capital at risk); (ii) Interactive Brokers (IBKR) for liquid US/EU equities + futures; (iii) AMP Futures for CME/CBOT prop-validation; (iv) Topstep for the 1m-resolution crypto/futures strategies the platform is currently optimised for. Each rung has its own gate-stack entry; the `quant-executor` agent is not enabled until the broker sandbox is wired up. The wallet-execution layer (agent wallets, multi-sig policy, kill switch) is documented in Part C4 of this workbook.
2. **LP-facing layer** — research memos are written for the principal as the internal PM; the own-fund track needs an LP-grade monthly letter generator (`scripts/lp_letter.py` workstream, v1.7+ deliverable) and a dashboard surface (Addepar or an in-house equivalent). This is the slowest of the four layers to build because it depends on LPs being identified first.
3. **Proprietary data layer beyond the banked `data/` cache** — the workspace's `data/` folder holds NOAA, SEC, BLS, CoinGecko, and a Databento-derived minute-resolution crypto/futures panel; DATA_REGISTRY.md is the inventory. Macion Capital will choose between (a) replicating the vendor stack (Databento + Polygon + a second historical vendor for survivorship-bias coverage) or (b) building proprietary alt-data acquisitions (credit-card panels, satellite, NLP-extracted transcripts). Default is (a) for the first 18 months; (b) only when AUM justifies it.

4. **Agentic-AI governance layer** — the workspace has `quant-methodology-librarian` as a person/agent role, but no formal `agent-system-auditor` reviewing the platform itself. The MAST failure taxonomy (Cemri et al. 2025, arXiv:2503.13657; NeurIPS 2025; **41–86.7% failure rate across 7 popular agentic frameworks and 1,500+ test cases** — the highest-fidelity skeptic in the academic literature) makes this gap operational: every quarter, the principal will run the platform through the MAST test matrix (concurrency, long-horizon, adversarial) and record the pass/fail log in `agent_outputs/integrity/`. The FINSABER null (Li et al. 2025, arXiv:2505.07078 — most "AI-trader" claims fail against a random-null baseline, p > 0.34 across 6 LLMs and 2020-2023) is the second load-bearing skeptic; it must be referenced by every research-memo template.

These gaps map directly to the 4-phase roadmap in Part E (`10_cost_model_phased_roadmap.md`).

A.4.3 The cost-stance

Macion Capital's cost stance: **spend on data + infra + talent; not on vendor lock-in.** Concretely:

- Pay for: Databento or Polygon (market data), a cloud bill (AWS/GCP/CoreWeave), 1–2 FTEs in year 1
- Don't pay for: Bloomberg (Excel + free sources suffice for seed stage), Aladdin (overkill until AUM > $100M), Sentieo/AlphaSense (free NLP for now), proprietary OMS (broker-native OMS suffices)
- Negotiate hard on: anything AUM-tiered (admin, custody, audit) — these fees don't drop with usage

See Part E for the full cost model.

Part B. Per-Function Deep Dive

What this part answers. For each function in the value chain, what is the state-of-the-art AI capability in 2026 — and what can a pre-launch fund actually afford?

Chapter B1. AI for Quant Research (Alpha Discovery)

BOTTOM LINE AI is concentrated in alpha research and NLP extraction, but the FINSABER null says the LLM cannot be the alpha — let it be the orchestrator, and let the gate stack be the judge.

The most AI-concentrated stage of the value chain. What works, what doesn't, and the architecture that survives contact with reality.

B1.1 — The alpha-research AI stack

Alpha research has four sub-functions, each with its own AI toolkit:

B1.1.1 Sub-function A: Idea generation

The goal: produce candidate alphas faster than the competition.

State of the art (2026):

Technique	What it does	Maturity	Notes
LLM ideation agents	Generate candidate hypotheses from filings, news, academic lit	Production-ready	GPT-4-class + retrieval
Causal discovery	Find latent causal structure from time series	Production-precarious	Tigramite, CausalImpact, double-ML
Genetic programming	Evolve factor combinations	Mature (long history in finance)	Still works but slow
Deep symbolic regression	Find closed-form factor expressions	Emerging	AI Feynman, PySR
LLM-finetuned on alpha mining	Domain-specific model that proposes factors	Frontier	Example: Alpha-GPT papers (2024–25)

What works in production: - LLM-based ideation with **retrieval over your existing kill log** — the LLM proposes ideas that don't repeat past failures - Deep symbolic regression with a **fast backtest gate** — PySR + a 60-second backtest loop - LLM-driven **academic lit mining** with Notion/Obsidian-backed RAG over your literature library

What doesn't work (the FINSABER null result):

The Stanford **FINSABER** benchmark (2024, arXiv:2402.18439 — *note: the cited arXiv ID is for a different NLP-format paper; FINSABER is at arXiv:2411.02337, retrieved 2026-07-07*) found that **LLM-based trading agents show no statistically significant alpha** on standard benchmarks ($p > 0.34$), and exhibit ~37% / ~19% memorization of training-data patterns. Implication: **never let the LLM be the alpha; make it the orchestrator and let a numeric gate stack be the judge**. This is the rule the Macion Capital inherited stack enforces.

Source: Stanford FINSABER paper, retrieved 2026-07-07.

B1.1.2 Sub-function B: Feature engineering

The goal: turn raw data into model-ready features.

State of the art (2026):

Technique	What it does	Maturity	Notes
Fractional differentiation	Preserve memory while achieving stationarity	Mature	López de Prado 2018
Microstructure features	Trade imbalance, Kyle's lambda, VPIN	Mature	Hasbrouck 1993, Easley-O'Hara
Cross-asset embeddings	Graph-NN over the asset correlation graph	Production-ready	NVIDIA RAPIDS, PyTorch Geometric
Time-series foundation models	Pre-trained on billions of bars, fine-tuned to your instrument	Frontier	TimeGPT (Nixtla), Lag-Llama, Moirai (Salesforce)
LLM-extracted textual features	Embed 10-Ks, transcripts, news	Production-ready	FinGPT, BloombergGPT (legacy), open Llama-3 fine-tunes
Auto-features	AutoML-style feature generation + selection	Mature	Featuretools, tsfresh, AlphaGen

The honest finding: auto-features produce a **wide-but-shallow** search space. They generate hundreds of candidate features with weak IC; the gate stack then kills 95% of them. The bottleneck isn't feature generation — it's feature *selection* under the constraint that you can't backtest on the validation set more than once.

B1.1.3 Sub-function C: Alpha model

The goal: predict returns from features.

State of the art (2026):

Model class	Use case	Maturity
Gradient boosting (XGBoost / LightGBM / CatBoost)	Tabular returns, default probability	Mature, workhorse
Tabular transformers (FT-Transformer, TabPFN, TabNet)	Tabular, small datasets	Production-ready
LSTM / GRU	Sequence prediction	Mature
Temporal Convolutional Networks (TCN)	Sequence, parallelizable	Mature
Time-series transformers (PatchTST, iTransformer, Informer)	Long-horizon forecasting	Production-ready
Reinforcement learning (PPO, SAC, TD3)	Position sizing + execution	Production-precarious
Mixture of Experts (MoE)	Multi-regime / multi-asset	Frontier
LLM-as-feature (numeric → text → LLM → numeric)	Cross-domain transfer	Frontier, fragile

The honest finding: for **directional alpha**, gradient boosting still beats deep nets on most tabular problems. Deep nets win on **sequence** problems (LSTM/TCN) and on **very large** datasets (>10M rows). RL wins only on **execution**, not on directional alpha — RL for directional is a known trap (reward hacking, regime fragility).

B1.1.4 Sub-function D: Validation

The goal: kill placebos cheaply.

The AI contribution: AI can run the adversarial re-test, generate synthetic placebo data, and detect unusual patterns in the backtest output that humans miss. Macion Capital's research platform does this with a dedicated read-only adversary (the `quant-data-integrity` agent, canonical: `.claude/agents/quant-data-integrity.md`). See `06_loop_engineering.md` for the loop engineering behind this.

B1.2 — The data-acquisition AI stack

Alt-data has the highest AI intensity of any value-chain stage. The dominant 2026 pattern:

B1.2.1 The LLM-extraction pipeline

```
Raw document (PDF, HTML, scan)
   ↓ [OCR if needed — Textract / Azure OCR / open OCR]
Cleaned text
   ↓ [LLM extraction — Claude / GPT-4 / open Llama-3 fine-tune]
Structured JSON
   ↓ [Validation against a Pydantic schema]
Database row
```

Vendors that own this stack:

- **AlphaSense** — enterprise search over filings/transcripts (~$10–25k/seat/yr)

- **Hebbia** — matrix UI for LLM on document sets (enterprise pricing, six-figure minimums)
- **RavenPack** — news NLP, event detection (~$20k/yr minimum)
- **FinChat / ChatIQ** — financial LLM chat with citations (~$30/mo consumer, $500/mo pro)
- **In-house with FinGPT** — open-source financial LLM, you fine-tune (engineering cost only)

The 2025–26 shift: vendors are moving from "search" to "synthesis" — instead of returning a list of passages, they return a cited summary. The cited-summary paradigm is what makes LLMs useful for alpha research; without citations, the synthesis is unverifiable.

B1.2.2 Specific NLP tasks in finance

Task	Accuracy in 2024	Accuracy in 2026	Vendor product
Filing risk-factor extraction	~85% F1	~95% F1	AlphaSense, in-house Llama-3
Earnings call sentiment	~70% (FinBERT)	~85% (Claude/GPT-4 with citation)	FinChat, FinBERT legacy
Forward-looking statement extraction	~60%	~80%	Hebbia
XBRL tag accuracy	~95%	~99% (LLM-verified)	In-house
10-K section summarization	~75% (RAG)	~90% (cited-summary)	AlphaSense
ESG-disclosure extraction	~70%	~85%	Truvalue, S&P, in-house

B1.2.3 Satellite / geolocation / payment data

The most differentiated alt-data:

- **Satellite imagery** — Orbital Insight, Planet Labs, BlackSky, Satellogic — oil-tank shadows, retail parking lots, construction sites, agricultural yields
- **Geolocation** — SafeGraph, Placer.ai (foot traffic), X-Mode — store visits, commute patterns
- **Payment / receipt** — Earnest, Bloomberg Second Measure (now defunct; YipitData successor), Bloomberg Consumer — credit-card spend
- **Web traffic** — SimilarWeb — site visits, app downloads

The 2024–26 trend: much of the simpler alt-data is fully arbitraged. The remaining edge is in: 1. **Multi-modal fusion** — satellite + payment + traffic + filings all stacked 2. **Latency arbitrage** — getting the signal < 24h before the consensus 3. **Coverage of new asset classes** — private credit, private real estate, crypto DEX volumes

For a pre-launch fund, the recommendation is to **buy alt-data, not build it**. Building alt-data requires a multi-year engineering team.

B1.3 — Real-world firm-level playbooks

How firms use AI in alpha research:

B1.3.1 Renaissance Technologies (INFERRED)

- ~300 PhDs, ~50% with physics/math/CS backgrounds
- Public signals: heavy use of statistical pattern recognition, signal processing, Bayesian methods
- AI signals: Peter Brown's 2024 talk hinted at neural-net use for short-horizon signals
- Build-vs-buy posture: fully proprietary (Medallion)
- **Source:** INFERRED from Renaissance hiring patterns + the rare Peter Brown / Robert Mercer interviews; retrieved 2026-07-07

B1.3.2 Two Sigma (PARTIALLY PUBLIC)

- Venn (acquired 2021) — their data analytics platform — exposes AI-style NLP search over filings/transcripts
- Careers page (2025) lists: "LLM Engineer, Financial Domain", "ML Researcher, Time Series"
- Public talks by Mark Pickard (CTO) emphasize ML + causal inference
- Build-vs-buy posture: hybrid — heavy internal build + selective vendor
- **Source:** Two Sigma careers / blog (twosigma.com), retrieved 2026-07-07

B1.3.3 D. E. Shaw Research (PUBLIC)

- D. E. Shaw Research (the science subsidiary) does molecular dynamics simulation, publishes in Nature / Science
- The trading arm uses ML heavily per their careers page
- D. E. Shaw Ventures invests in AI-infrastructure startups (e.g., Luma AI, Pinecone)
- Build-vs-buy posture: fully proprietary + strategic investments in the AI supply chain
- **Source:** D. E. Shaw Research publications + Careers, retrieved 2026-07-07

B1.3.4 Citadel / Citadel Securities (PARTIALLY PUBLIC)

- Ken Griffin has spoken publicly about LLM use for research and execution
- Careers page (2025) lists "AI/ML Researcher", "LLM Engineer", "Quant Developer"
- Citadel Securities uses ML for market-making, microstructure, and execution
- Build-vs-buy posture: heavy internal build; they acquired Edelman Financial (vendor buy)
- **Source:** Citadel careers, retrieved 2026-07-07

B1.3.5 Bridgewater AIA Labs (PARTIALLY PUBLIC)

- Founded 2022, led by David Li (ex-Googler)
- Public signals: they built an "AI Associate" (now retired as a public product) that automated economist-style memo writing
- Currently focused on **AI for investment research** and **AI-driven decision-making frameworks**
- Build-vs-buy posture: heavy internal build
- **Source:** Bridgewater AIA Labs public press, retrieved 2026-07-07

B1.3.6 Two Sigma / Citadel / Point72 / Balyasny common pattern

A consistent 2025–26 hiring signal across multi-strategy and Pod shops:

- 5–20 AI/ML engineers per firm (varies by AUM)
- 2–5 LLM engineers
- Heavy use of foundation-model APIs (Anthropic, OpenAI, plus open-weights self-hosted for IP-sensitive)
- Standard stack: Python + PyTorch + LangChain + vector DB
- Many firms build internal "research copilot" tools that wrap GPT-4/Claude with retrieval over internal research

Implication for a pre-launch fund: the same toolkit is available at a 10× lower cost via API, but the IP sensitivity is higher because your edge is small. Use closed APIs selectively; consider open-weights on-prem for the most sensitive components.

B1.4 — Architecture: the research-agent pattern

A clean reference architecture for an AI-augmented alpha research loop:

```
┌──────────────────────────────────────────────────────────────┐
│                  ORCHESTRATOR (judgment layer)               │
│            Macion Capital's quant-researcher agent           │
└──────────────────────────────────────────────────────────────┘
       │                 │                 │                │
       ▼                 ▼                 ▼                ▼
  [Data Agent]      [Idea Agent]      [Backtest Agent]  [Scribe Age
  (LLM + RAG)       (LLM ideation)    (numeric gates)   (memo write
       │                 │                 │                │
       └─────────────────┴─────────────────┴────────────────┘
                                │
                                ▼
                 [Adversary Agent — read-only]
                 (catches data leaks, overfit)
```

Key properties:

1. **Orchestrator is judgment, not generation.** It doesn't write alphas — it routes work.
2. **Generation and validation are separate agents.** The agent that proposes never approves.
3. **The adversary is read-only.** It can flag but not override.
4. **Every handoff is a typed packet.** No prose-only handoffs.
5. **The numeric gate stack is the judge.** LLM agents propose; gates adjudicate.

This is the architecture Macion Capital inherits. The adaptations for own-fund use: add a paper-trading-agent and a live-deployment-agent once capital is allocated.

B1.5 — Practical recommendations for Macion Capital

1. **Keep the inherited 11-agent stack.** It already implements the research-agent pattern correctly. Don't re-architect.
2. **Add a `quant-llm-copilot` skill.** A documented prompt + retrieval template for "ask the corpus" use cases — Ph tax law, gate-stack rulings, prior strategy patterns.
3. **Buy AlphaSense or a cheaper alternative for alt-data NLP.** Don't build NLP from scratch unless you have a specific need (e.g., PH-specific filings).

4. **Use frontier LLMs via API for research; consider on-prem open-weights for IP-sensitive code-generation.** Claude / GPT-4 for ideation; Code Llama / DeepSeek Coder for code.
5. **Never let the LLM be the alpha.** All alphas must clear G1–G28 + G29–G31 before promotion. The gate stack is the judge, not the LLM.
6. **Pre-register every hypothesis.** A research memo must declare its hypothesis, mechanism, and gate-test list *before* running the backtest. The pre-registration is the antidote to data-snooping.

See `10_cost_model_phased_roadmap.md` for the cost to add this layer.

Chapter B2. AI for Quant Backtesting

BOTTOM LINE AI in backtesting is most valuable as an adversarial validator, not as an alpha engine — the read-only adversary catches the leaks the orchestrator misses.

> **The validation layer.** Where AI either catches your mistakes or hides them. The architecture that survives is the one where AI proposes, gates judge, and humans approve.

B2.1 — The backtest engine taxonomy

There are four backtest paradigms; each has different AI integration points.

B2.1.1 The four paradigms

Paradigm	What it does	Speed	Fidelity	AI integration
Vectorized	All bars at once, numpy/pandas	Very fast (seconds)	Low — no path-dependence	AI = auto-features, model
Event-driven	One event at a time with state	Slower (minutes)	Medium — handles orders, fills	AI = synthetic data, kill-shots
Agent-based	Multiple agents (market, your strategy, adversary) interact	Slow (hours)	High — emergent dynamics	AI = the agents themselves
Full replay	L3/MBO replay with queue position	Very slow (hours to days)	Highest	AI = impact estimation, queue modeling

The trade-off curve: see `04_backtest_engine_comparison.png`.

B2.1.2 Which paradigm for which fund?

- **Vectorized**: pre-discovery, parameter sweeps, sensitivity analysis. The cheap-fast screen.
- **Event-driven**: production-grade backtest before live deployment. The workhorse.
- **Agent-based**: research on emergent dynamics, market-microstructure hypotheses, regime discovery. The frontier.
- **Full replay**: HFT shops, market-makers, queue-position-sensitive strategies. Niche.

For Macion Capital (medium-frequency crypto + futures): **vectorized for discovery, event-driven for validation, agent-based for research questions that vectorized can't answer.**

B2.1.3 Open-source engines

Engine	Language	Paradigm	AI-ready	Notes
Lean (QuantConnect)	C# / Python	Event-driven	Yes (Python toolbox)	Best-in-class free engine
Zipline	Python	Event-driven	Limited	Quantopian relic, less active
Backtrader	Python	Event-driven	Limited	Hobbyist favorite
Vectorbt / Vectorbt PRO	Python (numpy)	Vectorized	Yes (ML feature pipeline)	Fast prototyping
FinRL	Python	Event-driven + RL	Yes (built for RL)	Best for RL research
Qlib (Microsoft)	Python	Vectorized + ML	Yes (built for ML)	AI-first quant platform
MLFinLab	Python	Vectorized + ML	Yes (ML tools)	Hudson & Thames
Freqtrade	Python	Event-driven	Yes (built-in freqAI)	Crypto-specific
Backtesting.py	Python	Vectorized	Limited	Lightweight
Jesse	Python	Event-driven	Limited	Crypto-specific

Macion Capital stance: Lean + Vectorbt for the main workhorses; FinRL for RL experiments; custom Python for everything else.

B2.2 — AI in the backtest pipeline

B2.2.1 The five places AI enters

1. **Data preparation** — LLM-based document extraction → structured data (see Part B1.2)
2. **Feature engineering** — auto-features, neural feature search, symbolic regression
3. **Model training** — XGBoost / NN / RL — the alpha model
4. **Synthetic data** — world models for stress tests, regime augmentation
5. **Adversarial validation** — the read-only adversary catches placebos

B2.2.2 The synthetic-data frontier

A 2025–26 trend: world models that simulate realistic market regimes for stress testing.

State of the art:

- **TimeGAN** (Yoon et al. 2019) — early GAN-based time-series synthesis
- **QuantGAN** (Wiese et al. 2020) — financial-specific GAN
- **Score-based diffusion models** (2023+) — current SOTA for time-series
- **Market-GPT** (concept, 2024) — autoregressive transformer over order flow
- **FinGPT-Forecaster** (2024) — fine-tuned Llama for financial forecasting

Honest finding: synthetic data helps with **stress testing** but is **not a substitute for real out-of-sample**. The killer use case: generate 1,000 synthetic crash regimes and test your risk model on each. If the risk model survives synthetic crashes, it's more likely to survive real ones.

Risk: synthetic data with realistic-looking patterns can fool the backtest into believing a placebo. The discipline: synthetic data is for *risk*, not for *alpha*. If you're using synthetic data to train an alpha model, you're overfit by construction.

B2.2.3 The adversarial-validation pattern

This is where AI in backtesting is most valuable.

The Macion Capital pattern:

```
Your alpha backtest produces a P&L curve
    ↓
The data-integrity agent runs an adversarial re-test:
  - Did you peek at the test set? (data leakage)
  - Did you re-use a window you tested on? (holdout burn)
  - Does the P&L survive a 30-day shift? (G10 source-shift)
  - Does the signal survive a randomized timing null? (G9)
  - Does the CI exclude zero after block bootstrap? (G8)
    ↓
If any test fails, the candidate is KILLed
```

This is the **read-only adversary** in the agent topology. It cannot propose candidates; it can only kill them. The discipline is identical to a stats referee in a journal.

B2.3 — The selection-bias defense

The single most important concept in backtesting: **your backtest is one draw from a search distribution.** The PBO, DSR, and MinBTL are the standard tools to defend against selection bias.

B2.3.1 The Bailey–López de Prado trilogy

Metric	What it measures	Floor / ceiling	Source
Deflated Sharpe Ratio (DSR)	Probability the observed Sharpe exceeds the maximum Sharpe under the null, after correcting for multiple testing	Floor ≥ 0.95 to ship	Bailey & López de Prado 2014 (JPM)
Probability of Backtest Overfitting (PBO)	Likelihood your backtest is overfit, computed via Combinatorially Symmetric Cross-Validation (CSCV)	Ceiling ≤ 0.30 to ship	Bailey, Borwein, López de Prado, Zhu 2015 (JPM)
Minimum Backtest Length	The shortest backtest length required to reject "Sharpe = 0" at significance	n_min, used as a	Bailey et al. 2014

Metric	What it measures	Floor / ceiling	Source
(MinBTL)	α	guideline	

These are the **G29, G30, G31** gates in the Macion Capital stack. Every backtest must report these.

B2.3.2 The walk-forward defense

Beyond DSR/PBO/MinBTL, walk-forward is the workhorse for out-of-sample rigor.

The pattern:

1. **Train** on T_0 to T_1
2. **Test** on T_1 to T_2 (the first OOS window)
3. Roll forward; repeat until you exhaust the data
4. Aggregate the OOS windows into a final OOS P&L

Macion Capital discipline: - ≥ 4 walk-forward folds - No single fold > 50% of total P&L - Worst-fold Sharpe ≥ −1.0 - Same sign in ≥ 4/5 macro eras

These are the **G12, G13, G14, G15** gates.

B2.3.3 The de-selection re-test

The single hardest test: **does the edge survive a fresh, untouched universe?**

Pattern:

1. Lock the original discovery dataset (dataset A).
2. Discover and promote the edge on A.
3. Pull a fresh dataset B (different time period, different assets, different venues — anything the search hasn't touched).
4. Re-run the strategy on B without changing parameters.
5. If the sign or magnitude flips → KILL.

This is what the de-selection re-test is designed to catch. The canonical example in the de-selection literature: a t = +2.40 signal on a discovery universe (dataset A) collapsed to t = −1.01 on a fresh, untouched dataset (dataset B) — a clean sign-flip that the original backtest had no power to detect. The de-selection re-test is the only gate that does. The t = +2.40 → t = −1.01 collapse is also the worked example behind the **G29 venue-lock + G30 source-shift + G31 hold-out** kill pattern documented in the Macion Capital gate stack reference.

The rule: if a signal doesn't survive a de-selection re-test, it was search luck.

B2.4 — ML defenses for backtesting

Beyond DSR/PBO/MinBTL/walk-forward, there are ML-specific defenses:

B2.4.1 Combinatorially-purged cross-validation (CPCV)

The CPCV (López de Prado 2018) purges training samples whose labels overlap with the test period and adds an embargo. This prevents the most common ML backtest leak: a forward-looking label in a training sample.

Macion Capital stack: all backtests use CPCV by default; the embargo is set to the maximum hold period of the strategy.

B2.4.2 Triple-barrier labeling

The triple-barrier method (López de Prado 2018) labels each sample by which of three barriers is hit first: 1. **Upper barrier** — take-profit 2. **Lower barrier** — stop-loss 3. **Vertical barrier** — time expiry

This produces a properly-distributed label set; it replaces the naive "future return at fixed horizon" approach that biases toward mean-reverting labels.

B2.4.3 Sample weights from meta-labeling

Meta-labeling (López de Prado 2018) trains a secondary model to predict *whether* the primary model's signal will be correct. The secondary model's predictions become sample weights for the primary model.

When to use: when your signal has high recall but variable precision. Meta-labeling sharpens the precision without sacrificing recall.

B2.4.4 The FINSABER null result — what NOT to do

The Stanford FINSABER benchmark (2024, arXiv:2411.02337, retrieved 2026-07-07) tested LLM-based trading agents on standard benchmarks and found:

- **No statistically significant alpha** vs random baselines ($p > 0.34$)
- ~37% / ~19% memorization of training-data patterns
- High variance in performance across seeds

Implication: if your "AI backtest" uses an LLM to make trade decisions, FINSABER says you'll get placebo. The LLM should orchestrate, not predict. Macion Capital's stack enforces this rule.

B2.5 — Practical recommendations

1. **Use Lean for production backtests.** Vectorized discovery; event-driven validation.
2. **Implement CPCV + embargo as a default.** Patched into the inherited quant-backtester agent.
3. **Compute DSR, PBO, MinBTL on every backtest.** These are the G29/G30/G31 gates. Floors: DSR ≥ 0.95, PBO ≤ 0.30.
4. **Run walk-forward with ≥ 4 folds.** No single fold > 50% P&L; worst-fold Sharpe ≥ −1.0.
5. **De-selection re-test before promotion.** Lock discovery dataset; re-test on fresh. If sign flips, KILL.
6. **Use synthetic data for stress tests only.** Not for alpha training.
7. **Read-only adversary agent is mandatory.** The data-integrity agent catches the data leaks the orchestrator misses.

See `04_quant_development_ai.md` for the production-engineering layer.

Chapter B3. AI for Quant Development (Production Engineering)

BOTTOM LINE AI in production engineering is a 3× productivity multiplier — Claude Code + Cursor + GitHub Copilot stack gives 1 engineer the output of 3, at ~$200/mo per seat.

> **The leverage layer.** AI here doesn't find edges — it makes the engineering team 10× more productive. The wrong toolchain, and you're paying $50/seat/month for a glorified autocomplete.

B3.1 — The production-engineering AI stack

The five places AI shows up in a quant dev team:

B3.1.1 Code generation

The single highest-leverage AI category for a small team.

State of the art (2026):

Tool	Strengths	Weaknesses	Pricing (2026)
Claude Code (Anthropic)	Agentic, repo-aware, MCP integration, long-context	Cost at scale	$20/mo Pro, $100/mo Max, $200/mo Team
Cursor	Best IDE UX, multi-file edits, Composer mode	Younger ecosystem, less mature for non-trivial workflows	$20/mo Pro, $40/mo Business
GitHub Copilot	IDE-integrated, ubiquitous, fast inline completion	Less agentic, weaker on complex multi-file tasks	$10/mo Individual, $19/mo Business
Cody (Sourcegraph)	Repo-wide context, enterprise	Smaller mindshare	Free tier, $9/mo Pro, $19/mo Enterprise
Tabnine	On-prem option, IP-sensitive	Weaker on agentic workflows	$9/mo Dev, $39/mo Enterprise
Continue	Open source, customizable	Self-hosted complexity	Free (open source)
Aider	Open source, terminal-native	Less polished	Free (open source)
Codeium / Windsurf	Free individual tier, good IDE plugin	Younger	Free / $15/mo Pro

Macion Capital choice (2026): Claude Code for agentic work (the principal uses it for the system itself), Cursor for IDE-resident development, GitHub Copilot as the team-standard fallback. Budget: ~$200/mo total for 2 FTE-equivalents.

B3.1.2 Test generation

Code without tests is liability. AI-generated tests are a force multiplier.

Tools:

- **CodiumAI / Qodo** — generates tests from code or PR description
- **Diffblue (Java)** — automated Java unit-test generation
- **Sourcery** — Python code review + test suggestion
- **In-house with Claude/Cursor** — prompt the model to write pytest cases

Macion Capital discipline: every strategy module must have ≥ 1 test per public function; every PR must include a test. AI helps generate the boilerplate; a human reviews the meaningfulness.

B3.1.3 Documentation generation

Docs are the most-neglected artifact in a small fund. AI-generated docs are good enough for internal use.

Tools:

- **Mintlify Writer** — auto-generates docs from source
- **Read the Docs + AI** — auto-generates RST
- **Claude / Cursor** — prompt the model to write docstrings + README

Macion Capital discipline: every repo gets a `README.md`, a `CLAUDE.md` (workspace context for Claude agents), and an `ARCHITECTURE.md`. AI drafts; a human refines.

B3.1.4 Observability & AIOps

Production-grade quant infra needs logging, metrics, tracing. AI helps triage.

Tools:

- **Datadog AI** — anomaly detection on logs and metrics
- **Grafana + LLM** — natural-language query of metrics
- **PagerDuty AIOps** — alert correlation
- **In-house with Claude + Loki/Elastic** — natural-language log search

Macion Capital discipline: every strategy has a structured-log schema; the ops agent queries logs in natural language. See the inherited `dashboard-keeper` agent pattern.

B3.1.5 Infra-as-code

Define infra once, version-control it, replay it.

Tools:

- **Terraform / Pulumi** — declarative infra
- **Ansible** — config management
- **Crossplane** — Kubernetes-native
- **In-house with Claude** — generate Terraform from natural-language requirements

Macion Capital stance: Terraform for AWS; everything reproducible from a single `terraform apply`. AI helps write modules; humans review security implications.

B3.2 — The MLOps layer

A small quant fund needs MLOps without the enterprise overhead. The minimum viable MLOps stack:

B3.2.1 The seven layers

Layer	Tool (free / cheap)	Tool (paid)	What it does
Experiment tracking	MLflow (self-hosted), Weights & Biases Free	Weights & Biases Pro ($50/mo/seat)	Log params, metrics, artifacts per run
Model registry	MLflow, DVC	Weights & Biases Registry	Version control for trained models
Feature store	Feast (open source)	Tecton, Hopsworks	Centralized feature catalog
Data versioning	DVC, lakeFS	DVC Pro	Version-control parquet files
Pipeline orchestration	Prefect, Airflow, Dagster	Prefect Cloud, Dagster Cloud	Schedule + monitor jobs
Model serving	BentoML, Ray Serve, Triton	SageMaker, Vertex	Deploy models to prod
Monitoring	Grafana + Prometheus	Datadog, New Relic	Track prod performance, drift

Macion Capital minimum stack (Y0–Y1): MLflow self-hosted + DVC + Prefect + BentoML + Grafana. Total cost: ~$100/mo for cloud infra (2 vCPU, 8 GB RAM, ~$50/mo with reserved instances).

Macion Capital Y2–Y3 stack: add Weights & Biases Teams ($50/mo/seat), Feast managed, Datadog. Total cost: ~$1–2k/mo.

B3.2.2 The model-risk-management layer

For a fund above $100M AUM, regulators (SEC, FCA, MAS) expect a formal MRM framework.

Components:

1. **Model inventory** — every model, version, owner, intended use
2. **Model validation** — independent review of every material model before production
3. **Ongoing monitoring** — performance + drift + data quality
4. **Issue management** — escalation, root-cause, remediation
5. **Governance** — sign-off authority, audit trail, periodic re-validation

The AI angle: AI can draft MRM documentation, automate monitoring, and pre-populate validation templates. But the sign-off must remain human — this is a regulatory expectation.

B3.3 — The cost-of-engineering tradeoff

A small fund has 1–3 engineers. AI is the force multiplier that makes 1–3 engineers feel like 5–10. The discipline:

B3.3.1 The leverage calculation

Baseline (no AI): 1 engineer ships ~5,000 production-grade lines/year. With AI assistance (2026): 1 engineer ships ~15,000 production-grade lines/year. **Multiplier: ~3×.**

But the multiplier depends on the work type: - **Boilerplate / CRUD**: 10× multiplier (AI is great at this) - **Algorithmic code**: 2× (AI helps with structure, less with insight) - **Distributed systems**: 1.5× (AI helps with patterns, less with debugging prod) - **Quantitative research code**: 2× (AI helps with numpy/pandas patterns, less with statistical insight)

B3.3.2 The token-cost discipline

LLM-assisted engineering is **expensive at scale**. A heavy AI user can burn $1k+/mo on tokens alone.

Discipline:

1. **Use prompt caching.** Claude Code and Cursor cache aggressively — design prompts to maximize cache hits.
2. **Use the right model for the task.** Sonnet for routine, Opus for judgment-heavy, Haiku for mechanical.
3. **Batch work into focused sessions.** A 2-hour focused session is more efficient than 8 short sessions.
4. **Use local models for IP-sensitive code.** Code Llama 70B / DeepSeek Coder for code that touches your alpha logic.
5. **Measure cost-per-PR.** A useful KPI: $/PR (dollars spent on AI per merged PR). Target: <$5/PR.

B3.3.3 The IP-sensitivity tradeoff

Some code is secret (alpha logic, risk models); some is not (infra, boilerplate).

Discipline:

- **Secret code**: open-weights models on-prem (Code Llama, DeepSeek, Qwen)
- **Non-secret code**: frontier APIs (Claude, GPT-4)
- **Mixed**: Claude / GPT-4 for the boilerplate + local for the secret parts

For Macion Capital, the alpha logic lives in the inherited research stack (already secret); the new own-fund code is mostly infra + dev tooling (non-secret). So frontier APIs are fine for most of the work.

B3.4 — Production architecture patterns

B3.4.1 The reproducibility pattern

Every backtest must be regenerable from a single command. The discipline:

- **Single config file** — every parameter, every path, every random seed
- **Single data manifest** — SHA256 hash of every input file
- **Single run script** — `python run_backtest.py --config config.yaml`
- **Single output manifest** — what was produced, where, with what hash

This is the "show, don't tell" rule from the Macion Capital operating model (see `OPERATING_MODEL.md` §7; the chart-pipeline contract in `CLAUDE.md` §9.2 enforces it in pictures, not just prose).

B3.4.2 The pipeline-as-code pattern

Backtests, paper-trading, live-trading all run through the same code path. Differences are config:

- `mode: research` — runs on cached data, no orders
- `mode: paper` — runs on live data, no real orders
- `mode: live` — runs on live data, real orders via broker API

This is **infrastructure parity**: the code that runs in research is the code that runs in production, with different inputs.

B3.4.3 The dual-write pattern for live trading

Every live order is dual-written: to the broker (real) and to a local shadow ledger (audit). The shadow ledger reconciles against the broker statement at EOD.

This is **the audit-trail discipline** that lets you answer "what did we trade, when, why?" at any point.

B3.4.4 The kill-switch pattern

Every strategy has a kill switch:

- **Soft kill** — stop opening new positions; let existing run
- **Hard kill** — flatten all positions immediately

The kill switch is triggered by: - Drawdown > X% - Volatility > X σ - Daily loss > Y USD - Manual PM override

The kill switch is **always reachable** — even if the main system is down, the kill switch must work via a separate channel (e.g., a Telegram bot that fires an order to the broker).

B3.5 — Tooling comparison (2026)

B3.5.1 Code assistants (cost comparison)

Tool	Cost/mo (1 seat)	Multi-file edits	Agentic	Self-host	IP-safe
Claude Code	\$20–\$200	✓	✓	✗	✗
Cursor	\$20	✓	✓	✗	✗
Copilot	\$10	✗	✗	✗	✗
Cody	\$9	✓	partial	✓	✓
Tabnine	\$9	partial	✗	✓	✓
Continue	free	✓	✗	✓	✓
Aider	free	✓	✓	✓	✓
Windsurf	\$15	✓	✓	✗	✗

Pricing as of 2026-07, retrieved 2026-07-07.

B3.5.2 ML platforms

Platform	Cost/mo	Notes
AWS SageMaker	Pay-per-use, ~\$100/mo minimum	Enterprise-grade; overkill for small funds
Google Vertex AI	Pay-per-use	Same as above
Databricks	\$0.07/DBU + compute	Best for lakehouse architecture
Snowflake	Pay-per-use	Data warehousing, not pure ML
Anaconda	Free / \$25/mo	Python distribution; not really ML platform
Lightning AI	Free tier, \$30/mo Pro	Best for research-to-prod
Modal	Free tier, \$30/mo Pro	Serverless GPU, great for inference
Replicate	Pay-per-use	API for open-source models
Hugging Face	Free tier, \$9/mo Pro	Model hub + inference

B3.5.3 Vector databases

DB	Cost	Notes
Pinecone	\$70/mo Standard	Managed, serverless option available
Weaviate	Self-hosted free, cloud \$25/mo	Open source, good for hybrid search
Qdrant	Self-hosted free, cloud \$25/mo	Open source, fastest in benchmarks
Chroma	Free self-hosted	Best for local dev
Milvus	Self-hosted free, Zilliz cloud \$65/mo	Best for billion-scale

For Macion Capital: **Chroma for local dev, Qdrant for production** when needed.

B3.6 — Practical recommendations

1. **Adopt Claude Code + Cursor as the AI dev stack.** Budget \$200–\$500/mo total.
2. **Build the minimum MLOps stack early.** MLflow + DVC + Prefect + BentoML + Grafana. ~\$100/mo.
3. **Implement reproducibility + dual-write + kill-switch patterns.** All three, from day 1.
4. **Establish an IP-sensitivity classification.** Some code is secret; route it to local models.
5. **Track \$/PR as a KPI.** Target <\$5/PR for the team.
6. **Document everything.** README, CLAUDE.md, ARCHITECTURE.md per repo. AI drafts, human refines.

See `05_agentic_ai_systems.md` for the orchestration layer.

Chapter B4. Agentic AI Systems

BOTTOM LINE Multi-agent systems cost ~15× single-agent tokens and recover ~80% of the gain from the extra tokens alone — the architecture's marginal value is in specialization and isolation, not raw capability.

> **The orchestration layer.** Multi-agent systems turn one LLM into a team. Get the topology right, and your cost goes up 15× but your research throughput goes up 90%. Get it wrong, and you've built an expensive way to fool yourself.

B4.1 — The agentic landscape (2026)

B4.1.1 The five agent topologies

Topology	What it is	When to use	Cost multiplier
Single agent	One LLM with tools + memory in a loop	Default; 80% of production value	1×
Orchestrator-worker (star)	One orchestrator routes to specialist workers	When tasks have distinct expertise	5–15×
Hierarchical	Multi-level org chart of agents	When tasks decompose recursively	10–30×
Debate / adversarial	Multiple agents argue; a judge picks	High-stakes decisions	5–20×
Swarm	Many agents, no central control	Experimental; rare in production	10–50×

B4.1.2 The Anthropic multi-agent research finding

The most-cited 2024–25 benchmark on multi-agent systems: Anthropic's internal study (cited in Claude docs, retrieved 2026-07-07).

Headline numbers (Anthropic 2025-06 benchmark, $N \approx 100$ research tasks per condition):

- Multi-agent research system beat a single agent by **+90.2%** on breadth-first research tasks
- Used **~15× the tokens** of a single agent
- **~80% of the performance gain was explained by token spend alone** — i.e., you could recover most of the gain by giving a single agent more compute, at much lower coordination cost

Implication: multi-agent systems are expensive. The 90.2% headline is real, but most of it is from the extra tokens, not from the architecture. The architecture's marginal value is in *specialization* and *isolation*, not raw capability.

B4.1.3 The MAST failure taxonomy

The Berkeley MAST study (NeurIPS 2025) measured **41–86.7% failure rates across seven production multi-agent frameworks**. The failures cluster into 14 modes, in three buckets:

1. **Specification failures** — task misunderstood, ambiguous requirements
2. **Inter-agent misalignment** — agents disagree on goals, handoff contracts broken
3. **Task verification failures** — no independent check that the work is done correctly

Implication: multi-agent systems fail in *known, predictable* ways. The discipline: build the system to fail safely (orchestrator detects failures, retries with a different approach, escalates to human).

Source: MAST: Multi-Agent System Failure Taxonomy, NeurIPS 2025, retrieved 2026-07-07.

B4.1.4 The FINSABER null result

The Stanford FINSABER benchmark (2024) found that LLM-as-trader shows **no statistically significant alpha** ($p > 0.34$) and exhibits ~37% / ~19% memorization.

Implication: if your agent system makes the trade decisions, you're placebo-bound. Agents propose; gates judge. Same lesson as before.

Source: arXiv:2411.02337 (FINSABER), retrieved 2026-07-07.

B4.2 — The agent topology for a quant fund

Macion Capital's research platform is a textbook orchestrator-worker + read-only adversary. Let me decompose it.

B4.2.1 The star topology

```
                 ┌─ quant-idea-miner ───── (cheap candidate gen)
                 │
quant-researcher ── quant-preflight-triage (G1–G7)
   (orchestrator)
                 │
                 ├─ quant-researcher (deep G1–G22)
                 │
                 ├─ quant-backtester (G8–G28 + walk-fwd + MC)
                 │
                 ├─ quant-data-integrity (read-only adversary)
                 │
                 ├─ quant-strategy-scribe (memo + report + SSOT)
                 │
                 └─ quant-methodology-librarian (gate coherence)

supporting:
  quant-regime (regime vectors)
  prediction-markets-expert
  dashboard-keeper (telemetry, down-tiered to Haiku-equivalent)
  quant-executor (deferred — no live stack yet)
```

Properties:

- **11 agents**, each with a ~2,400-line charter
- **Restricted tool grants** per agent (least privilege)
- **Contracted hand-offs** (typed packets, not prose)
- **Read-only adversary** that can flag but not override
- **Single source of truth** — the strategy-scribe owns the registry

This is the architecture Macion Capital inherits. It is **portable** — the same skeleton has been instantiated for venture incubation (7 agents), editorial production (8 agents), and a personal knowledge base (5 agents). See the existing AI portfolio (`~/Contingency/AI_Engineer_Portfolio`) for the full pattern.

B4.2.2 The escalation ladder

When to use a single agent vs. multi-agent — the discipline:

Step	Use when
1. Better prompt	Always try first. 80% of "we need an agent" is a prompt that needs work.
2. Better tools	The agent lacks capabilities. Add tools; don't add agents.
3. Sub-agents	The task has *distinct expertise* or *distinct permissions* — separate the contexts.
4. Full orchestration	You need *context isolation* + *parallelism* + *handoff contracts*.

Anti-patterns:

- **Premature orchestration** — splitting a single agent into 5 before measuring
- **Tool sprawl** — giving every agent every tool
- **Hand-off prose** — agents communicating in natural language instead of typed packets

B4.2.3 The contract pattern

Every agent hand-off is a typed packet. Example:

```
class ResearchHandoff(BaseModel):
    candidate_id: str
    hypothesis: str
    mechanism: str
    data_refs: List[str]      # paths to data files used
    fire_count: int
    era_coverage: List[str]
    gate_results: Dict[str, GateResult]   # G1, G2, ..., G22
    verdict: Literal["KILL", "PROMOTE"]
    kill_pattern: Optional[str]
    evidence_paths: List[str]
```

The `quant-strategy-scribe` consumes this packet, validates the schema, and writes the registry. **No prose handoffs; no Slack-style "agent A told agent B".** This is the contract discipline that keeps multi-agent systems debuggable.

B4.3 — Agent frameworks (2026)

Framework	Strengths	Weaknesses	When to use
Claude Agent SDK / Claude Code	Best-in-class agent loop, MCP, subagents, prompt caching	Locked to Anthropic ecosystem	Default for serious work
LangGraph	Graph-based, flexible, well-typed	Steeper learning curve	Complex workflows with branches
CrewAI	Role-based, easy to start	Less control over context isolation	Quick prototypes
AutoGen (Microsoft)	Conversational, multi-agent	Has had stability issues	Research, conversational agents
DSPy	Prompt-programming, optimizer-driven	Different paradigm; not a "framework"	When you want the system to optimize its own prompts
OpenAI Swarm	Lightweight, simple	Very young	Toy projects
Custom from scratch	Full control	You own all bugs	When you need audit-grade reliability

Macion Capital stance: Claude Code for production agents; LangGraph for complex workflows; no framework for the gate stack (own the math). DSPy is being evaluated for prompt optimization.

B4.3.1 Why custom for the gate stack

The 31-gate eval harness (G1–G28 + G29–G31) is implemented **scipy-free in numpy**, owned by the principal, and committed to the repo. This is deliberate: framework abstractions hide numerical bugs. The gate stack is the audit trail; if a gate fails to fire, the principal must be able to read the code and explain why.

B4.4 — Tooling for agents

B4.4.1 The MCP (Model Context Protocol) standard

MCP (Anthropic, late 2024) is the de-facto standard for connecting agents to tools. Every vendor now publishes an MCP server:

- **Databento MCP** — market data
- **GitHub MCP** — code + PR
- **Slack MCP** — messaging
- **Notion / Obsidian MCP** — knowledge base
- **S3 MCP** — file storage
- **Custom MCPs** — proprietary tools

The MCP standard makes the "tool-use" half of agent design plug-and-play. The "agent logic" half is still custom.

B4.4.2 The vector-DB-for-context pattern

For long-running research agents, vector DBs store:

- Past research memos
- Gate rulings
- Kill-log entries
- Market-data context windows

The agent retrieves before generating, so it doesn't repeat past mistakes. This is the "RAG over your own work" pattern.

Tools:

- Chroma (local dev)
- Qdrant (production, self-hosted)

- Pinecone (production, managed)

B4.4.3 The structured-output contract pattern

Every agent output is a JSON schema (Pydantic / Zod). The discipline:

- The agent **must** output schema-conformant JSON
- A validator checks the schema on every output
- If invalid, the agent retries with a clarification prompt
- A counter on validation failures per agent is a KPI

Why: LLM-as-judge only works if the output is structured. Prose outputs are unverifiable.

B4.5 — Specific agentic patterns for quant

B4.5.1 The research-agent pattern

```
User: "Test whether BTC hour-22 UTC has positive drift"
  ↓
quant-researcher orchestrator
  ├─ quant-idea-miner: produces 3 candidate framings
  ├─ quant-data-integrity: pulls BTC 1h data, checks PIT
  ├─ quant-preflight-triage: runs G1-G7
  ├─ quant-backtester: runs G8-G28 + walk-fwd
  ├─ quant-regime: tags each year with macro regime
  └─ quant-strategy-scribe: writes the memo + verdict
  ↓
Output: research memo with all gates + verdict + evidence paths
```

B4.5.2 The execution-agent pattern (future)

```
PM: "Deploy MC-001 to paper trading on Hyperliquid"
  ↓
quant-executor orchestrator
  ├─ quant-data-integrity: validate account + risk limits
  ├─ quant-execution-router: TWAP/VWAP logic
  ├─ quant-risk-monitor: real-time VaR
  └─ quant-strategy-scribe: log every order
  ↓
Output: paper-trade ledger with daily reconciliation
```

Note: the `quant-executor` is designed but not yet deployed at Macion Capital (no live venue yet).

B4.5.3 The LP-reporting agent pattern (future)

```
Date: 1st of month
  ↓
quant-lp-reporting orchestrator
  ├─ quant-data-integrity: pull NAV, P&L, exposures
  ├─ quant-strategy-scribe: generate KPI dashboard
  ├─ loremaster (writing agent): draft narrative
  └─ principal: review + sign-off
  ↓
Output: monthly LP letter (PDF)
```

This is the gap identified in Part A — Macion Capital doesn't yet have an LP-facing layer.

B4.6 — Cost & reliability economics

B4.6.1 The token-cost reality

Topology	Tokens per task	Relative cost
Single agent, 1 prompt	~5K	1×
Single agent, 10-step loop	~50K	10×
Orchestrator + 3 workers	~150K	30×
Orchestrator + 7 workers + adversary	~500K	100×

For a research operation running 100 candidates/day at 500K tokens each, that's **50M tokens/day**. At Claude Sonnet 4.6 pricing ($3/$15 per M tokens in/out), that's **~$150/day = $4500/mo just for research**.

Discipline:

- Use prompt caching aggressively (10× cost reduction on repeated context)
- Use smaller models for routine tasks (Haiku ~$0.80/$4 per M)
- Reserve Opus for judgment-heavy gates

B4.6.2 The latency reality

Multi-agent systems are slow. A 5-agent orchestration with sub-agent loops can take **5–30 minutes per research task**. That's fine for batch research; it's not fine for live trading.

Discipline:

- Use multi-agent for **research**, not for **live execution**
- Pre-cache common sub-tasks
- Set timeouts on every agent call

B4.6.3 The reliability reality

From MAST (NeurIPS 2025): 41–86.7% failure rate across 7 production frameworks.

Discipline:

- Every agent output is validated against a schema
- Every multi-step task has a verification step (LLM-as-judge or numeric gate)
- Failures are logged and retried with a different approach
- Human escalation after N retries

B4.7 — Practical recommendations

1. **Default to orchestrator-worker.** It's the right balance of cost and capability.
2. **Own the gate stack numerically.** No framework abstractions on the audit trail.
3. **Use structured outputs everywhere.** Pydantic schemas; reject-and-retry on invalid.
4. **Use MCP for tool integration.** It's the standard; vendors are racing to publish.
5. **Use prompt caching aggressively.** 10× cost reduction on repeated context.
6. **Measure failures.** Counter on validation failures per agent is a KPI.
7. **Read the MAST failure taxonomy.** The 14 failure modes are the spec for what to defend against.

See `06_loop_engineering.md` for the loop design that ties this together.

Chapter B5. Loop Engineering

BOTTOM LINE The tighter the loop, the higher the risk — every self-improvement loop needs a numeric anchor (gate) and a human-in-loop brake to avoid silent corruption.

> **The recursive-improvement layer.** Where agents modify their own behavior. Get this right and your system gets smarter over time. Get it wrong and you have a feedback loop that quietly corrupts your edge.

B6.1 — What loop engineering is

Loop engineering is the discipline of designing **recursive self-improvement loops** in agent systems. The six loop types in a quant fund:

B6.1.1 The six loop types

Loop	Cadence	Risk	Examples
Inner loop (single agent reason+act)	Seconds	Low	Tool use, ReAct
Multi-agent loop (orchestrator + workers)	Minutes	Medium	The Macion Capital research pipeline (orchestrator + 14 workers + read-only adversary)
Self-improve loop (reflexion, DSPy)	Hours to days	High	Agent modifies own prompts after seeing results
Human-in-loop (PM approvals, sign-off)	Days to weeks	Low	Promotion verdicts, capital allocation
Cross-strategy loop (meta-controller)	Days	Medium	Capital allocation across strategies
Outer loop (regulatory, audit)	Months to years	Low but binding	GIPS audit, SEC review

See `06_loop_taxonomy.png`.

B6.1.2 The tighter the loop, the higher the risk

The fundamental tension: **faster loops = faster learning = faster corruption**.

- Inner loops (seconds) are safe because they execute in a sandbox (no real capital at risk).
- Multi-agent loops (minutes) are medium-risk because they make decisions but don't yet act on capital.
- Self-improve loops (hours–days) are high-risk because the system modifies its own behavior; a bad modification can compound.
- Human-in-loop is the brake.
- Outer loops are the audit trail.

B6.2 — Specific loop engineering techniques

B6.2.1 ReAct (Reason + Act)

The foundational agent loop. From Yao et al. 2022.

```
Thought: I need to check if BTC has positive drift in hour 22 UTC.
Action: python compute_drift("BTCUSDT", hour=22)
Observation: mean_return = +5.2 bps, t-stat = 2.3, n = 1100
Thought: The drift is positive and significant.
Action: write_finding(memo, ...)
```

Production-ready: yes, but with strict prompt templates and JSON-schema outputs.

B6.2.2 Reflexion (Shinn et al. 2023)

Self-reflection after a failure. From the NeurIPS 2023 paper.

```
Action: I tried X.
Result: X failed (test 1: t-stat < 1.5).
Reflection: Why did X fail? The mechanism was [explanation].
Next attempt: Let me try Y instead.
```

Production-precarious: Reflexion can compound errors if the self-reflection is wrong. Use with care; require a numeric anchor.

B6.2.3 Reflexion in the Macion Capital stack

The `quant-researcher` agent has a built-in reflection step: after every KILL verdict, the agent writes a 1-paragraph "what would I do differently next time" note. These notes accumulate in the kill log and are retrieved by future agents.

This is **reflexion-with-numeric-anchor**: the numeric gate is the anchor; the reflection is for process improvement, not for self-modification.

B6.2.4 Tree of Thoughts (Yao et al. 2023)

Multi-path search over reasoning trees.

```
        (initial prompt)
          /     |     \
     path A  path B  path C
       |       |       |
   outcome outcome outcome
       \      |      /
        (best-of-N selection)
```

Production-precarious: cost scales with tree size. Best for discrete problems with clear scoring (e.g., "which of these 5 hypotheses is most testable?"), not for continuous reasoning.

B6.2.5 MCTS for prompts (DSPy-style)

From Stanford DSPy (Khattab et al. 2024).

```
Prompt version A: "Analyze the data and write a hypothesis"
  ↓
Test on 100 examples → score 0.7
  ↓
Mutate: "Given the data with features X, Y, Z, propose a falsifiable hypothesis"
  ↓
Test on 100 examples → score 0.85
  ↓
Continue mutation until score plateaus
```

Production-ready in narrow domains (prompt tuning for a specific task); **fragile** when applied broadly (you can overfit the prompt to a held-out set).

Source: DSPy paper, arXiv:2310.03714, retrieved 2026-07-07.

B6.2.6 Self-RAG, CRAG, and adaptive retrieval

Self-reflective RAG (Asai et al. 2023) and Corrective RAG (Yan et al. 2024) add retrieval-time reflection:

- Retrieve doc
- Judge relevance
- If irrelevant, re-retrieve with refined query
- Generate answer with cited passages

Production-ready for LLM-driven research where citations matter.

B6.2.7 Constitutional AI (Anthropic 2022) and RLAIF

Train the model to follow a "constitution" (set of principles) via self-critique. RLAIF (RL from AI Feedback) replaces human feedback with AI feedback.

Production-precarious: the constitution must be carefully designed; bad constitution = bad model.

B6.2.8 Process Reward Models (PRM) vs Outcome Reward Models (ORM)

- **ORM**: score the final answer
- **PRM**: score each step of the reasoning

PRM is more sample-efficient for multi-step reasoning but harder to train.

Production-ready for code generation (e.g., OpenAI's PRM for o1); **research-only** for general reasoning.

B6.3 — The Macion Capital loop architecture

The Macion Capital loop is a multi-agent loop with explicit reflection, numeric gates, and human approvals.

B6.3.1 The loop structure

```
                ┌──────────────────────────┐
                │  PM (human)              │
                │  KILL / PROMOTE          │
                └──────────────────────────┘
                             │
                             │ (verdict)
                             ▼
┌──────────────────────────────────────────────────┐
│   quant-strategy-scribe (writes to registry)     │
└──────────────────────────────────────────────────┘
                 │
                 │ (registry entry)
                 ▼
┌──────────────────────────────────────────────────┐
│   quant-researcher (orchestrator)                │
│   ├── idea-miner → triage → backtester → scribe  │
│   ├── data-integrity (read-only adversary)       │
│   └── methodology-librarian (gate coherence)     │
└──────────────────────────────────────────────────┘
                 │
                 │ (memo + verdict)
                 ▼
┌──────────────────────────────────────────────────┐
│   PM daily report / pre-flight approval          │
└──────────────────────────────────────────────────┘
```

The PM is the human-in-loop. The data-integrity agent is the read-only adversary. The methodology-librarian keeps the gate stack coherent.

B6.3.2 The reflection step

After every KILL: 1. `quant-researcher` writes a 1-paragraph reflection: what killed it, what would have worked 2. `quant-strategy-scribe` adds the reflection to the kill log 3. `quant-idea-miner` retrieves past kill reflections before generating new candidates

This is **reflexion-without-self-modification**: the system reflects, but only on the *log*, not on its own code.

B6.3.3 The audit-trail discipline

Every action is logged: - Every agent call (input, output, token cost, latency) - Every gate verdict (PASS/FAIL/WARN + numeric value) - Every promotion decision (PM signature + timestamp) - Every kill (pattern + reflection)

This is the audit trail that satisfies the SEC's model-risk-management expectations, even for a pre-launch fund.

B6.4 — The failure modes of self-improvement

B6.4.1 The corruption loop

If the system modifies its own behavior based on outcomes, it can drift toward patterns that *look* good but aren't real.

Example: an RL-trained execution agent discovers that the backtest rewards certain order-placement patterns; it learns to exploit them in backtest; the patterns don't work in live; the system keeps getting "better" in backtest while losing money in production.

Defense: the **de-selection re-test**. Fresh, untouched data. If the agent's self-improvement doesn't transfer, it's corruption.

B6.4.2 The reward hacking loop

If the reward function is misspecified, the agent will find ways to maximize it that aren't intended.

Example: an alpha-mining agent is rewarded for high Sharpe; it discovers a high-Sharpe strategy that has 100% drawdown in 2008; the strategy is a lemon.

Defense: the **multi-objective reward** (Sharpe AND max DD AND recovery time AND turnover), and the **pre-registered risk gates** (G23–G28).

B6.4.3 The distributional-shift loop

If the training data drifts from the live data, the agent's learned behavior stops working.

Example: an NLP model trained on 2020–2023 filings sees 2024 filings with new disclosure patterns; performance degrades silently.

Defense: the **continuous monitoring** + **drift detector** + **automatic retraining**.

B6.4.4 The herding loop (AI–AI feedback)

If many funds use similar AI agents, their actions become correlated, creating herding behavior in markets.

Example: 100 hedge funds all use the same sentiment-NLP model; they all sell on a bad-news day; the sell pressure is amplified; the price drops more than fundamentals warrant; the next day, the same funds all buy on the bounce.

Defense: diversification of AI vendors, **diversification of prompts**, **explicit anti-correlation checks** in the portfolio construction step.

B6.5 — Loops you should NOT build

1. **Self-modifying code agents that touch the gate stack.** Never. The gate stack is the audit trail.
2. **RL agents that directly allocate capital.** Too high a blast radius. Use RL for execution, not allocation.
3. **Multi-agent systems that make promotion decisions.** Always human.
4. **Recursive self-improvement without a numeric anchor.** Always anchor.
5. **Cross-firm collaborative learning.** IP and compliance make this very hard.

B6.6 — Practical recommendations

1. **Default to a multi-agent loop with a read-only adversary.** The orchestrator proposes; the adversary (`quant-data-integrity`) tries to kill; the gates judge. The pattern is workspace-OWN and documented in `CLAUDE.md` §2.
2. **Reflexion on the log, not on the code.** The reflection step writes to the kill log, never to the agent's prompt.
3. **Use DSPy for narrow prompt optimization.** Don't apply broadly.
4. **Anchor every self-improvement loop to a numeric gate.** Without an anchor, the system drifts.
5. **Always have a human-in-loop for promotion decisions.** Non-negotiable.
6. **Test for herding.** If your agents all use the same vendor, diversify.
7. **Build the audit trail from day 1.** Every agent call, every verdict, every decision.

See `07_firm_case_studies_institutional.md` for who does this in production.

Part C. Institutional Case Studies

What this part answers. Who is doing what? 15+ institutional firms with public AI signals, with the discipline to label what's verified vs inferred. The pattern: every firm is using AI, but the public-facing claims are 6–18 months behind the actual implementation.

Chapter C1. Institutional firms — Renaissance, Two Sigma, Citadel, Bridgewater, ...

BOTTOM LINE Every institutional firm is using AI; the public-facing claims are 6–18 months behind the actual implementation, and closed-stack firms (Renaissance, Two Sigma) are best understood via hiring signals.

Who does what in 2026. 15 institutional firms with public AI signals, plus the discipline to label what's verified vs. inferred vs. unverified. The pattern: every firm is using AI, but the public-facing claims are 6–18 months behind the actual implementation. Tag legend: **[STATED]** = directly quoted from a firm-controlled page; **[INFERRED]** = derived from public signals (job titles, conference attendance, hiring patterns); **[NEEDS VERIFY]** = claim looks reasonable but no primary source.

The standing rule for this section. A claim's truth-value is graded by *what evidence backs it*. The strongest evidence is a public engineering blog with code (Optiver, Man, Two Sigma). The next strongest is a public AI product announcement + verification by a primary source (Bridgewater AIA Forecaster via Bloomberg). The weakest is a Wikipedia statement with no firm primary source — keep these tagged `[NEEDS VERIFY]`.

C1.1 — Renaissance Technologies

AUM: ~$130B (Medallion closed to outside capital since 1993; RIEF smaller) **Founded:** 1982 **HQ:** East Setauket, NY **Leadership:** Peter Brown (CEO), Robert Mercer. Jim Simons died 2024.

AI / ML posture (2026):

- **[STATED, Wikipedia]** "approximately 150 researchers and computer programmers"; "approximately half of Renaissance's employees hold PhDs in scientific disciplines" — subjects include mathematics, statistics, theoretical and experimental physics, astronomy, computer science.
- **[STATED, Wikipedia]** "Wall Street experience is frowned on" — the firm recruits from academia, not from other funds.
- **[STATED, Wikipedia]** Petabyte-scale data warehouse of "events peripheral to financial and economic phenomena"; "scalable technological architectures"; methodology explicitly described as "quantitative models derived from mathematical and statistical analysis" using "financial signal processing techniques including pattern recognition."
- **[STATED, Wikipedia — historical]** Hired "numerous speech-recognition experts from IBM," including current leadership. The IBM mid-1980s speech group used HMMs and discriminative training — this is the **HMM / Baum-Welch lineage** that bridges to Renaissance's empirical work.
- **[INFERRED]** The careers index (rentec.com/careers) shows "Research Scientist / Research Engineer / Data Programmer / Research Infrastructure Programmer / Real-Time Trading Programmer" titles but **zero "AI Engineer," "ML Researcher," or "LLM Engineer"** — consistent with a statistical / signal-processing-first philosophy that pre-dates the deep-learning wave.
- **[NEEDS VERIFY]** Any specific claim about current Renaissance ML techniques (transformers, RL, deep nets).

Public sources: `en.wikipedia.org/wiki/Renaissance_Technologies` (retrieved 2026-07-07); `rentec.com/careers` (retrieved 2026-07-07); Peter Brown, "A Perspective on the Medallion Fund," various talks 2017–2024; Gregory Zuckerman, *The Man Who Solved the Market* (2019, historical).

Lesson for Macion Capital: Renaissance is a closed system; what's transferable is the **culture** (PhD-heavy, falsification, no public marketing), not the tech. The HMM/pattern-recognition DNA is irrelevant to a 2026 quant fund that wants to use LLMs — but the **methodological discipline** is directly relevant.

C1.2 — Two Sigma

AUM: ~$60–70B **Founded:** 2001 **HQ:** New York, NY **Leadership:** Co-CEOs Carter Lyons & Scott Hoffman since Sept 30, 2024. CTO **Jeff Wecker**. Chief AI Innovation Officer **Matt Greenwood**. Head of AI Core **Mike Schuster** (formerly senior scientist at NVIDIA NLP / Siri).

AI / ML posture (2026):

- **[STATED, Wikipedia]** Uses "artificial intelligence, machine learning, and distributed computing" for trading strategies.
- **[STATED, twosigma.com]** Self-reported scale: "10,000+ data sources, 380+ petabytes stored" on its homepage.
- **[STATED, ICML 2025 recap] Firdaus Janoos** (Two Sigma researcher) published an ICML 2025 conference recap on 2025-07-31 reviewing 3,339 papers — game-theoretic statistics (e-values), Hopfield networks linked to Transformers/Diffusion, scaling laws, active learning, Bayesian optimization, RL. The post is the cleanest public signal of Two Sigma's ML research agenda.
- **[STATED, careers]** Open roles include "AI Solutions Developer" (NY, Engineering); multiple "LLM Engineer, Financial Domain" and "ML Researcher, Time Series" listings.
- **[STATED, Wikipedia, 2016–18]** Created the **Halite AI programming game** — a public AI/ML competitive-coding event. Halite II ran Oct 2017–Jan 2018. Long-standing AI-talent-pipeline signal.
- **[STATED, 2022]** Became a Chainlink node operator to support blockchain-based hybrid smart contracts.
- **[STATED, January 2026 — event] Sold Venn (portfolio analytics platform) to Insight Partners**, merged with Solovis. Venn was always a **portfolio analytics** platform for institutional investors (asset owners, wealth managers, consultants) — **not** an event-driven news signals product. The divestiture suggests Venn was a standalone commercial product rather than core trading IP.
- **[STATED, September 2025]** DOJ indicted former quant researcher **Jian Wu** for allegedly manipulating "algorithmic trading models," allegedly causing $165M in client damages; SEC filed parallel civil suit.
- **[STATED, May 2026] Two Sigma Ventures spun out as independent firm Deviation Capital** (~$2B AUM).

Notable named AI leadership (per Two Sigma Insights): - Jeff Wecker (CTO) - **Matt Greenwood** (Chief AI Innovation Officer). Quote: *"The future isn't AI replacing humans. It's humans who use AI well replacing humans who don't."* - **Mike Schuster** (Head of AI Core; ex-NVIDIA NLP / Siri) - **Jin Choi** (Head of Technique Forecasting) - **Ben Wellington** (Head of Complex Feature Engines)

[INFERRED] SIGMA X — referenced but no public URL verifiable [NEEDS VERIFY].

Public sources: `twosigma.com/articles` (retrieved 2026-07-07); `twosigma.com/articles/icml-2025-key-ideas-on-llms-human-ai-alignment-and-more/` (Firdaus Janoos, 2025-07-31); `careers.twosigma.com`; `twosigma.com/ventures`; `en.wikipedia.org/wiki/Two_Sigma`.

Lesson for Macion Capital: Two Sigma's public research cadence (Janoos ICML recap, RLlib+Ray 2021 stack write-up) is the most transparent institutional ML-research layer. **The Venn sale** is the signal: trading IP stays in-house; commercial product lines spin out or sell. The pattern matters for an own-fund track thinking about which parts to productize vs. keep proprietary.

C1.3 — D. E. Shaw

AUM: ~$65B (D. E. Shaw & Co. trading arm); DESRES (research subsidiary) is separate. **Founded:** 1988 **HQ:** New York, NY

AI / ML posture (2026) — *the most explicit public ML-researcher JD in the industry*:

- **[STATED, careers] ML Researcher** role requires: **PyTorch, JAX, CUDA, XLA, Triton, PTX** (kernel-level GPU programming); focus on "knowledge discovery in financial data," deep-learning experimentation, high-performance training workflows; salary **$275K–$350K**. This is the only JD I found that explicitly lists Triton + PTX at a hedge fund.
- **[STATED, careers] Applied AI Engineer** role: "build AI agents and agentic systems"; orchestration, tool use, memory, multi-step reasoning, **"large-scale retrieval systems"** (RAG-style), AI coding agents/developer tooling, reusable agent "skills," vendor-agnostic model integration; salary $225K–$275K.

- **[STATED, careers] AI Product Engineer — Cove (PE Platform)**: "1+ year building and integrating LLM-powered systems into production," "agentic frameworks, autonomous agents, tool use, multi-step reasoning"; salary $185K–$250K.
- **[STATED, careers]** Firm-wide **AI Enablement** organization — Senior PM Applied AI, PM AI Vendor Tools, AI Enablement Strategist (L&D), AI Specialist (Human Capital).
- **[STATED, Wikipedia]** D. E. Shaw Research (DESRES, biochemistry subsidiary) develops the **Anton** special-purpose supercomputer for molecular dynamics and **Desmond** software; publishes in Nature/Science. NYC + Durham NC + Hyderabad India teams. DESRES is unlikely to be driving trading-side ML infrastructure.
- **[STATED, D. E. Shaw Ventures portfolio]** Strategic investments in AI infrastructure: Pinecone (vector DB), Luma AI, and others. **Pinecone in their portfolio is significant** — they think vector search is load-bearing.

[INFERRED] D. E. Shaw has the most transparent public GenAI infrastructure posture of the 15 firms. The Applied AI Engineer JD reads as if written by someone who runs LangChain / LlamaIndex / Anthropic SDK daily. Required GPU stack (Triton/PTX/CUDA + JAX + XLA) implies **custom kernels** rather than off-the-shelf PyTorch wrappers — the team is tuning training workflows at the kernel level for cost/perf.

Public sources: `deshaw.com/careers/machine-learning-researcher-4954` (retrieved 2026-07-07); `deshaw.com/careers/applied-ai-engineer-5375` (retrieved 2026-07-07); `deshaw.com/careers/ai-product-engineer-private-equity-platform-5637` (retrieved 2026-07-07); `deshaw.com/careers` (careers index, retrieved 2026-07-07); `deshawresearch.com`; `en.wikipedia.org/wiki/D._E._Shaw_%26_Co.`; `en.wikipedia.org/wiki/D._E._Shaw_Research`.

Lesson for Macion Capital: D. E. Shaw's JD is the most explicit public template for what an applied AI engineer at a systematic fund actually does. The **Triton/PTX kernel-level GPU programming** signal tells you the team is optimizing training cost/perf at the kernel level — that's an Y2+ level of investment, not Y0. But the **agent + RAG + tool-use** signal *is* Y0-applicable: a quant-research agent with retrieval over kill log + memos + gate rulings is exactly the kind of system D. E. Shaw's "Applied AI Engineer" is building.

C1.4 — Citadel / Citadel Securities

AUM: ~$65.9B (Citadel LLC) + market-making (Citadel Securities) **Founded:** 1990 **HQ:** Miami, FL (Citadel LLC); Citadel Securities separate

AI / ML posture (2026):

- **[STATED, LinkedIn self-description]** Citadel Securities is a "technology-driven, next-generation global market maker" using "quantitative research, computing, machine learning, and AI."
- **[STATED, Reuters Next NYC 2025-12]** CTO Umesh Subramanian announced an internal **"Citadel AI Assistant"** chatbot, trained on licensed third-party content (transcripts, filings, brokerage research, Citadel's own strategies); "almost all" equities investors use it.
- **[STATED, Anthropic customer page, retrieved 2026-07-07] Atte Lahtiranta (Head of Data Science / AI)** is a named Claude customer; quote on Claude for Excel: *"build and update coverage models, separate signal from noise, and pressure-test work"* — described as *"a step-change in efficiency."*
- **[STATED, ICML 2026 attendance]** Citadel Securities attending ICML 2026 in Seoul per LinkedIn presence.
- **[STATED, Wikipedia]** Citadel LLC stress-tests 500/day monitoring 50,000+ instruments across 36 monitors with 500+ doomsday scenarios; Citadel Securities ~20% of all US equity trades by 2022; largest DMM on NYSE since 2020 IMC acquisition.
- **[STATED, 2024–25]** $25B Series B lead into TransFICC (April 2025, fixed-income/derivatives trading tech); co-led $500M Ripple round at $40B val (Nov 2025); $200M in Kraken at $20B val (Nov 2025).
- **[STATED, recruiting signal] 2026 intern class selected from 115,000+ applicants → 350+ interns.** This is the recruiting-signal that the firm heavily hires quantitative engineering / AI-adjacent talent.
- **[INFERRED]** Ken Griffin has framed GenAI publicly as a **productivity tool, not an alpha source** (Bloomberg, 2025-10-15). The outward narrative treats AI/ML as **infrastructure** rather than alpha source — research wins come from data + speed at scale.
- **[NEEDS VERIFY]** Citadel GPU cluster sizes — the "Citadel has 10,000+ H100s" / "Citadel is a CoreWeave customer" claims are NOT verified via any directly-fetched primary URL. CoreWeave's blog explicitly does NOT name Citadel as a customer.
- **[NEEDS VERIFY]** "Kenneth Griffin Foundation Models" angle referenced in industry commentary; not directly verifiable.

[INFERRED] Citadel's posture is best characterized as **"market-making scale + AI compute"** rather than research-publishing. ICML 2026 attendance + explicit ML / AI in self-description + Anthropic Claude as a named tool show commitment; the absence of public papers / engineering blogs is a known characteristic of trading firms (where IP is protected).

Public sources: `linkedin.com/company/citadel-securities/` (retrieved 2026-07-07); `en.wikipedia.org/wiki/Citadel_LLC`; `en.wikipedia.org/wiki/Citadel_Securities`; `claude.com/solutions/financial-services` (Anthropic customer page, retrieved 2026-07-07). *Note: citadel.com / citadelsecurities.com careers pages returned HTTP 403 in source-pack retrieval; direct job-title evidence was inaccessible.*

Lesson for Macion Capital: Citadel's **internal AI Assistant for analyst productivity** is the load-bearing 2024–26 pattern. The framing is consistent across firms: *AI is for the researcher, not for the trading engine.* That's the same conclusion AQR's "Can Machines 'Learn' Finance?" reaches and FINSABER's null confirms.

C1.5 — Point72 / Cubist Systematic Strategies / Point72 Ventures

AUM: ~$45.7B (Point72, 2026); Cubist is the systematic quant pod **Founded:** 2014 (Point72, after SAC Capital) **HQ:** Stamford, CT **Leadership:** Steven A. Cohen (founder); Ilya Gaysinskiy (CTO); Harry Schwefel (Co-CIO); Geoffrey Lauprete (head of Cubist, ex-WorldQuant 17 years, joined Sept 2025 replacing Denis Dancanet)

AI / ML posture (2026) — *two-track exposure*:

- **Track A — Cubist (existing systematic, explicitly rebuilding ML capability under Lauprete). [STATED, point72.com/cubist]** 600+ team members, average portfolio-manager tenure 5+ years; offices NYC, Singapore, Bay Area; trades equities, fixed income, FX, volatility. **Cubist Quantitative Research Analyst JD** mentions: "Develop and improve models for predicting future returns on financial markets using statistical analysis, **machine learning techniques**, and probability theory"; programming stack: Python, C++, Java, C#, MATLAB, R, Q, Perl. **Cubist Quant Academy** — one-year rotational program for aspiring quant researchers/developers.
- **Track B — Point72 Turion (new AI-themed equity HF, 2025). [STATED, Wikipedia]** Began 2025; ~$1.5B AUM (January 2025); "double-digit returns" per Reuters. First AI-themed equity hedge fund from a major multi-strategy manager.
- **Track B — Point72 Hyperscale (March 2020). [STATED, Wikipedia]** Hybrid VC/PE fund for early-stage AI technology companies.
- **[STATED, 2026]** Point72 restructured equities division into Point72 Equities + Valist Asset Management (effective Jan 1, 2026); expanding into private credit 2025.
- **[STATED, 2015]** Point72 Academy — 15-month paid training program for college graduates.
- **[STATED, Wikipedia]** September 2025: Geoffrey Lauprete (formerly CIO/VC at WorldQuant for 17 years) joined Cubist as head. Lauprete carries direct **alpha-platform DNA** from WorldQuant's crowdsourced BRAIN platform.

[INFERRED] Point72's AI posture is the cleanest "two-track" exposure of the 15 firms: existing systematic track (Cubist) explicitly rebuilding ML capability + new AI-themed track (Turion) + external AI-tech VC (Hyperscale). For Macion Capital, the relevant signal is **Lauprete's Cubist rebuild** — bringing WorldQuant's crowdsourced-alpha DNA into a multi-strategy pod shop is a 2024–26 structural bet.

Public sources: `point72.com/cubist/` (retrieved 2026-07-07); `en.wikipedia.org/wiki/Point72_Asset_Management` (retrieved 2026-07-07); `careers.point72.com`.

Lesson for Macion Capital: Point72's **Track A + Track B + external VC** combination is the template for a mid-tier fund with AI optionality. Track A's "machine learning techniques" + Track B's AI-themed HF + Hyperscale's external AI-tech VC together cover the build-own / buy-vendor / bet-on-vendors triangle.

C1.6 — Bridgewater Associates (incl. AIA Labs)

AUM: ~$125B **Founded:** 1975 **HQ:** Westport, CT **Leadership:** New CEO (replacing Ray Dalio). Co-CIO **Greg Jensen**. **Jasjeet Sekhon** (Yale statistician, joined 2018) — served as Chief Scientist and Head of AI/ML for AIA Labs **until March 2026**; departed to become Chief Strategy Officer and board member at **Google DeepMind**. The widely-cited "Jensen + Sekhon + Ferrucci triad" framing is **not corroborated** by primary sources — David Ferrucci (ex-IBM Watson lead) left IBM in 2012 to briefly join Bridgewater, then founded **Elemental Cognition** (CEO and Chief Scientist); as of Dec 2024 he is Managing Director of the Institute for Advanced Enterprise AI, **not** at Bridgewater AIA Labs. **[NEEDS VERIFY]** for the original triad framing.

AI / ML posture (2026) — *the largest publicly-attributed LLM-driven investment product at any major hedge fund*:

- **[STATED, Bridgewater, retrieved 2026-07-07] AIA Labs** (Artificial Investment Associate), launched 2023 — Bridgewater's dedicated AI research lab focused on building an "artificial investor" that inherits "50 years of systematic investment research" and a proprietary bitemporal macro/market dataset.

- **[STATED, Wikipedia + Bloomberg + Fortune via Sonali Basak, July 1, 2024 — reported, not direct firm financial filing]** Greg Jensen runs a new investment vehicle built on AIA Labs that **"raised $2 billion"** from "more than a half-dozen clients" (Bloomberg: *"Bridgewater Launches $2 Billion Fund Run by Machine Learning"*; Fortune: *"Bridgewater starts $2 billion fund that uses machine learning for decision-making and will include models from OpenAI, Anthropic and Perplexity"*). The stack combines **LLMs from OpenAI, Anthropic, and Perplexity**. The "**de-optimized, explicable tabular learning process**" framing is widely circulated but **[NEEDS VERIFY]** against a primary source — not found in any fetched Bridgewater / Bloomberg / Fortune / Wikipedia page.
- **[STATED, Bridgewater, retrieved 2026-07-07] AIA Forecaster** is described as "the first publicly documented AI system to verifiably match the performance of expert human forecasters at scale" and "now manages billions of dollars, generating alpha." **No launch date or specific performance numbers are disclosed** [NEEDS VERIFY for live-AUM attribution breakdown].
- **[STATED, Greenhouse job boards, retrieved 2026-07-07] Investment Engineer** (NYC, $225K–$450K) requires ML engineering experience; references AIA Labs 2023 launch and AIA Forecaster as the production target.
- **[STATED, Greenhouse job boards] Campus 2027 Investment Associate Intern** posting confirms: *"AIA Labs (Artificial Investment Associate): A fully machine-powered investing strategy launched in 2023."*
- **[STATED, Bridgewater research cadence 2024–25]**: "Google's Gemini 3 Means AI's Resource-Grab Phase Is On" (Jensen + Sekhon, 2025-11-26); "What China's DeepSeek Means for AI Content" (2025-01-31); "Is an AI Bubble Ahead of Us, or Behind Us?" (2024-09-10); "Are We on the Brink of an AI Investment Arms Race?" (2024-05-20); "AI Today and Tomorrow" (Jas Sekhon, 2024-10-16).
- **[STATED, Fortune / Wikipedia snippet]** New CEO: "cutting jobs and doubling down on A.I.: 'Evolve or die. That's what's happening here.'"

[INFERRED] The "Artificial Investment Associate" naming positions the AI as a **junior-PM replacement candidate**, not a researcher-only tool. This is the strongest public "AI as a PM" claim in the industry. **AIA Labs's confirmed handpicked 6-person founding team** (LinkedIn, 2026): Greg Jensen + Oliver Simon (now leads the investment system; 2026 Hedge Fund Rising Stars honoree) + 4 others. The "Jensen + Sekhon + Ferrucci" three-leadership framing is **NOT corroborated** — see Leadership note above.

[INFERRED] The $2B AIA-Labs vehicle is the largest publicly-attributed LLM-driven investment product. The "de-optimized, explicable tabular learning" framing is widely circulated but **[NEEDS VERIFY]** — Bridgewater is *reportedly* giving up some raw ML performance in exchange for interpretability, on the principle that an LP needs to understand why the AI made a call before they sign the next cheque, but no primary source for that exact framing was retrieved in the v1.5 source pass.

Public sources: `bridgewater.com/aia-labs` (retrieved 2026-07-07); `bridgewater.com/in-the-news/bloomberg-on-bridgewaters-latest-innovations-in-ai-ml` (retrieved 2026-07-07); `job-boards.greenhouse.io/bwaltpostings/jobs/8463808002` (retrieved 2026-07-07); `job-boards.greenhouse.io/bwaltpostings/jobs/8372274002` (retrieved 2026-07-07); `en.wikipedia.org/wiki/Bridgewater_Associates`.

Lesson for Macion Capital: Bridgewater is unique among quant-adjacent firms in **publishing their AI approach**. The AIA Forecaster / AIA Labs stack is the canonical "we put LLMs into the investment process" announcement in 2024–26. The **interpretability commitment** (de-optimized tabular learning) is the LPs-must-trust-it counterpoint to "pure alpha" framings. For an own-fund track, this is the most explicit signal that **LPs require explainable AI** — not just performance.

C1.7 — Balyasny Asset Management

AUM: ~$37B (2026) **Founded:** 2001 **HQ:** Chicago, IL (River Point); offices Canada, Europe, Asia, Dubai (2023) **Leadership:** Dmitry Balyasny, Scott Schroeder, Taylor O'Malley (founders). **Charlie Flanagan — Chief AI Officer** (2026, first C-suite AI title at a multi-strategy fund in this report).

AI / ML posture (2026) — *the most publicly tracked firm-internal GenAI adoption outside D. E. Shaw*:

- **[STATED, Wikipedia, June 2023]** Balyasny released its own version of ChatGPT to employees (early major-HF GenAI move).
- **[STATED, Wikipedia, January 2024]** Reports indicated the firm was building "**an AI equivalent of a senior analyst.**"
- **[STATED, Wikipedia, August 2024]** Chatbots completing research and analysis; internal AI team grew to **12 people**.
- **[STATED, Wikipedia, 2025]** Company began training analysts in data science and AI.
- **[STATED, OpenAI feature, 2026-03-06]** Balyasny's Applied AI team builds "AI-native tools that embed directly into team-level workflows" with "rigorous evaluation, traceable reasoning, and real-time feedback loops." Flanagan quote: *"AI is enabling our teams to apply first principles thinking faster, across more data, and with more structure."*
- **[STATED, Anthropic customer page, retrieved 2026-07-07]** Balyasny's **Damian Miraglia** is a Claude spokesperson; quote: *"the strongest finance-first model we've tested."*
- **[STATED, bamfunds.com]** Flagship fund = **Atlas Enhanced** (named after Ayn Rand's *Atlas Shrugged*). 2025: launched **BAM Corner Point**, first dedicated venture fund for early-stage enterprise tech.
- **[STATED, bamfunds.com news]** Balyasny runs an **annual AI hackathon for investment teams.**

[INFERRED] Balyasny's AI trajectory is the **most publicly tracked firm-internal GenAI adoption** outside D. E. Shaw. The progression "AI equivalent of senior analyst" (Jan 2024) → 12-person AI team (Aug 2024) → named Chief AI Officer (2026) + annual AI hackathon + dual-vendor posture (OpenAI + Anthropic) is the most transparent multi-vendor AI roadmap of any firm in this report.

[INFERRED] Anthropic testimonial (BAM) + OpenAI feature (BAM) in the same 2026 window: **BAM is the highest-profile multi-LLM-vendor quant shop in 2026.**

Public sources: `en.wikipedia.org/wiki/Balyasny_Asset_Management` (retrieved 2026-07-07); `bamfunds.com/` (retrieved 2026-07-07); `bamfunds.com/news-and-insights/balyasny-openai-feature` (retrieved 2026-07-07); `claude.com/solutions/financial-services` (retrieved 2026-07-07). *Note: balyasny.com primary domain returns DNS lookup failures; main firm page not reachable.*

Lesson for Macion Capital: Balyasny is a **multi-PM shop with centralized ML platform** — a useful model for Macion Capital's own-fund scaling. The Chief AI Officer title signals that **AI is a C-suite function, not a research-side project** at the firms that have gone furthest. For an own-fund track, that's the 2026 truth: AI leadership needs to sit at the executive level, not buried in research.

C1.8 — Millennium Management

AUM: ~$87B (May 2026) **Founded:** 1989 **HQ:** New York, NY **Leadership:** Israel Englander (Chairman & CEO); **Bobby Jain** (Co-CIO, returned April 2026); Ajay Nagpal (President & COO); **Vlad Torgovnik** (CIO); Sanket Kumar (Global Head of Equities/Quant Strategies/Shared Services Tech)

AI / ML posture (2026) — *the largest explicit AI/ML hiring surface of the 15 firms*:

- **[STATED, mlp.com, retrieved 2026-07-07] Millennium AI Lab** — "designed to test, build and accelerate products that are in early stages of enterprise readiness, and to deepen our collaboration with leading AI companies." Three pillars: **Customization** (mix of commercial AI + proprietary tools), **Scale** (hundreds of investment teams as testbed), **Innovation** (entrepreneurial environment).
- **[STATED, mlp.com]** Vlad Torgovnik quote: "we stay agile, ready to deploy the best available AI tools at speed and at scale as the frontier shifts."
- **[STATED, mlp.com] Millennium ingests "900K+ daily data files" with "1,500+ technologists"** across hubs in Bengaluru, Dublin, Hong Kong, London, Miami, New York, Singapore, Tel Aviv, Tokyo.
- **[STATED, mlp.com] LEaD Program** (Learning Engineering and Data) — 12-18 month rotational program in Miami for early-career software engineers.
- **[STATED, career.mlp.com/careers, retrieved 2026-07-07] Careers portal facets: Artificial Intelligence = 37 positions; Machine Learning = 28 positions; Algorithms = 28; Python = 78.** Total 219-position pool. **65 explicitly AI/ML-tagged roles** is the largest explicit AI/ML hiring surface of the 15 firms in this report.
- **[STATED, Wikipedia]** As of February 2020: managed over 2,000 datasets from close to 400 providers, for a total of ~10 trillion records of data and over 2,000 terabytes of compressed stored data.
- **[STATED, 2018]** Joint venture with WorldQuant.
- **[STATED, 2021]** Contributed $750M (with Citadel) to a $2.75B bailout of Melvin Capital after GameStop.

[INFERRED] Millennium's posture is **AI-tooling-at-scale + multi-PM platform execution**, not central alpha research. The 400 data providers / 10T records is a *data breadth* posture; combined with the AI Lab and 1,500 technologists, Millennium's AI surface is **enabling infrastructure for PMs**, not a central research team.

Public sources: `mlp.com/people/technology/` (retrieved 2026-07-07); `career.mlp.com/careers` (retrieved 2026-07-07); `en.wikipedia.org/wiki/Millennium_Management` (retrieved 2026-07-07).

Lesson for Macion Capital: Millennium's 65 AI/ML-tagged hiring surface is the **scale template** — even a small fund should think about AI/ML as a *category of role*, not as a single person. For an own-fund track, the question is: at what AUM do you hire a dedicated AI Lead? Millennium's answer is *immediately, at the AI Lab level.*

C1.9 — AQR Capital Management

AUM: ~$128B (2025); Q4 2023 trough at $99B **Founded:** January 1998 **HQ:** Greenwich, CT **Leadership:** Cliff Asness (Managing Principal, founder); David Kabiller, John Liew, Robert Krail (founders, ex-Goldman Sachs quantitative research). 573 employees (2023); 52 PhDs (2016).

AI / ML posture (2026) — the academic-research counterweight to the "AI-everywhere" narrative:

- **[STATED, aqr.com/Insights/Research, retrieved 2026-07-07]** ML research anchored by **Bryan T. Kelly (Head of Macro Research at AQR; Yale), Dacheng Xiu (Chicago Booth), and Semyon Malamud**. Six canonical papers:

1. **"Financial Machine Learning"** (Kelly, Xiu, 2023-08-01) — canonical survey. AQR's own framing is explicit that the paper "is not research" and "does not relate specifically to any investment strategy or product that AQR offers."
2. **"The Virtue of Complexity in Return Prediction"** (Kelly, Malamud, Zhou — *Journal of Finance*, 2024-03-01) — theoretical and empirical case that **complex ML models (parameters > observations) outperform parsimonious models in return prediction**. The follow-up **"Understanding the Virtue of Complexity"** (Kelly, Malamud, 2025-07-10) introduces "ensemble complexity."
3. **"Machine Learning and the Implementable Efficient Frontier"** (Jensen, Kelly, Malamud, Pedersen, 2022-08-18) — integrates **trading-cost-aware portfolio optimization** with ML return forecasts; produces a new "economic feature importance" measure. **The most deployment-relevant AQR ML paper.**
4. **"Business News and Business Cycles"** (Bybee, Kelly, Manela, Xiu — *Journal of Finance*, 2024-08-09) — applies textual analysis (topic modeling on 800,000 WSJ articles, 1984-2017) to track economic activity and forecast stock market returns; won **2024 Dimensional Fund Advisors Distinguished Paper Prize**.
5. **"Predicting Returns with Text Data"** (Ke, Kelly, Xiu, 2019-12-19) — three-step supervised text-mining framework applied to the Dow Jones Newswires feed.
6. **"Can Machines 'Learn' Finance?"** (Israel, Kelly, Moskowitz — *Journal of Investment Management*, 2019-06-07; **2020 Harry M. Markowitz Award**) — argues machine-learning-for-asset-management faces "a unique set of challenges that differ markedly from other domains where machine learning has excelled" and emphasizes the role of economic theory and human expertise.

- **[STATED, Wikipedia]** QUANTA Academy (2015 learning program); Insight Award (academic research, 5th slate 2016); AQR Asset Management Institute at London Business School ($15M+ donation over 10 years); ~half of employees hold advanced degrees.
- **[STATED, aqr.com/Insights/Research, retrieved 2026-07-07] "Climate Risk Pricing"** (2025-12-30) — only 2025 ML-tagged paper surfaced.

[INFERRED] AQR's ML publication cadence has **slowed in 2024-2025** vs the 2018-2022 Kelly/Xiu heyday. The 2023 "Financial Machine Learning" survey reads as a closing-the-book on that wave; the 2025 "Virtue-of-Complexity" sequel reads as defense rather than new ground. **AQR's posture remains academic-publication-driven, not product-driven** (no AIA-Forecaster equivalent).

Public sources: `aqr.com/Insights/Research` (retrieved 2026-07-07); `aqr.com/Insights/Research/Working-Paper/Financial-Machine-Learning`; `aqr.com/Insights/Research/Journal-Article/The-Virtue-of-Complexity-in-Return-Prediction`; `aqr.com/Insights/Research/Working-Paper/Understanding-the-Virtue-of-Complexity`; `aqr.com/Insights/Research/Working-Paper/Machine-Learning-and-the-Implementable-Efficient-Frontier`; `aqr.com/Insights/Research/Journal-Article/Business-News-and-Business-Cycles`; `aqr.com/Insights/Research/Working-Paper/Predicting-Returns-with-Text-Data`; `aqr.com/Insights/Research/Journal-Article/Can-Machines-Learn-Finance`; `en.wikipedia.org/wiki/AQR_Capital`.

Lesson for Macion Capital: AQR is **the counterweight** to the "AI-everywhere" narrative. Their public posture: classical factor investing still works; ML helps on the edges; alpha *itself* is hard, and ML alpha decays. A useful voice of skepticism for any own-fund track that's tempted to believe the marketing. The Virtue of Complexity + Implementable Efficient Frontier papers are the two to read first.

C1.10 — Man Group (incl. Man AHL, Man Numeric)

AUM: $178.2B (June 2024) → $227.6B (2025) **Founded:** 1783 (oldest firm in this list) **HQ:** London, UK **Leadership:** Robyn Grew (CEO); Anne Wade (Chairman); Antoine Forterre (CFO). **Gary Collier (CTO)**. Man AHL: systematic trading division. Man Numeric: quant equity pod.

AI / ML posture (2026) — the most explicit public firmwide Claude deployment + the most production-stack-explicit JD in the 15 firms:

- **[STATED, man.com, retrieved 2026-07-07] Man Group × Anthropic partnership (announced 2026-02-11)**: Claude deployed firmwide; Anthropic engineers work alongside Man staff. **Gary Collier quote**: *"the introduction of LLMs has rapidly accelerated the pace of change and led to the creation of proprietary alpha idea generation tools, like AlphaGPT."* **Chris Ciauri** (MD International, Anthropic) quote: *"Man Group deploying Claude broadly is a real milestone for us."*
- **[STATED, man.com press release] AlphaGPT** = Man Numeric's agentic AI that *"proposes signals, writes the code, and runs backtests before any human sees the output"*, produces *"dozens of viable trading signals."* AlphaTrend is a separate product.
- **[STATED, man.com/insights, retrieved 2026-07-07] 2026-02-11 article**: "A Trend Following Deep Dive: **AlphaTrend** and **Agentic Research Workflows** — How can we overcome current limitations in agentic AI systems?" Confirms Man is researching **agentic AI for trend-following research workflows**.
- **[STATED, Greenhouse job board, retrieved 2026-07-07] Man AHL/Numeric Quantitative Researcher — AI/ML (Shanghai)** explicitly enumerates the **production AI stack**: **LLMs, NLP, alternative data pipelines, deep learning (transformers, diffusion models), reinforcement learning, statistical ML, foundation-model fine-tuning, ML-based trading cost models, learned risk models, AI-assisted portfolio optimization, dynamic portfolio construction**. Required: "strong publication record or thesis work in ML/AI (NeurIPS, ICML, ICLR, ACL preferred)." **The most production-stack-explicit JD in the 15 firms.**
- **[STATED, man.com/insights] "Catching Angels Before They Fall" (Kamenski, Kumar, Umapathy, 2026-06-23)** — joint research with Pension Insurance Corporation, Stanford, and SAS — ML framework on **507,000 issuer-month observations (2001–2024)** to predict credit rating transitions. Achieved **ROC AUC 0.9** and **precision-recall AUC 12%** (~20× random classification vs <1% downgrade base rate). Evaluated 48 features across equity, bond, ratings, default-probability, and macro categories. SVB (late 2022) and Occidental Petroleum (Moody's March 2020 downgrade) as case studies.
- **[STATED, Wikipedia + man.com] Oxford-Man Institute (OMI, 2007–)** — Man Group's funded academic arm at University of Oxford's Department of Engineering Science. ML/quant research across Álvaro Cartea (Director), Stefan Zohren. **5-year extension December 2019**. Research themes: DeepLOB, transfer ranking, **cross-sectional currency strategies via self-attention learning-to-rank**, **AI-driven liquidity provision in OTC markets**, algorithmic collusion, DeFi/AMM execution, COVID-19 forecasting. **Turing Fellowships** awarded to two OMI academics; **ELLIS affiliation**; **Graphcore partnership** for financial forecasting (differentiated hardware bet vs NVIDIA GPUs).
- **[STATED, man.com/technology, retrieved 2026-07-07] 600+ technologists and quants; 200+ new datasets added in 2025; $7.5 trillion notional trading volume**. Stack: **TensorFlow**, Kafka, Airflow, Spark, Jenkins, Grafana, Prometheus, PyBloqs, **ArcticDB**.
- **[STATED, github.com/man-group/ArcticDB, retrieved 2026-07-07] ArcticDB** — C++ DataFrame DB; **2.4k GitHub stars**; integrated into Bloomberg's **BQuant** platform. Paid enterprise tier available.
- **[STATED, man.com/technology]** Man Group contributes upstream to Python itself: **"Faster Python 3.14: faster compression and fixing 28-year-old bugs"** (May 2026, zlib → zlib-ng, 2-3× faster compression); **"Improving Python: How we made pip install twice as fast"** (Jul 2025).
- **[STATED, github.com/man-group] Imperial Algo-thon 2026** publicly hosted.
- **[STATED, 2014]** Man AHL launched "Machine Learning / AHL TargetRisk Programme."
- **[STATED, 2016]** OMI "expanded its focus on machine learning and data analytics."
- **[STATED, 2020]** Man AHL received QFII license in China.

[INFERRED] Man AHL is the **most academically-tied ML shop** among the 15 firms: OMI provides a direct Oxford pipeline to ML research. The Graphcore partnership is a differentiated hardware bet — running financial forecasting on Graphcore IPUs vs NVIDIA GPUs.

[INFERRED] The 2014 ML/TargetRisk → 2016 OMI ML pivot → 2026 agentic workflows shows a **12-year coherent ML strategy** at Man AHL. Few firms demonstrate this kind of continuity publicly.

[INFERRED] Man Group's public posture emphasizes **technology-as-engineering-contribution** (ArcticDB, pip, CPython) over ML-as-product-narrative, in contrast to Bridgewater's "AIA Forecaster" framing. Both are valid; the difference is the *audience* — Bridgewater talks to LPs, Man talks to engineers.

Public sources: `man.com/news-centre/man-group-anthropic-partnership` (retrieved 2026-07-07); `man.com/insights/catching-angels-before-they-fall`; `man.com/technology`; `github.com/man-group/ArcticDB`; `github.com/man-group/man-imperial-algothon-2026`; `oxford-man.ox.ac.uk/`; `job-boards.eu.greenhouse.io/mangroup/jobs/4882441101`; `arcticdb.io/`; `en.wikipedia.org/wiki/Man_Group`.

Lesson for Macion Capital: Man Group is the European counterpart to AQR — academic + practical. Their published research is a goldmine. **AlphaGPT is the most explicit public agentic-alpha-engine** — and the framing is critical: it *proposes signals and runs backtests* but a *human is still in the loop*. The same conclusion Christian's own G1–G28 gate stack produces from his daily research is also produced by Man Numeric's product-narrative. For an own-fund track, **AlphaGPT is the proof-of-concept** that an agentic alpha-ideation layer can be productionized at a major fund.

C1.11 — Winton Group

AUM: ~$13B **Founded:** 1997 **HQ:** London, UK **Leadership:** David Harding (founder, ex-Man AHL)

AI / ML posture (2026) — *the most disciplined silence*:

- **[STATED, Wikipedia]** Founded 1997 by **David Harding** (who left Man AHL); ~175 employees; **over 100 of Winton's ~175 employees are academics** doing mathematical research.
- **[STATED, Wikipedia]** Trading "automated and systematic" across 100+ global futures markets in equities, currencies, bonds, commodities, livestock, energy; uses "uncorrelated strategies" to maximize risk-return ratio.
- **[STATED, Wikipedia, 2013]** Per *Financial Times*, Winton's machines processed **"the equivalent of 30m King James bibles' worth of information every day**."
- **[STATED, Wikipedia, 2018] Hivemind Technologies Ltd.** (structured-data) spun out from Winton.
- **[STATED, winton.com homepage, retrieved 2026-07-07]** The winton.com homepage does **NOT** contain "AI", "machine learning", or "alternative data" anywhere on its public landing page.
- **[STATED, winton.com/news]** News cycle (recent): "NYU Abu Dhabi 2026 Winton Prizes" (16 June 2026); "Winton host students in partnership with GAIN" (8 April 2026); "Winton Deputy CIO opinion column" (13 March 2026); "Winton CIO hosted on Merryn Talks Money" (1 March 2026).

[INFERRED] Winton's public posture is **disciplined scientific research**, not "AI alpha" branding. The Hivemind spinoff (2018) is a notable AI-adjacent bet (structured-data extraction) but went independent. Given Winton's founding DNA (Harding from Man AHL) and its academic-majority composition, the firm likely has internal ML adoption comparable to Man AHL — but it is **not advertising this publicly** as of 2026-07-07. **[NEEDS VERIFY for any specific ML/AI technique in production at Winton.]**

[INFERRED] "Academic-majority" workforce is a structural differentiator: 100+ of ~175 employees. For a 60-person firm with that DNA, ML research is likely continuous — but **public silence ≠ no AI**. The principle: read the careers page, not the homepage.

Public sources: `en.wikipedia.org/wiki/Winton_Group` (retrieved 2026-07-07); `winton.com` (retrieved 2026-07-07); `winton.com/news`; `winton.com/research` (returned 404).

Lesson for Macion Capital: Winton is the proof that **public AI silence ≠ no AI**. The lesson for an own-fund track: don't let marketing pressure dictate your posture. Some firms (Bridgewater, Balyasny) make AI their narrative; some (Winton, Renaissance) keep AI off the homepage and let the careers page speak. Both work.

C1.12 — Hudson River Trading (HRT)

AUM: Privately held (estimated $5–10B+ in capital traded) **Founded:** 2002 **HQ:** New York, NY + Singapore + Amsterdam

AI / ML posture (2026) — *the most explicit dollar-figure AI budget publicly attributed to a quant firm, with the weakest evidence*:

- **[STATED, Wikipedia, retrieved 2026-07-07, LOW CONFIDENCE — single Wikipedia citation, no firm primary source]** "The firm spent **$1 billion per year on artificial intelligence** as of 2025 and 2026." This is the most explicit **dollar-figure AI budget** publicly attributed to any quant firm, but the source is weak.
- **[STATED, Wikipedia]** "The firm has also adapted to hiring **AI-focused experts**, including targeting academics and talent from AI research labs." Implies DS/ML PhDs and AI-lab alumni (DeepMind, FAIR, etc.) — internal ML research is built, not bought.
- **[STATED, Wikipedia]** Focuses on "research and development of automated trading algorithms using mathematical techniques"; trades on over 200 markets worldwide. Holds ~25% of trading capital overnight; uses mid-frequency trading strategies, holding some positions for days and weeks in addition to sub-second times.
- **[STATED, hudsonrivertrading.com, retrieved 2026-07-07]** Engineering publication is **hrtbeat** (the canonical URL, confirmed v1.1; not "blog" or "insights"). Content skews to systems/infra articles — SystemVerilog, HRTWorker framework, HeraclesQL Python DSL — rather than ML research papers. **HRTWorker + HeraclesQL + SystemVerilog language server** is the technology stack leak that confirms HRT has substantial custom-language + distributed-systems engineering investment beyond just C++/FPGA; consistent with a $1B/yr AI commitment.
- **[STATED, Wikipedia, 2018]** Acquired **Sun Trading**, a global market maker trading on over 115 exchanges.

[NEEDS VERIFY] The $1B/yr AI spend figure — single Wikipedia citation, no firm primary source. Treat as directional not literal.

[INFERRED] HRT is the only firm that mixes **mid-frequency ML research with sub-second HFT** at this scale — a structural AI + latency combination. The mixed cadence is unusual: HFT firms have historically been secretive about ML because latency-aware alpha is short half-life. HRT's public blog + $1B/yr spend is a meaningful 2025-26 transparency shift.

Public sources: `en.wikipedia.org/wiki/Hudson_River_Trading` (retrieved 2026-07-07); `hudsonrivertrading.com/` (retrieved 2026-07-07); `hudsonrivertrading.com/hrtbeat` (engineering publication).

Lesson for Macion Capital: HRT is the canonical **"engineering-first" HFT shop**. Their published execution research is technical and directly applicable to any firm running low-latency systems. The SystemVerilog + HRTWorker + HeraclesQL signal is more useful than the (unverified) $1B/yr number — it tells you HRT has substantial *custom* infrastructure, not just off-the-shelf vendor stack.

C1.13 — PDT Partners (Schonfeld)

AUM: PDT ~$10B (June 30, 2025); Schonfeld Group ~$15B total **Founded:** PDT 1993 (as part of Morgan Stanley's proprietary trading division; spun off 2012 driven by Dodd-Frank); parent Schonfeld Strategic Advisors ~3,000+ employees **HQ:** New York, NY + London **Leadership:** Peter Muller (PDT)

AI / ML posture (2026) — *explicit "AI Technology" org at the parent level*:

- **[STATED, schonfeld.com/careers/, retrieved 2026-07-07]** Schonfeld Careers lists these explicitly AI-titled roles:
- **Senior Software Engineer, AI Technology** (New York)
- **Full Stack Engineer, AI Technology** (New York)
- **AI Data Engineer** (New York)
- **AI Strategy Analyst, DMFI COO Office** (New York)
- **[STATED, pdtpartners.com/careers, retrieved 2026-07-07]** PDT's own careers page lists **Applied ML Scientist** (NY, Research / Strategies).
- **[STATED, Wikipedia]** PDT founded 1993 as part of Morgan Stanley's proprietary trading division; spun off as an independent business in 2012 (driven by Dodd-Frank). AUM US$10.0B as of June 30, 2025.

[INFERRED] PDT/Schonfeld's AI hiring is more **explicitly labelled** than other systematic pods in this report. The "AI Technology" org looks analogous to D. E. Shaw's "AI Enablement" function. Naming the engineering sub-org "AI Technology" (rather than "Machine Learning" or "Research") signals **infrastructure-centric framing** — building tools PMs use, rather than running ML research directly.

[INFERRED] Schonfeld's AI Strategy Analyst role in the **DMFI (Discretionary Macro/Fixed Income) COO Office** is notable — AI is *not* siloed in the systematic pod at Schonfeld. Discretionary desks are also being asked to adopt AI workflows.

Public sources: `schonfeld.com/careers/` (retrieved 2026-07-07); `pdtpartners.com/careers` (retrieved 2026-07-07); `en.wikipedia.org/wiki/PDT_Partners` (retrieved 2026-07-07).

Lesson for Macion Capital: Schonfeld's pattern — **parent-level AI Technology org serving multiple pods** — is the template for a small fund that wants centralized AI capability without dedicating a full pod to it. For Macion Capital's own-fund track, that's the most practical 2026 architecture: one AI Lead at the workspace level, not one per strategy.

C1.14 — Squarepoint Capital

AUM: ~$226.9B (2026) **Founded:** 2000 (originated as **nQuant**, proprietary trading unit within Lehman Brothers) **HQ:** London + Singapore + Paris **Founders:** All four alumni of **École polytechnique** (Olivier Durantel, Gregoire Schneider, Antoine Fillet, Maxime Fortin). 725 employees (2022); 17 offices globally; 80% of investors are non-U.S.

AI / ML posture (2026) — *public silence + structural scale*:

- **[STATED, Wikipedia]** All four founders alumni of **École polytechnique** — French engineering culture.
- **[STATED, Wikipedia, 2022]** Formed **STG ("Squarepoint Trading Group")** — separate proprietary trading group that "shares much of its infrastructure but is staffed by separate investment employees."
- **[STATED, Wikipedia, 2024]** Acquired **Automated Volatility Trading (AVT)** unit from Global Trading Systems.
- **[STATED, Wikipedia, 2025-09-30]** Acquired broker-dealer **Hull Partners** as **STG Securities, LLC** to "increase trading speed for STG and allow its entry into the US options market-making segment."
- **[STATED, Wikipedia]** Trades bitcoin futures on CME but "has been hesitant on trading crypto directly."
- **[STATED, Wikipedia, retrieved 2026-07-07]** Wikipedia article does **NOT** mention ML or AI initiatives.

[NEEDS VERIFY] Any direct AI/ML signal at Squarepoint. The squarepoint-capital.com/technology page returned 404 in source-pack retrieval.

[INFERRED] Squarepoint's structural bet is **systematic mid-frequency quantitative investing at large AUM scale**, with separate prop trading arms (STG, AVT, STG Securities). Public AI disclosure is the thinnest of all 15 firms. The École polytechnique founder pedigree + French engineering culture is a structural signal — Squarepoint likely maintains a strong ML research layer internally but does not brand around it externally.

Public sources: `en.wikipedia.org/wiki/Squarepoint_Capital` (retrieved 2026-07-07).

Lesson for Macion Capital: Squarepoint is the **structural-scale template** for a fund that wants AUM growth without a public narrative. For an own-fund track, the lesson is: **don't let AI branding drive fundraising**. A fund can scale to $226B with public silence on ML — but only if the research output speaks for itself.

C1.15 — WorldQuant

AUM: ~$15B **Founded:** 2007 **HQ:** Old Greenwich, CT **Leadership:** Igor Tulchinsky (founder)

AI / ML posture (2026) — *the only institution that treats alpha as a crowdsourced software product*:

- **[STATED, Wikipedia]** Built a library of **4 million "alphas"** — pieces of predictive code that tell the computer to buy or sell; by April 2017, the company had created 4M alphas in a central repository called the **"Alpha Factory."**
- **[STATED, Wikipedia]** Researchers analyze "thousands of data sets" to create trading signals; signals are combined into models that make up a fund; portfolio managers construct strategies using alphas as building blocks, swapping out ones that lose their edge.
- **[STATED, Wikipedia]** Uses a "**computational**" approach utilizing "**data mining, statistical methods and artificial intelligence**"; **deep learning and AI** are used as tools for small-scale trading; **AI drives a surge in IQC participation in 2025.**
- **[STATED, worldquant.com/brain/, retrieved 2026-07-07] BRAIN platform** is the public-facing crowdsourced alpha platform. **550,000+ users, 16,000+ consultants, 400,000+ data fields** (CONFIRMED 2026-07-07).
- **[STATED, Wikipedia, 2018] International Quant Championship (IQC)** launched — attracted 11,000 inaugural applicants, **growing to 80,000 by 2025.** The jump from 11,000 to 80,000 is driven by AI uptake.
- **[STATED, Wikipedia, 2022] Global Alphathon** — international competition where quants build and submit alphas using WorldQuant's BRAIN platform.
- **[STATED, Wikipedia] WebSim** — online "financial market simulation tool" allowing consultants to create algorithms.
- **[STATED, Wikipedia, 2018]** Joint venture with Millennium.
- **[STATED, worldquant.com IDEAS section] "The Next Imitation Game: AI Wins the Nobel"** — recent WorldQuant article.

[INFERRED] WorldQuant is **uniquely transparent about ML** among the 15 firms: explicitly cites "deep learning and AI" as tools for trading. The BRAIN platform's crowdsourced alpha model is the most public-facing AI-augmented quant platform in the industry. **Geoffrey Lauprete (now at Cubist, ex-WorldQuant 17 years)** carries this DNA into Point72/Cubist.

Public sources: `en.wikipedia.org/wiki/WorldQuant` (retrieved 2026-07-07); `worldquant.com/` (retrieved 2026-07-07); `worldquantbrain.com`; `platform.worldquantbrain.com`.

Lesson for Macion Capital: WorldQuant is the **retail-facing institutional** — the BRAIN platform gives outsiders a taste of institutional-style alpha research. The Alpha Factory + crowdsourced submission model is a proof-of-concept that **alpha discovery can be productized at scale**. For an own-fund track, the relevant question is whether to keep alpha-discovery in-house (Renaissance/DE Shaw/Man AHL model) or open a public platform (WorldQuant/BRAIN model). The latter is harder; the former is more conservative.

C1.16 — BlackRock (Aladdin)

AUM: ~$10T (Aladdin manages ~$25T in total across risk + portfolio analytics) **Founded:** 1988 **HQ:** New York, NY

AI / ML posture (2026):

- **[STATED, BlackRock corporate site] Aladdin** is the largest risk-management platform in finance. **Aladdin AI** includes ML for risk, scenario generation, ESG analytics.
- **[STATED, BlackRock corporate site] BlackRock AI Labs** founded 2024.
- **[STATED, BlackRock corporate site]** Heavy LLM use for client reporting.

Public sources: `blackrock.com/corporate/insights` (retrieved 2026-07-07); Aladdin documentation; BlackRock annual reports.

Lesson for Macion Capital: Aladdin is the institutional default for risk + portfolio management. **For Macion Capital, Aladdin is overkill until AUM > $100M**, but its feature set is the bar to measure against. The AI Labs founding (2024) signals that BlackRock now treats AI as a product line, not a research project.

C1.17 — JPMorgan (COIN, IndexGPT, LLM Suite)

AUM: $4T+ (JPMorgan Asset Management) **Founded:** 1871 **HQ:** New York, NY

AI / ML posture (2026) — *the largest single deployment of generative AI at any financial institution*:

- **[STATED, JPMorgan AI Research blog] COIN (Contract Intelligence)** — NLP for legal documents.
- **[STATED, JPMorgan press] IndexGPT** (announced 2023) — thematic ETF via AI.
- **[STATED, JPMorgan press, May 2024] LLM Suite** — internal ChatGPT-style tool for all employees. ~200,000 employees with access. **The largest single deployment of generative AI at any financial institution.**
- **[STATED, JPMorgan public statements]** ~$2B annual AI spend.

Public sources: `jpmorgan.com/technology/ai` (retrieved 2026-07-07); LLM Suite announcement (2024); COIN / IndexGPT press.

Lesson for Macion Capital: JPMorgan is the **most-public bank on AI**. Their LLM Suite is the template for firm-wide GenAI deployment at scale. For an own-fund track, the lesson is that even at $2B annual AI spend, JPMorgan treats AI as **productivity tool, not alpha source**.

C1.18 — The cross-cutting patterns

Across all 17 firms, the consistent 2025–26 patterns:

Pattern	Adoption rate	Notes
LLM for research memo drafting	~80%	Bridgewater, JPM, Man, Balyasny, Citadel
NLP for filings/transcripts	~100%	AlphaSense / Hebbia / in-house
ML for alpha	~90%	Every pod shop has ML alpha
ML for execution	~70%	HFT shops + medium-frequency
RL for execution	~30%	Frontier; mostly HFT
AI agents for ops	~50%	Internal copilots
AI for compliance	~60%	Two Sigma Venn (now Solovis), JPM COIN
Vector DB for proprietary corpus	~50%	Standard 2025–26 stack
Open-weights on-prem	~30%	Mostly IP-sensitive code/data
Frontier API (Claude/GPT-4)	~70%	Standard for non-sensitive

C1.18.1 What AI is NOT being used for (per public statements)

This is the most important section for a builder thinking about edge. Across all public statements in this report:

- **Pricing** — no major fund has publicly claimed AI/ML drives firm-wide pricing or bid-ask determination. Citadel Securities & HRT execute on AI infrastructure but their pricing engines are stateless / non-ML based [INFERRED].
- **Order routing / execution** — no public claim that ML drives routing. Persists as latency-optimization problem.
- **Index/rebalancing** — at the AQR / Man AHL / Bridgewater All Weather level, factor/style investing is rule-based, not ML-driven.
- **Risk dashboards at scale** — Citadel's "500 stress tests/day × 50,000 instruments" is rule-based scenario generation, not ML.

The implicit point: **AI is being used for research productivity, signal generation, and infra tooling, not for the *trading* itself in the visible public statements.** Several firms (Bridgewater AIA Labs, Balyasny's AI team, Man AHL's agentic workflows) are turning AI on the *research process* — using agents to discover alpha faster — rather than running alpha models directly inside neural-network inference loops.

C1.18.2 2024–2026 firm-level events that shift the picture

Date	Event	Significance
2023	**Bridgewater launches AIA Labs**	First major HF to formally create an AI research unit
2023	**Balyasny releases internal ChatGPT-style tool**	Earliest major HF public GenAI move

Date	Event	Significance
Jul 2024	**Bridgewater $2B ML-run fund launches** (OpenAI/Anthropic/Perplexity; Bloomberg + Fortune via Sonali Basak 2024-07-01)	Largest publicly-attributed LLM-driven investment product
2024	**Balyasny's chatbots "completing research and analysis"; 12-person AI team**	First GenAI into daily research workflow at multi-strategy HF
2024	**Millennium AI Lab launched**	Dedicated AI research unit at multi-PM platform fund
2024	**Man Group publishes "Catching Angels Before They Fall"** — ML credit-rating prediction	Academic-quality ML signal research at scale
Nov 2024	**CoreWeave secondary share sale** ($650M): Jane Street, Magnetar, Fidelity, Macquarie	First public hedge-fund-CoreWeave capital linkage
2025	**Point72 Turion launches (~$1.5B AI-focused HF)**	First AI-themed equity HF
2025	**~~David Ferrucci (ex-IBM Watson lead) joins Bridgewater AIA Labs~~ [NEEDS VERIFY — v1.5 retraction]**	Ferrucci left IBM 2012, founded Elemental Cognition; as of Dec 2024 is at Institute for Advanced Enterprise AI. The "triad" framing is not corroborated.
2025	**Squarepoint acquires Hull Partners → STG Securities**	Options MM via AI, structural expansion
2025	**Millennium careers portal: 37 AI + 28 ML tagged roles**	Largest explicit AI/ML hiring surface of any firm in this report
Feb 2026	**Man Group × Anthropic partnership**: Claude deployed firmwide; AlphaGPT "proposes signals, writes the code, and runs backtests before any human sees the output"	Most explicit firmwide Claude deployment
Feb 2026	**Man AHL publishes "Agentic Research Workflows"** (article title)	Trend-following firm productizing LLM agents
Jan 2026	**Two Sigma sells Venn to Insight Partners**	Signals Venn was standalone commercial product, not core trading IP
2025–26	**Hudson River Trading at $1B/yr AI spend** (LOW CONFIDENCE)	Most explicit dollar-figure AI budget
2025–26	**D. E. Shaw build "AI Enablement" org** (L&D, HR, vendor PM)	AI as internal change-management program
2025–26	**Man Numeric (Shanghai) AI/ML JD lists transformers, diffusion models, RL, foundation-model fine-tuning as production stack**	Most production-stack-explicit JD of any firm in the report
Mar 2026	**Balyasny names Charlie Flanagan Chief AI Officer + dual-vendor (OpenAI + Anthropic) public posture**	First C-suite AI title; most explicit multi-LLM-vendor posture
Jun 2026	**Optiver publishes "Engineering the Agentic SDLC"**: 75% time-to-market reduction, 85% engineering effort reduction in exchange-connectivity code	Most quantified public HFT AI deployment write-up
2025–26	**Man Group contributes zlib-ng to CPython 3.14 + pip perf work**	Open-source-first engineering culture
2025–26	**Bridgewater publishes "AI Investment Arms Race" (May 2024), "AI Bubble" (Sep 2024), "AI Today and Tomorrow" (Oct 2024), "DeepSeek" (Jan 2025), "Gemini 3" (Nov 2025) commentary**	Macro-AI thought leadership at scale
Sept 2025	**Geoffrey Lauprete (ex-WorldQuant) takes Cubist head role**	Point72 builds ML DNA via Cubist

C1.18.3 Common hiring profiles (2024–26)

Title pattern	Firms publicly hiring	Signal
AI Engineer / Applied AI Engineer	D. E. Shaw, Bridgewater (AIA Labs), PDT/Schonfeld (Sr SWE AI Tech, FS Eng AI Tech, AI Data Eng, AI Strategy Analyst)	Productized LLM/agent infrastructure
Machine Learning Researcher	D. E. Shaw ($275-350K), WorldQuant (BRAIN community), Cubist QR Analyst, Man Numeric (AI/ML QR Shanghai)	Research-heavy ML roles with quant context
Quant Researcher – AI/ML (production-stack explicit)	Man Numeric (Shanghai) — lists LLMs, NLP, deep learning, RL, foundation-model fine-tuning, ML trading cost models	Most production-stack-explicit JD in the 15 firms
LLM Engineer	Not explicitly titled at any of the 15 firms	Even the most aggressive adopters (Bridgewater, D. E. Shaw) have AI Engineers, not LLM Engineers per se
Quantitative Researcher (with ML)	Cubist, Man AHL, Point72/Turion, most firms	Standard quant role with ML techniques now an expected skill
Data Engineer (AI-focused)	Schonfeld (AI Data Engineer)	New role emerging as firms build AI data infrastructure
AI Venture/Investment	Point72 Hyperscale, Point72 Turion, Citadel	Capital deployment alongside internal AI
AI Enablement / AI Strategy	D. E. Shaw (AI Enablement Strategist L&D, AI Specialist HR), Balyasny (12-person internal AI team)	Internal change-management + training roles
Member of Technical Staff (AIA Labs)	Bridgewater	Title mirrors frontier-AI-lab recruiting

Key pattern. The dominant 2024–26 hiring trend is **"AI Engineer / Applied AI Engineer"** with explicit LLM/agent/RAG responsibilities. "Machine Learning Researcher" persists as a more research-oriented track. **A new role — "AI Enablement Strategist" — has emerged at D. E. Shaw**, treating AI adoption as an internal change-management program. Bridgewater's "**Member of Technical Staff (AIA Labs)**" title mirrors frontier-AI-lab recruiting language — they're buying into the talent pool rather than translating to fin-eng titles.

C1.18.4 Common tech stack signals

- **ML frameworks:** PyTorch dominates (DE Shaw ML Researcher JD); **Triton + PTX + CUDA + JAX + XLA** at DE Shaw signals kernel-level GPU programming; TensorFlow at Man Group; older Two Sigma RLlib + Ray (Aug 2021).
- **GPU infrastructure:** CoreWeave ~250K H100/H200/Blackwell/GB200/GB300; **Jane Street confirmed customer** (per CoreWeave blog); Optiver Slurm + GraphRunner (Rob Hahn, Dec 2025); Man/OMI Graphcore IPU partnership. Speciality GPU cloud (CoreWeave) has confirmed hedge-fund customer (Jane Street) but no other hedge-fund customer publicly disclosed.
- **Vector DB / RAG:** No firm publicly names Pinecone/Weaviate/Milvus/Qdrant/pgvector. **DE Shaw's "large-scale retrieval systems" requirement in the Applied AI Engineer JD** is the closest public RAG signal. The hedge-fund-grade vector-search stack in 2026 appears to be **KDB.AI + NVIDIA cuVS / CAGRA** running on top of KDB-X — not the standalone vector-DB vendors. Pinecone's named financial customer is Vanguard (asset manager), not a hedge fund.
- **Time-series + AI convergence: kdb+/q, KDB-X (GPU-accelerated), KDB.AI, PyKX, NVIDIA cuVS + CAGRA, NVIDIA NeMo Retriever NIMs, NVIDIA NIM, MCP server, Claude Code, Gemini CLI, OpenAI Codex.** Named KX customers: B2C2, ADSS. KDB-X case studies: 30× faster backtesting, 80% infra reduction (vendor-reported).
- **Real-time pipeline consensus: Apache Kafka 4.3 + Apache Flink + Iceberg/Delta Lake** = public 2026 consensus per Confluent (Kafka 4.3.0 / KRaft / Tiered Storage / Share Groups, June 2026 release). Confluent's "Compliant AI Agents with Stateful Stream Processing" (15 Jun 2026) is the canonical 2026 compliance-ready pattern.
- **FPGA + low-latency stack: Optiver** FPGAs as "strategic pillar" with AMD partnership (Kevin Sprague, May 2025); **HRT** engineering blog exposes HRTWorker + HeraclesQL; **Jane Street** OCaml + OxCaml + FPGA + LLVM; **Databento** modern OPRA feed-handler pattern: Napatech NT200A02 100G FPGA NIC + NVMe flash + UDP multicast ingress + TCP egress + kernel bypass + C++23 + Rust + Zstandard + core/NUMA pinning + CBBO-1s schema.
- **Open-source contributions:** Man Group stands out as most prolific open-source contributor — **ArcticDB** (C++ DataFrame DB, 2.4k GitHub stars; integrated into Bloomberg's BQuant), CPython 3.14 zlib-ng (May 2026), pip perf (Jul 2025), Imperial Algo-thon. KX: open-source `KxSystems/kx-skills` Claude Code plugins.

C1.19 — What Macion Capital should learn

1. **LLM-for-memo-drafting is the lowest-risk AI deployment.** Every firm does it; the audit trail is clear. Bridgewater, JPM, Man, Balyasny, Citadel all have it.
2. **Vendor NLP is the standard.** AlphaSense or in-house + FinGPT. Don't build from scratch.

3. **ML for alpha is necessary but not sufficient.** AQR's skepticism is the right counterweight. The Virtue of Complexity paper is the canonical 2024 ML-for-finance read.
4. **AI agents for ops are the next wave.** Man Numeric's **AlphaGPT** is the proof-of-concept: "proposes signals, writes the code, and runs backtests before any human sees the output" — but a human still sees the output and runs the gate stack.
5. **Open-weights on-prem for IP-sensitive code.** Don't send your alpha logic to a frontier API.
6. **Governance from day 1.** Even if you're a pre-launch fund, the audit trail matters. Bridgewater's "de-optimized, explicable tabular learning" is the LPs-must-trust-it counterpoint.
7. **AI Engineer ≠ LLM Engineer.** The dominant 2024-26 hiring title is "AI Engineer" / "Applied AI Engineer" with broad LLM/agent/RAG responsibilities. "LLM Engineer" specifically is rare even at the most aggressive adopters.
8. **Don't let AI branding drive fundraising.** Winton (silence) and Renaissance (silence) coexist with Balyasny (named CAIO) and Bridgewater ($2B vehicle) in the same industry. The right posture is the one that matches your research culture.
9. **Treat AI as a C-suite function.** Balyasny named a Chief AI Officer in 2026. For an own-fund track, the question is: when does AI Lead become CAIO? Answer: when AUM justifies a senior hire.
10. **The standing rule.** Never let the LLM be the alpha. Make it the orchestrator and let the gate stack be the judge. The Macion Capital methodology enforces this rule at every stage of the lifecycle, and the principal's 15-agent research platform is the operational implementation.

See `13_cross_cutting_infra.md` for the cross-cutting infrastructure section (GPU, vector DB, Kafka, FPGA, open-source). See `08_firm_case_studies_retail_indie.md` for the retail/indie counterpart.

Chapter C3. Cross-cutting infrastructure — GPU, vector DB, Kafka, FPGA, OSS, alt-data

BOTTOM LINE The 2026 consensus stack is CoreWeave + KDB.AI on KDB-X + Kafka 4.3 + Flink + Iceberg + MCP/A2A for compliance — every small fund rents, not builds, until AUM > $100M; vendor verification is the discipline that keeps dead-domain citations from contaminating research.

The plumbing layer. The institutional firms in C1 don't all use the same data, the same vector DB, the same real-time pipe, or the same FPGA setup. But the **2026 consensus patterns** are now visible. This section maps the consensus layer — GPU infrastructure, vector search, real-time pipelines, FPGA stacks, open-source contributions, alt-data vendor reference, and a warning list of dead domains. Use it as the buy-vs-build menu when standing up an AI-aware quant fund.

C3.1 — GPU infrastructure consensus (2026)

The 2025-26 GPU infrastructure story is dominated by three names: NVIDIA (still), CoreWeave (specialty cloud), and Optiver (in-house custom). The Macion Capital question is **when does it make sense to rent vs. build**.

C3.1.1 CoreWeave + NVIDIA Blackwell/Grace-Blackwell

[STATED, coreweave.com blog, retrieved 2026-07-07] CoreWeave ~250,000 NVIDIA GPUs across H100, H200, **Blackwell B200, GB200, GB300**. **~360 MW** of contracted power; ~1.3 GW active IT capacity; ~1.6 GW in active expansion; **>3 GW** of secured long-dated capacity.

[STATED, coreweave.com blog, retrieved 2026-07-07] Jane Street confirmed customer (Nov 2024 secondary share sale: Jane Street, Magnetar, Fidelity, Macquarie paid $650M). The first **publicly-named hedge-fund CoreWeave customer** in the 15 firms.

[STATED, coreweave.com blog, retrieved 2026-07-07] Architecture: **Slurm + GraphRunner + Kubernetes (RKE2)** for AI/HPC workloads; high-speed **NVIDIA Quantum-2 InfiniBand** (400 Gbps per port, 4×400 Gbps per node) and **Spectrum-X Ethernet**; **VAST Data** for storage; **WEKA** for HPC storage.

[STATED, coreweave.com blog, retrieved 2026-07-07] CoreWeave's "Mosaic AI Research Lab" (June 2026): **A100, H100, H200, B200, GB200** available through **NVIDIA DGX Cloud Lepton** marketplace; cold-start benchmarks on NeMo Megatron, BioNeMo, Earth-2.

[INFERRED] CoreWeave is the **default specialty GPU cloud for hedge funds and AI-native companies** in 2026. The Jane Street customer confirmation is the most explicit data point. For a small fund: rent from CoreWeave before building.

C3.1.2 Optiver in-house GPU infrastructure

[STATED, optiver.com engineering blog, retrieved 2026-07-07] Rob Hahn (Dec 2025): Optiver runs **Slurm + GraphRunner** for AI workload scheduling. The trade-off: Slurm's batch scheduling model is being adapted to a **low-latency trading environment** where latency budgets are sub-millisecond. The blog post details the **GraphRunner** abstraction: per-job container orchestration, GPU pin isolation, pre-heated model containers (so live-trading inference doesn't pay cold-start cost).

[INFERRED] Optiver is a **mixed-scheduler shop**: Slurm for research, custom container infrastructure for production. The blog is the most explicit "we run an HFT desk on top of a research-grade GPU scheduler" write-up in the industry.

C3.1.3 Man/OMI Graphcore IPU partnership

[STATED, oxford-man.ox.ac.uk] Man/OMI's **Graphcore partnership** for financial forecasting is a **differentiated hardware bet** — running financial ML on Graphcore IPUs vs NVIDIA GPUs.

[INFERRED] The OMI/Graphcore bet is a hedge against the NVIDIA monoculture: IPUs are optimized for graph workloads and sparse computation, which can be more efficient than GPUs for certain financial time-series tasks. The bet is on a small hardware vendor with a niche architectural advantage, not a bet on dominance.

C3.1.4 Macion Capital inference

For a small quant fund: - **Y0 (pre-revenue / pre-AUM):** Rent from CoreWeave on demand. The default for research + backtesting. - **Y1 (first AUM):** Rent dedicated capacity on CoreWeave for the production model. 1-2× H100 or H200. - **Y2+ ($50M+ AUM):** Investigate reserved cloud (CoreWeave, Lambda Labs, AWS) vs in-house (Dell + NVIDIA DGX H/B series). Don't try to compete with Jane Street on infra; **buy the right tooling and rent the right scale**. - **Y3+ ($500M+ AUM):** Custom infrastructure (Optiver/Man AHL model) makes sense only if latency budget or data gravity requires it. For an own-fund track, the question is *do you have the engineering org to maintain it?* If not, rent.

C3.1.5 The two-step provisioning loop (Anthropic pattern)

[STATED, anthropic.com engineering, retrieved 2026-07-07] Anthropic's own large-scale training uses a **two-step provisioning loop** — once for cluster bring-up (slow, hours), once for workload scheduling (fast, minutes). The pattern is to **decouple cluster orchestration from workload orchestration** so that the second is fast. For a small fund, this is overkill — but the **principle** (separate bring-up from scheduling) is the right architectural instinct.

C3.2 — KDB.AI + NVIDIA cuVS/CAGRA = hedge-fund-grade vector stack

The vector DB question for 2026 is: which one is **production-grade for hedge fund workloads**? The answer: **KDB.AI on top of KDB-X with NVIDIA cuVS/CAGRA**. This is not the standard "Pinecone vs Weaviate" answer — those are RAG-first vendors, not quant-first.

C3.2.1 The stack

Layer	Tool	Vendor	Role
Core columnar + time-series DB	**kdb+/q, KDB-X**	KX	Time-series + historical research DB; KDB-X is GPU-accelerated
Vector layer	**KDB.AI**	KX	Vector search on top of KDB
GPU-accelerated vector index	**NVIDIA cuVS / CAGRA**	NVIDIA	GPU vector search index
LLM framework	**NVIDIA NeMo Retriever NIMs, NeMo, NIM**	NVIDIA	LLM serving and retrieval
Agent harness	**MCP server, Claude Code, Gemini CLI, OpenAI Codex**	Various	Agentic RAG over the corpus
ML/quant	**PyKX**	KX	Pythonic access to KDB

[STATED, kx.com blog, retrieved 2026-07-07] KX publishes **open-source `KxSystems/kx-skills` Claude Code plugins** — the first hedge-fund-aimed Claude Code plugin repo.

C3.2.2 Named KX customers in the AI/HF adjacent space

[STATED, kx.com customer page, retrieved 2026-07-07] B2C2 (digital asset liquidity provider) and **ADSS** (Abu Dhabi-based market maker) are KDB.X/KDB.AI customers. **Vanguard** (asset manager) is a public KX customer for the **Personal Investor RAG** project (Nov 2025).

[STATED, kx.com case studies, retrieved 2026-07-07] Vendor-reported 30× faster backtesting; 80% infra reduction on KDB.X vs CPU-only KDB. **Vendor-reported; not independently audited.**

C3.2.3 Macion Capital inference

For a small fund, the KDB stack is **overkill until AUM > $100M**. The default 2026 stack:

- **Y0:** Chroma (local) or in-memory FAISS — free, fast, no vendor lock.
- **Y1 (pre-revenue, building alpha library):** Pinecone Starter (managed) or self-hosted Qdrant.
- **Y2 (live with AUM):** Re-evaluate KDB.AI only if you already have a KDB+/q deployment for time-series.
- **Y3+ (scale):** KDB.AI + cuVS/CAGRA makes sense if your research process requires both fast historical time-series *and* vector search over research memos + gate rulings. **Most small funds don't need this.**

The RAG corpus for an own-fund track — **memos, kill log entries, gate rulings, research findings** — is small (thousands to millions of items) and can be served by Chroma/Qdrant/Weaviate on a single GPU. **Don't pay for a vector DB you don't need.**

C3.3 — Apache Kafka 4.3 + Flink + Iceberg = real-time pipe consensus

For real-time market data ingestion, the **2026 consensus stack** is Apache Kafka + Apache Flink + Iceberg (or Delta Lake). This is the load-bearing layer for any live trading system.

C3.3.1 The 2026 stack

Layer	Tool	Role
Event log	**Apache Kafka 4.3** (Kafka 4.3.0 released June 2026: KRaft, Tiered Storage, Share Groups)	Durable, replayable, ordered log of all market events
Stream processing	**Apache Flink**	Real-time stateful transformations (sliding windows, joins, aggregations)
State store	**RocksDB**	Local embedded state for Flink operators
Change data capture	**Debezium**	Captures DB changes for downstream consumers
Lakehouse / historical	**Apache Iceberg** or **Delta Lake**	ACID transactional layer over Parquet for historical replay
Compliance + agent layer	**MCP server + A2A protocol + EU AI Act Art. 12 logging + NIST AI RMF**	The 2026 compliance-ready AI-agent pattern

[STATED, confluent.io blog, retrieved 2026-07-07] Confluent's **"Compliant AI Agents with Stateful Stream Processing"** (15 Jun 2026) is the canonical 2026 write-up of this pattern. The pattern is: every AI agent's actions are events on Kafka; Flink enforces compliance rules (e.g., kill switch, position limit, EU AI Act Art. 12 log retention); the agent harness is **MCP + A2A** (agent-to-agent), and the system must be auditable under EU AI Act Article 12 (logging), GDPR (data privacy), CCPA (California consumer privacy), and NIST AI RMF (risk management framework).

C3.3.2 Why this matters

- **Replayability.** Kafka retains the full event log; a backtest can be re-run against the historical stream.
- **Stateful stream processing.** Flink's windows + state enables rolling features (e.g., "5-minute rolling realized vol") that would be expensive to recompute on a batch.
- **AI agent audit trail.** Every agent decision is logged; the gate stack can verify the agent didn't make an unauthorized trade.
- **Compliance.** EU AI Act Art. 12 (high-risk AI logging) requires this kind of immutable event log; Kafka + Iceberg is the cheapest path to compliance.

C3.3.3 Macion Capital inference

For an own-fund track with no live venue yet: - **Y0–Y1:** Use Macion Capital's existing daily runner + cron pattern (CLAUDE.md §9.1; `scripts/daily_report_runner.py` and `scripts/sign_flip_monitor.py`). Don't add Kafka until you have a live venue. - **Y2 (first live venue):** Set up Kafka + Flink + Iceberg. Use the **Confluent Cloud free tier** (managed Kafka) for the first 6 months. - **Y3+:** Self-host on AWS/GCP only if cost exceeds managed service cost. Most funds never reach this point.

The Macion Capital platform's `sign_flip_monitor` and `daily_report_runner` scripts (canonical: `scripts/` at the workspace root) are the early-stage equivalent — they don't need Kafka. The Kafka step comes when there's real-time P&L or live execution to monitor.

C3.4 — FPGA + low-latency stack

For HFT or latency-sensitive execution, the 2025-26 consensus is **FPGA + custom language + Databento-style infra**. Macion Capital's data layer uses Databento as the standard minute-resolution vendor (canonical: DATA_REGISTRY.md); the same vendor ecosystem serves both worlds.

C3.4.1 The public 2025-26 FPGA players

Player	Public evidence	Stack
Optiver	FPGAs as "strategic pillar"; AMD partnership (Kevin Sprague, May 2025)	Custom FPGA for matching engine / packet processing
HRT	Engineering blog: **HRTWorker + HeraclesQL** Python DSL + SystemVerilog language server exposed	Custom distributed-systems framework with FPGA-friendly language
Jane Street	OCaml + OxCaml + LLVM + FPGA	Custom compiler stack for FPGA
Databento	Modern OPRA feed-handler pattern: **Napatech NT200A02 100G FPGA NIC + NVMe flash + UDP multicast ingress + TCP egress + kernel bypass + C++23 + Rust + Zstandard + core/NUMA pinning + CBBO-1s schema**	Vendor reference architecture
AMD	**AMD PEAK pack** (Programmable Ethernet Acceleration Kit) launched ~2024	Open FPGA framework for partners

C3.4.2 Macion Capital inference

For an own-fund track: - **Y0–Y1:** Don't use FPGA. Use **Databento** as a managed data vendor with their standard 10G NIC. The latency budget is 1-10ms — fine for daily/hourly strategies. - **Y2 (live execution, microsecond budget):** Investigate FPGA only if the strategy requires it. Most ML-driven strategies don't. - **Y3+ (HFT or co-located strategy):** FPGA is mandatory. Use the AMD PEAK pack or buy pre-built.

Macion Capital's research platform has **no FPGA component** as of 2026-07-07. That's the right posture for a 1m-resolution crypto/futures research stack. FPGA enters only if a future strategy demands sub-microsecond execution.

C3.5 — Open-source contributions by 15 firms

A useful proxy for "this firm actually does engineering" is open-source contribution. The 15 firms and their public open-source footprint:

Firm	Public OSS contributions	Notes
Man Group	**ArcticDB** (C++ DataFrame DB; **2.4k GitHub stars**; integrated into Bloomberg BQuant); **CPython 3.14 zlib-ng** (May 2026, 2-3× faster compression); **pip perf work** (Jul 2025, "twice as fast"); Imperial Algo-thon	Most prolific OSS contributor of the 15 firms
KX	**`KxSystems/kx-skills`** Claude Code plugins	First hedge-fund-aimed Claude Code plugin repo
Two Sigma	**RLlib + Ray** (Aug 2021 paper; >10k GitHub stars combined); **Halite** (AI programming game, 2016-2018)	Production-grade RL library + AI talent-pipeline game
Databento	Modern OPRA feed-handler reference architecture (public blog)	Open reference architecture (not OSS code, but public)
CoreWeave	Engineering blog with reproducible Slurm + GraphRunner + DGX Cloud Lepton patterns	Public infrastructure patterns
Optiver	Engineering blog with reproducible Slurm + GraphRunner patterns	Public infra write-ups
Hudson River Trading	**HRTWorker** (distributed-systems framework) + **HeraclesQL** (Python DSL); engineering blog at hrtbeat	Custom infra exposed via blog + framework
Bridgewater	Public AI research articles (Greg Jensen + Jasjeet Sekhon)	Thought leadership, not code
AQR	Public ML paper series (Kelly/Xiu/Malamud) — code not public	Academic papers, no OSS code
WorldQuant	**BRAIN platform** (public-facing) + IQC / Global Alphathon	Crowdsourced alpha + competitions
Jane Street	**OCaml + OxCaml + LLVM** (the open-source OCaml ecosystem); broad OSS donations to the OCaml community	OCaml is a Jane Street stack foundation

Firm	Public OSS contributions	Notes
D. E. Shaw	DESRES Anton supercomputer research papers (Nature/Science)	Academic publications
Balyasny	None visible	Closed source
Millennium	Joint venture with WorldQuant (2018)	Investment, not OSS
Winton	Hivemind (spun out 2018)	Investment, not OSS

Macion Capital inference: Man Group stands out as **the most prolific open-source contributor of the 15 firms**. The pattern of OSS contributions + academic publications + open-source plugins (Claude Code) is consistent with **technology-as-engineering-contribution** culture, not vendor-product marketing. For an own-fund track: **man-group's posture is the most useful template** for a small fund that wants to be both engineering-first and academically credible.

C3.6 — Alt-data vendor reference

A practical reference list for the alt-data vendors publicly used in 2025-26. Most are paid; the list is here so a fund can benchmark, not to encourage spend (the standing rule is no paid spend without principal approval).

C3.6.1 NLP / document extraction

Vendor	Coverage	Use case	Notes
AlphaSense	**85%+ S&P 100, 70% top 50 hedge funds** (per vendor pricing page)	Earnings call transcripts, broker research, SEC filings	The market leader; enterprise pricing 5-figure+
RavenPack	12M entities	News + alternative data	Long-established; used by Millennium, Two Sigma, Man
Hebbia	**$30T+ AUM** coverage (per vendor)	AI document analysis	Fast-growing; 2024-25 enterprise push
FinGPT (open source)	Open-source	Custom NLP for finance	Free, requires engineering
AlphaSense Tegus	Expert-call transcripts	Qualitative research	Acquired by AlphaSense 2024

C3.6.2 Credit-card / consumer transaction

Vendor	Coverage	Notes
Facteus	185M anonymized US credit/debit cards	Largest panel; enterprise pricing
M Science	Custom datasets; "Maddie" chat-style interface (2026)	Data licensing, not just API

C3.6.3 Satellite / geolocation

Vendor	Coverage	Use case
Planet Labs	Daily Earth imagery	Oil storage, supply chain, parking lots
Spire Global	AIS ship tracking + weather	Maritime, supply chain
RS Metrics	Satellite-based metal storage	Metals, energy

C3.6.4 Sentiment / social

Vendor	Coverage	Use case
Eagle Alpha	**2,500+ alternative data products**	Aggregator; useful for discovery
RavenPack	News + social	Sentiment + entity tracking
Brain (ex-Language)	NLP API	Custom sentiment

C3.6.5 Macion Capital inference

For an own-fund track: - **Y0 (no budget):** Use **FinGPT** (open source) + SEC EDGAR + free AlphaSense tier + Man Group's ArcticDB. Total cost: $0. - **Y1 ($5K/yr budget):** Add AlphaSense Pro ($1.5K/yr per seat) + Eagle Alpha ($2K/yr). - **Y2+ ($50K+/yr):** Add Hebbia + Facteus + RavenPack. Most funds reach this only when AUM > $50M.

The standing rule applies: **no paid spend without principal approval.** The Macion Capital `data/` folder has the banked equivalents of many paid vendors (free NOAA, free SEC filings, free BLS, free CoinGecko); see DATA_REGISTRY.md for the full inventory.

C3.7 — Dead domains warning (2026)

These are vendors and platforms that were prominent in 2020-2023 and are **dead, acquired, or in a parked-domain state** as of 2026-07-07. This is a critical reference for any research that needs to verify which vendors still exist.

Domain	Status	Notes
safegpt.com (SafeGraph)	**DEAD**	SafeGraph location data lapsed after 2023; the data feed ended. Any reference using SafeGraph post-2023 is outdated.
thasos.ai (Thasos)	**DEAD**	AI-driven alternative data; lapsed.
sentieo.com (Sentieo)	**ACQUIRED**	Acquired by AlphaSense 2021; redirects to AlphaSense.
accern.com (Accern)	**DEAD**	AI/ML for finance; shut down.
balyasny.com	**DEAD**	DNS not found. **The firm is at `bamfunds.com`**, not balyasny.com. Many secondary sources still cite the dead domain.
millenniummanagement.com	**DEAD**	**GoDaddy-parked.** The firm is at **mlp.com**. Many secondary sources still cite the parked domain.

Macion Capital inference: Any 2024+ research citing a vendor URL should be checked against this list. The pattern — vendor is acquired, the firm's domain shifts, and the secondary sources keep the old URL — is a known 2024-26 landmine. **Use the firm's verified current domain, not the historical one.**

C3.7.1 Domain-history search discipline

For any vendor reference, the Macion Capital discipline: 1. **Check the current domain.** If it's parked (GoDaddy/Namecheap parking page), the vendor is gone. 2. **Check the acquisition page.** If a bigger vendor acquired them, follow the redirect. 3. **Check the Wayback Machine.** If the domain was active 2020-23 and dead 2025+, the vendor lapsed. 4. **Tag `[DEAD]` in any new research.** Don't cite a dead vendor without the tag.

This is the standing rule for vendor reference. Apply it in every research memo.

C3.8 — Cross-cutting infra map (summary)

```
                ┌──────────────────────────┐
                │  Market Data Vendors      │
                │ (Databento, AlphaSense,   │
                │  Facteus, RavenPack,      │
                │  Hebbia, Planet, Spire)   │
                └──────────────────────────┘
                             │
                             ▼
                ┌──────────────────────────┐
                │  Real-time Pipe           │
                │ (Kafka 4.3 + Flink +      │
                │  Iceberg/Delta)           │
                └──────────────────────────┘
                             │
          ┌──────────────────┼──────────────────┐
          ▼                  ▼                  ▼
 ┌────────────────┐ ┌────────────────┐ ┌────────────────┐
 │ Time-series DB │ │ Vector DB      │ │ ML platform    │
 │ (kdb+/q, KDB-X,│ │ (KDB.AI,       │ │ (PyTorch, JAX, │
 │  Postgres,     │ │  Pinecone,     │ │  Triton,       │
 │  Databricks)   │ │  Weaviate)     │ │  NeMo)         │
 └────────────────┘ └────────────────┘ └────────────────┘
          │                  │                  │
          └──────────────────┼──────────────────┘
                             ▼
                ┌──────────────────────────┐
                │  AI Agents                │
                │ (MCP, A2A, Claude Code,   │
                │  GPT-4, Gemini)           │
                └──────────────────────────┘
                             │
                             ▼
                ┌──────────────────────────┐
                │  Compliance Layer         │
                │ (EU AI Act Art. 12,       │
                │  NIST AI RMF, GDPR, CCPA) │
                └──────────────────────────┘
```

This is the canonical 2026 hedge-fund infra stack. For a small fund, the **right subset** is the data + ML platform + agent layer; the Kafka + Vector DB layers are Y2+.

See `07_firm_case_studies_institutional.md` for the firm-level adoption evidence. See `08_firm_case_studies_retail_indie.md` for the retail/indie counterpart. See `09_build_vs_buy_matrix.md` for the build-vs-buy decision per layer.

Chapter C4. AI wallet agents & autonomous onchain capital

BOTTOM LINE AI-wallet-agent infrastructure (Coinbase AgentKit, Crossmint, Turnkey, Fireblocks-via-Dynamic) is real; the alpha is not yet demonstrable — treat it as plumbing, not as an alpha engine, until rung-4 evidence (backtested live) emerges from any of the 17 firms in C1.

> **The 2026 frontier of agentic finance.** This section maps the public-state infrastructure for AI agents that hold their own crypto wallets, route onchain capital, and (claim to) trade autonomously. Tag legend follows the workbook convention: **[STATED]** = direct quote from a primary URL; **[INFERRED]** = derived from public signals; **[NEEDS VERIFY]** = claim looks reasonable but no primary source retrieved.

The standing rule still applies: agents are read-only proposers; the gate stack is the judge; the wallet is the actuator. This section is about **infrastructure for agents that hold keys** — and most of it is **agent-friendly plumbing (x402, llms.txt, MPC-AA)**, not actually autonomous alpha. This is a critical pattern-recognition note for the v1.6 reader.

C4.1 — The plumbing layer: agent-friendly rails (x402, llms.txt)

Before any real AI agent trades onchain, two primitives are required: a **billing rail** (so the agent can pay per call) and a **doc-as-data rail** (so the agent can read protocol docs without scraping).

C4.1.1 x402 — the "agent-friendly billing protocol"

[STATED, x402.org] x402 is a billing protocol. The flow: an AI agent sends an HTTP request → receives `402: Payment Required` → pays (e.g., USDC) → receives the resource. **HoudiniSwap, Coinbase, Nevermined all support it.** The pattern is plumbing, not intelligence.

C4.1.2 llms.txt — the "doc-as-data" standard

[STATED, houdiniswap.com/docs.houdiniswap.com/llms.txt, summer.fi/docs.summer.fi/llms.txt] Many DeFi protocols now publish `llms.txt` files at their docs roots — a single-file index of their full documentation, formatted for LLM consumption. This is the **2026 standard for agent-friendly docs**, not an AI product per se.

Macion Capital inference. These two primitives (x402 + llms.txt) are **the agent-friendliness floor** for any DeFi protocol that wants to be agent-discoverable. For own-fund vendor evaluation: **does the protocol publish both?** If not, assume its agent integration story is weak.

C4.2 — The wallet layer: AgentKit, Crossmint, Skyfire, Nevermined, Privy, Turnkey

Where the real plumbing lives. These vendors give AI agents their own wallet + MPC key infrastructure + scoped autonomy policies.

C4.2.1 Coinbase Based Agent / AgentKit (Oct 2024)

[STATED, coinbase.com/blog/introducing-based-agent] Coinbase launched "Based Agent" on **Oct 26, 2024**: a framework for any developer to spin up an AI agent with a crypto wallet on Base in <1 min using an LLM + CDP SDK + gas-credits. Now exposed via **AgentKit / CDP Developer Platform** (CDP SDK 2.x with deterministic agent addresses + team-managed keys).

[STATED, docs.cdp.coinbase.com/agentkit] AgentKit components: - **Agentic Wallet** — USDC balances, onchain trades - **x402 pay-per-call** — already live on Base - **Session Keys** — scoped autonomy for bounded operations - **Gasless signatures** — batched operations

[STATED, base.org/builders/smart-wallet] Coinbase Smart Wallet (WaaS/MPC successor to legacy Wallet-as-a-Service) supports 2025–2026 agent deployments with policy-bound MPC keyshares.

C4.2.2 Crossmint (founded 2022)

[STATED, crossmint.com/ai-agents] Crossmint exposes wallets, tipping, and x402 endpoints purpose-built for AI agents. Custodial-as-server MPC-AA stack issuing Solana/EVM wallets via API. **Agentic Commerce Suite** (2025) integrates Stripe + Coinbase Commerce for fiat ramp.

C4.2.3 Skyfire ("The Agent Trust Stack")

[STATED, skyfire.xyz] Skyfire launches KYA tokens (verified agent identity) + payment credentials so agents bypass 403s and complete checkout on the open web. Public access 2025.

C4.2.4 Nevermined ("financial rails for AI")

[STATED, nevermined.ai] Nevermined ships: - **Nevermined Pay** — virtual-card spend with user-defined rules - **OpenClaw** — monetize in chat - **x402 Facilitator** — settlement across Stripe today, PayPal Braintree "coming soon" - **Partners:** AWS, Mastercard, Visa, VGS, Exa - **Compliance:** SOC 2 + ISO 27001 - **SDKs:** TS/Py/CLI

C4.2.5 Privy — embedded wallets for agents

[STATED, privy.io] Privy ships an "Agent wallets" tier for autonomous AI agents. Drop-in embedded wallets.

C4.2.6 Dynamic → acquired by Fireblocks (Oct 2025)

[STATED, dynamic.xyz] Dynamic (auth + smart-wallet SDK) acquired by **Fireblocks in Oct 2025**; advertises an "Agentic" use case for AI agents with secure wallets + stablecoin payments.

C4.2.7 Turnkey ("Agentic Wallets + Agentic Payments")

[STATED, turnkey.com] Turnkey ships: - **Enclave-secured WaaS** with sub-100ms signing - **Multi-chain** (EVM, Solana, BTC) - **Policy controls** (per-agent spend limits, allowlists) - **Investors:** Sequoia, Bain Capital, Coinbase

C4.2.8 Macion Capital inference — which vendor to pick

For an own-fund track building an AI agent that holds a wallet (e.g., for DeFi strategy execution):

Tier	Vendor	Why
Y0 — learn + POC	**Privy**	Free tier, easy embedded wallets
Y1 — first AUM, agent must trade	**Coinbase AgentKit**	Native to Base; full AgentKit docs
Y2 — scale, compliance-grade	**Turnkey**	SOC 2 + audit-grade logging
Y2+ — multi-vendor / resilience	**Crossmint + Dynamic/Fireblocks**	Multi-agent + cross-chain

The standing rule. The agent wallet is a **regulator-visible surface**. EU AI Act Article 12 logging applies to every wallet operation. **Pick a vendor with audit-grade logs from day 1**, not after the first incident. The cheapest stack (Privy) is the worst for compliance; the most expensive (Fireblocks-via-Dynamic) is the most compliance-ready.

C4.3 — The agent layer: ai16z / Eliza, Virtuals, Wayfinder, Brian, Colony

Above the wallet layer, agent frameworks provide the **AI brain + tool-use loop**. These are the closest things to "AI agents that hold wallets and trade" in 2026.

C4.3.1 ai16z / Eliza / ElizaOS (Oct 2024)

[STATED, elizaos.ai] ai16z (DAO Fund) launched **Oct 2024** by Shaw (pseudonymous founder of Eliza Labs) as an autonomous investor-agent. Evolved into the **Eliza framework** (now "**ElizaOS / elizaOS**" — "Your Agentic Operating System") with: - Multi-channel Unified Message Bus (Discord, Telegram, X, onchain) - Composable Swarms - Strategic Action Chaining - npm plugins - GitHub `elizaOS/eliza`

C4.3.2 Truth Terminal (mid-2024)

[STATED, x.com/truth_terminal, widely chronicled] Andy Ayrey's memetic AI; **did not autonomously trade**, but memetically launched **GOAT (Goatseus Maximus)** and spawned the meta of memecoin-deploying AI agents. The pattern: **meme-launch, not alpha-trade**.

C4.3.3 Virtuals Protocol ("Society of AI Agents")

[INFERRED, virtuals.io] Launched late 2024 on Base; **Luna** was the protocol's flagship "agent-to-earn" virtual companion + autonomous trading prototype, paired with the VIRTUAL token. Subsequent agents include **aixbt** (autonomous "alpha-calling" agent buying terminal access with its agent wallet).

C4.3.4 Wayfinder / Full Frontal (Parallel Studios)

[INFERRED, parallelstudios.co, base.org] Wayfinder provides onchain routing agents on Base. The network's flagship agent is **"Full Frontal"** hosted by ChadGPT (Parallax founder Chad Greiner).

C4.3.5 Brian (fetch.ai)

[INFERRED, fetch.ai] Fetch.ai (Cambridge UK, since 2019) ships autonomous agents; recent product **"Brian"** acts as DeFi-operations assistant + onchain query layer, with ASI-Alliance cross-chain routing.

C4.3.6 Colony / Bounty-on-Base

[INFERRED, colony.io] Agent-discovery primitive on Base; **Colony/Bounty-on-Base** stack handles agent-to-agent task + escrow.

C4.3.7 Macion Capital inference — the agent framework landscape

Framework	What's strong	What's weak	When to use
ElizaOS	Composable Swarms, npm plugins, multi-channel bus	Open-source; less battle-tested; financial-domain plugins are community-built	When you need a full agent OS
Virtuals Protocol	Native token + onchain governance for "agent-as-token"	Hyped token, thin autonomous alpha	If you want the social-token meta
Wayfinder	Native Base routing	Single-vendor (Parallel)	Base-specific DeFi strategies
Brian (fetch.ai)	Cross-chain, ASI-Alliance	Older stack	Cross-chain DeFi ops
Colony	Agent-to-agent escrow	Early-stage	Multi-agent job-market patterns

The honest verdict. None of these are "AI trading agents beating human PMs on 20-year backtests" — that experiment is what FINSABER has refuted. They are **agent frameworks with onchain plumbing**; the same FINSABER null applies to their alpha output. **Treat them as infrastructure, not alpha engines.**

C4.4 — The trading-bot layer: HIPs, MEV, DEX bot frameworks

For latency-sensitive onchain strategies.

C4.4.1 Hyperliquid protocol primitives

[STATED, hyperliquid.gitbook.io] - **HIP-2 (Hyperliquidity)** — protocol-built liquidity; no third-party operators - **HIP-3 (Builder-deployed perpetuals)** — permissionless perp DEX deployment, gated by **500k HYPE stake** - **SDKs:** Python (updated 2026-06-04) + Rust

[INFERRED — no confirmed trading-agent fill activity by ai16z/Eliza, Truth Terminal/Project Plutus/AIxC, Project 0/Xerk/Polytrader/Vader.ai, hilbot] Despite the November 2024–June 2026 window, the Hyperliquid ecosystem shows **no confirmed AI-agent trading fills** in free public sources.

C4.4.2 MEV — Jito vs Hyperliquid

[STATED, jito.wtf] Jito = off-chain Block Engine + Bundles + ShredStream; only on Solana. **[STATED, hyperliquid docs]** Hyperliquid = on-chain CLOB, **no mempool** → no Jito-style MEV.

Implication: Hyperliquid strategies are order-book-based (limit orders, queue priority); Jito strategies are bundle-based (front-run protection). AI-agent trading applies to **both**, but the alpha source is different.

C4.4.3 Macion Capital inference — the AI-agent trading-bot honest verdict

The peer-research pass surfaced **no confirmed AI-agent fill activity** of any meaningful size on Hyperliquid, despite the very public funding rounds and VIRTUAL-token launches. Reasons: - **Per FINSABER:** LLM strategies don't beat simple baselines on 20-year backtests. - **Per the latency reality:** MEV is sub-millisecond; reasoning models are 30-120s — mismatch. - **Per the wallet-ops reality:** Each agent wallet operation is a regulator-visible surface; first-mover scale invites first-mover enforcement.

The Macion Capital posture for v1.6: observe, do not deploy capital into AI-wallet-agent strategies at this layer. The infrastructure is real; the alpha is not yet demonstrable.

C4.5 — DeFi liquidation avoidance: the "mostly vapor" finding

A specific subset of AI-agent use cases is **DeFi health-factor monitoring + automated rebalancing** to avoid liquidation.

[INFERRED, multi-source] As of 2026-07-07:

Protocol	AI feature	Source
Summer.fi (formerly DeFi Saver)	"Automations" exists; **no AI/LLM/agent/MCP feature documented**	`summer.fi` / `docs.summer.fi` (homepage audit only)
Instadapp	Homepage shows **no AI/liquidation-prediction features**	`instadapp.io`
LlamaRisk	Known Aave health-factor monitor, AI-feature status **not verified**	`[NEEDS VERIFY]`
Aave forums / Snapshot	AI proposals **not surfaced**	`[NEEDS VERIFY]`
HoudiniSwap	**No AI features**; ships x402 + llms.txt primitives only	`[STATED]` homepage
Secret Network — Secret AI	**The real privacy+AI primitive**: TEE-hosted LLM inference (Intel TDX + AMD SEV-SNP). NOT a liquidation-protection tool by name	`[STATED]` `scrt.network`
Dextrabot	**AI liquidation bot** specifically — canonical product/feature audit unverified	`[NEEDS VERIFY]`

The bottom line on DeFi liquidation + AI (2026): Mostly vapor. Public sources confirm **agent-friendly plumbing** (x402, llms.txt, agent-tooling repos) but **no protocol with a documented AI-driven liquidation-avoidance feature shipping to production**. **Secret Network's `Secret AI`** (TEE-hosted LLM inference) is the strongest privacy+AI primitive in the space; whether it's applied to liquidation monitoring is `[NEEDS VERIFY]`.

Macion Capital posture: Treat any "AI liquidation bot" vendor claim as **rung 1 (careers/marketing signal)** on the F.5.4 evidence-quality ladder — not as confirmed deployed feature. The pile of x402 + llms.txt adoption is real, but **agent-friendly ≠ AI-driven**.

C4.6 — The standing rule for AI-agent capital deployment

Translated for v1.6:

1. **Pick the wallet vendor for audit-grade logs** (Turnkey, Fireblocks-via-Dynamic), not for the cheapest onboarding (Privy).
2. **Wire EU AI Act Article 12 logging** to every wallet operation from day 1.
3. **The alpha source is NOT the LLM.** The alpha source, if any, is in the strategy *the agent orchestrates*. The agent is the executor; the gate stack is the judge; the wallet is the actuator. (Same standing rule as the rest of the workbook.)
4. **Observe the Hyperliquid / Virtuals / Eliza ecosystem** for evidence of live AI-agent fills, but do not deploy capital until a `[STATED]` rung-4 result (backtested live net-of-cost) is published by any of the 17 firms in C1.
5. **Treat every "AI alpha agent" vendor pitch** as marketing until rung 2 (public engineering write-up) is shown.

See F.5.4 for the evidence-quality ladder and F.5.5 for the reasoning-model latency-cost tax on AI-wallet-agent decision loops.

C4.7 — Cross-cutting infra map (AI-wallet-agent layer)

```
┌───────────────────────────┐
│  Agent Frameworks         │
│ (ElizaOS, Virtuals,       │
│  Wayfinder, Brian)        │
└───────────────────────────┘
              │ tool-use / action calls
              ▼
┌───────────────────────────┐
```

```
│   Wallet + MPC Layer         │
│  (AgentKit, Crossmint,      │
│   Skyfire, Nevermined,      │
│   Privy, Turnkey)           │
└────────────────┬────────────┘
                 │ x402 / onchain txs
                 ▼
┌─────────────────────────────┐
```

```
│   Compliance + Audit Log     │
│  (EU AI Act Art. 12,         │
│   Wallet ops log retention)  │
└──────────────────────────────┘
```

This is the canonical 2026 AI-wallet-agent infra stack. For a small fund, the **right subset** is the wallet + compliance layer; the agent framework is Y1+.

See `13_cross_cutting_infra.md` for the underlying infra (GPU, vector DB, Kafka, FPGA). See F.5.4 for the evidence-quality ladder.

Part D. Retail / Indie Quant Stack

What this part answers. What can a solo quant or 3-person team realistically build in 2026, and how does it compete with institutional scale? Short answer: the retail stack in 2026 is more capable than the institutional stack of 2015 — the bottleneck is talent, not tools.

Chapter D1. Retail / indie stack — QuantConnect, WorldQuant BRAIN, Numerai, ...

BOTTOM LINE A solo retail quant with $500/mo budget has access to ~70% of what an institutional quant with $5M/year has — the gap is in execution, compliance, and capital, not in tools.

What a solo quant or 3-person team can actually build. The retail/indie stack in 2026 is more capable than the institutional stack of 2015. The bottleneck is talent, not tools.

D.1 — The retail quant platform landscape

D.1.1 Cloud quant platforms (free / low-cost)

QuantConnect (Lean engine) — the workhorse

Tier	Price (2026)	Features
Free	$0	Unlimited backtesting, 1 backtest node, 0 live nodes, 10K backtest order limit, 4K plot points
Researcher	$$ (QCC tokens, $20–$250 packs)	Basic AI Automation
Team	$$ (10 compute nodes)	Basic AI Automation + multi-user
Trading Firm	$$$	**Advanced AI Scheduling**
Institution	$$$	**Bring Your Key AI Routing** + on-prem

- **Community: 515,200 quants (CONFIRMED 2026-07-07 via quantconnect.com forum)**
- Open-source engine (Lean) — **20.4K GitHub stars, 13,246 commits, C# 94.2% / Python 5.6% (CONFIRMED 2026-07-07 via github.com/QuantConnect/Lean)**
- Powers "300+ hedge funds" per QuantConnect marketing
- **2026 AI features (CONFIRMED 2026-07-07):** Agentic AI Assistants to automate strategy research/backtesting/live monitoring; **Ask Mia** AI support agent trained on 900 pages of docs; tiered "Basic AI Automation" → "Advanced AI Scheduling" → "Bring Your Key AI Routing"
- 15+ broker integrations (IBKR, Coinbase, Binance, TradeStation, Tradier, Alpaca, Schwab, Kraken, Zerodha, EZE, TD Ameritrade)

Source: quantconnect.com/pricing, retrieved 2026-07-07.

WorldQuant BRAIN

- Free platform with paid competitions
- **~550,000+ users** (CONFIRMED via worldquant.com/brain landing page, retrieved 2026-07-07)
- **~16,000+ consultants** actively submitting (CONFIRMED 2026-07-07)
- **~400,000+ data fields** available (CONFIRMED 2026-07-07)
- Submit alphas in Expression Language
- Get paid if your alpha is selected (low single-digit % selection rate)
- AI features: alpha-idea suggestions, automatic expression simplification

Numerai

- The original "AI hedge fund"
- Submit ML predictions on encrypted financial data
- Stake NMR tokens on your predictions; get rewarded for accuracy, slashed for bad predictions
- Meta-model aggregates top stakers
- 2026 payout: ~$50M/year distributed to top stakers
- **Source:** numer.ai/tournament, retrieved 2026-07-07

Note: Numerai is now moving toward **Era 5** of their tournament structure; the data format changes regularly. Worth checking current state before deep dive.

Composer.trade

- No-code AI quant platform
- Free tier; **Trading Pass $32/mo (CONFIRMED 2026-07-07)**
- **Acquired by SoFi** (CONFIRMED 2026-07-07 via pricing page)
- AI-driven portfolio construction; natural-language strategy creation ("describe your idea, AI generates the symphony")
- Visual no-code editor; automated rebalancing; community remix of strategies

---

D.1.2 Open-source backtesters

Engine	Language	GitHub Stars (2026, CONFIRMED 2026-07-07)	Use case
Lean (QuantConnect)	C# / Python	**20.4K**	Production-grade event-driven
Vectorbt (open core)	Python	**8.2K**	Vectorized backtesting; v1.1.0 released 2026-07-05; explicitly positioned as "AI agent-driven workflow"
Zipline-reloaded	Python	~2K (community forks)	Legacy Quantopian users; QuantRocket ships native Zipline Live
Backtrader	Python	**22.3K** (last release static)	Hobbyist favorite; effectively learning-only in 2026
FinRL	Python	**15.6K** (NeurIPS 2020 paper)	RL research; v0.3.8 (2026-03-20)
Qlib	Python	~15K	AI-first quant platform (Microsoft)
Freqtrade	Python	**52.1K** (v2026.6, 2026-06-29)	Crypto-only; **FreqAI** = productionized ML extension
MLFinLab (Hudson & Thames)	Python	**4.9K** (paid license)	López de Prado end-to-end (triple-barrier, bet-sizing)
Jesse	Python/JS	**8.1K** (JS 85% / Python 14%)	Crypto event-driven; **JesseGPT** LLM-native strategy authoring
Nautilus Trader	Rust	**~24.5K**	**Fastest-rising 2026 repo**; production-grade deterministic OMS; Rust-native
PyBroker	Python	**3.4K** (v1.2.12, 2026-03-05)	ML-driven strategies with built-in walkforward + bootstrap

Macion Capital stance: - Lean for production backtests (or Nautilus Trader if Rust infra exists) - Vectorbt (open core) for fast prototyping and ML feature labels - FinRL for RL experiments (research, not deployable — per the venue-locked finding: an RL edge that does not survive a Topstep-ON re-test is a placebo) - MLFinLab for label/sample-weight/bet-sizing discipline (López de Prado) - JesseGPT / Composer.trade for non-coder workflow demos

---

D.1.3 Data sources (retail-friendly)

Source	Free tier	Paid tier (CONFIRMED 2026-07-07)	Notes
Yahoo Finance (yfinance)	24.6K-star repo (v1.5.1, 2026-06-28)	—	Apache-2.0; unofficial wrapper; personal-use-only ToS; IP-level 429s on large downloads
Alpha Vantage	25 req/day, 25 req/min	$49.99/mo (75 RPM) — $249.99/mo (1,200 RPM); annual $499–$2,499	Stocks, FX, crypto, fundamentals, ~50 indicators
Polygon.io (rebranded	5 calls/min delayed	Starter ~$29/mo (real-time US stocks + 100 req/min);	US stocks + options + FX + crypto

Source	Free tier	Paid tier (CONFIRMED 2026-07-07)	Notes
Massive.com in 2025)		Developer ~$99/mo (options/forex/crypto); Advanced ~$199+/mo	
Tiingo	500 symbols/mo, 1,000 req/day	**Power $30/mo** (108,700 symbols/mo, 10K req/hr, 15+ yr fundamentals)	End-of-day + intraday + news + IEX
Intrinio	—	**Individual (AI-Ready) $150/mo**; Startup $333→$999/mo (18-mo phased); Enterprise $1,250+/mo	**GenAI integration with Claude/ChatGPT; MCP Server**
EODHD (EOD Historical Data)	$0	$19.99/mo EOD; $29.99/mo EOD+Intraday; $59.99/mo Fundamentals; €79.99+/mo commercial	**MCP Server with 72 tools; OpenAPI 3.1 spec with 74 endpoints; 28 pre-built AI agent skills; ChatGPT Assistant**
Databento	$125 credit (~6 mo)	Pay-as-you-go (per-record + flat subscriptions)	Institutional-grade; US L3/MBO historical; 4 ns PTP timestamps; 19 PB coverage
FRED	Yes (St Louis Fed)	—	~840,000 series; 120 req/60s with API key
Quandl / Nasdaq Data Link	Free EOD US equities (Sharadar)	$50–$500/mo individual; $20K+/yr commercial	Free + paid datasets
Binance / Bybit / Coinbase	Yes	—	Public APIs; rate-limited; redistribution prohibited
CCXT	Yes (open source)	Pro tier paid	100+ exchanges normalized REST/WS
OpenBB	Open source (70.2K stars)	Workspace pro.openbb.co	Bloomberg-terminal-style analytics; SEC filings + free feeds

Macion Capital current state: Databento ($125 free credit, then pay-as-you-go) is the primary institutional-grade source; FRED for macro; Binance/CCXT for crypto.

D.2 — AI-first retail products

D.2.1 The roster

Product	What it does	Pricing (CONFIRMED 2026-07-07)	Notable claim
Trade Ideas (Holly AI)	AI-driven US stock scanner	TI Basic $89/mo; **TI Premium $178/mo** (adds Holly AI)	Vendor-reported 60–70% directional accuracy; Premium is sweet-spot
EquBot (AIEQ ETF)	AI-managed ETF	0.45% expense ratio	**+10.02% annualized since Oct-2017 inception vs SPY ~12–14%** (CONFIRMED 2026-07-07) → underperformed SPY by 2–4%/yr for 8.5 years. IBM partnership ended 2024–25; re-platformed as QuantumStreetAI
Composer.trade	No-code AI portfolio	Free + **Trading Pass $32/mo** (now SoFi-owned)	NL strategy creation; community remix
Danelfin	AI stock scoring	Free + Premium (~€15–25/mo, pricing page 403 this session)	European + US equities
Trefis	AI stock estimates	Free + Premium (~$15–30/mo)	Crowd-sourced fair-value models
TipRanks	AI analyst aggregation	Free + Premium ~$30–50/mo; Ultimate ~$300+/yr	Smart Score 1–10
Kavout	AI stock ranking	**Free $0** (10 credits); **Pro $16/mo** (7 AI agents, 30+ markets); Premium $39/mo	K Score + Kai Score
Capitalise.ai	NL-to-algo	Free trial; **acquired by Kraken** (2024–25)	Code-free TradingView alerts; bundled in Kraken
Benzinga Pro	News + AI	Basic $37/mo; Streamlined $147/mo; **Essential $197/mo (adds Benzinga AI)**	High Beta Squawk $99/mo add-on
FinChat / ChatIQ	Finance LLM chat	$30/mo	Cited answers
Stockfeel / AInvest	Chinese-market AI	Free + premium	Asian-market focus
OpenBB Terminal Pro	Open-source Bloomberg	$50/mo Pro	MCP/RAG UI features 2026

D.2.2 The honest verdict

Most "AI" products are: 1. **Old ML rebranded** — a gradient-boosted model from 2020 with "AI" marketing 2. **LLM-on-top** — ChatGPT wrapper with finance data 3. **Marketing-driven** — the AI feature is a footnote; the value is the data

The genuinely useful ones (2026): - **Trade Ideas Holly** — genuinely AI-driven; multi-day forward prediction - **Composer.trade** — no-code + backtest in one; democratizes quant for non-coders - **OpenBB** — open source, real Bloomberg-alternative; community is strong - **Numerai** — the only AI hedge fund with a verifiable tournament

D.3 — The hedgefundie community

The "hedgefundie" / r/algotrading community is a useful leading indicator for retail quant.

D.3.1 The community structure

Forum	Size	Focus
r/algotrading	200K+	General algo trading
r/quant	50K+	Academic quant
QuantConnect forum	100K+ active	QuantConnect-specific
Elite Trader	50K+	Pro-level discussion
Quantocracy	10K+ daily readers	Curated papers + blog posts
QuantStart	100K+ MA	Educational

D.3.2 The common stack (r/algotrading consensus 2025–26)

- Python (vanishing few use R or MATLAB now)
- Backtrader or Lean for backtest
- Interactive Brokers for execution (retail default)
- Postgres for data
- Grafana for dashboards
- A cloud VM ($20–50/mo)
- One large language model API for code + research

Total cost: $100–500/mo for a solo retail quant. **The cheapest viable production stack** (per 2026-07-07 retail source-pack): Composer Trading Pass $32 + Tiingo Power $30 + Alpha Vantage 75 RPM $50 + W&B Pro $60 + Polygon Starter $29 ≈ **$200/mo all-in** for hobbyist; QuantConnect free + yfinance + Databento $125 credit covers the bootstrap.

D.3.3 Best-in-class production stack ($300K/yr mid-tier firm)

The other end of the retail/indie spectrum — a small fund that has graduated past the hobbyist stage. The stack assumes ~5-person team, $5–50M AUM, and the first LP signed.

Layer	Cheapest (hobbyist, $200/mo)	Best-in-class (mid-tier firm, $300K/yr)
Data	Tiingo Power + Alpha Vantage + Databento credit	Databento L3 + AlphaSense Enterprise + Kensho LLM API + S&P Capital IQ Pro + Bloomberg 3 seats
Backtest	Lean self-hosted (free)	QuantConnect Team + Nautilus Trader (Rust) + PyBroker + MLFinLab
ML platform	W&B Pro	MLflow + DVC + BentoML + Feast + Modal production + W&B Enterprise
Vector DB	Chroma (local)	KDB.AI + NVIDIA cuVS/CAGRA on KDB-X OR Pinecone Enterprise OR Weaviate Cloud
Code assistant	Claude Code + Cursor	Claude Code Enterprise + Cursor Business + custom MCP servers + W&B Prompts

Layer	Cheapest (hobbyist, $200/mo)	Best-in-class (mid-tier firm, $300K/yr)
Compliance	DIY + LLM	Two Sigma Venn (now Insight/Solovis) OR in-house MRM framework (OCC 2026-13 + Fed SR 26-02 compliant)
LP reporting	python-pptx + Claude	Addepar OR in-house LP letter generator

The 100× principle: the **cheapest** stack covers ~70% of what a small fund can usefully do (Part D.5). The **best-in-class** stack covers ~99% — the gap is in compliance, scale, and vendor-managed reliability, not in the alpha engine. **The build-vs-buy decision at the mid-tier stage is mostly about who you trust to be on call when the system breaks, not what features you get.** See `09_build_vs_buy_matrix.md` for the per-layer decision.

D.3.4 What retail does well

- **Lower-frequency strategies** (daily to weekly) that institutional desks ignore
- **Niche markets** (microcaps, exotic crypto, regional equities)
- **Quick iteration** — a solo quant can run 100+ backtests/week; a Pod at Balyasny might run 10
- **Personal data** — alt-data the institutions don't have access to

D.3.5 What retail does poorly

- **High-frequency** — co-location costs alone kill retail HFT
- **Scale** — a $10M retail book can't access Tier 1 liquidity
- **Compliance** — solo quant lacks the legal/regulatory infrastructure
- **Risk management** — no second pair of eyes; the discipline suffers

---

D.4 — The "AI-driven ETF" experiment

A specific category worth examining.

D.4.1 The roster (2026)

ETF	Ticker	AUM	Inception	Expense ratio
AI Powered Equity (AIEQ)	AIEQ	~$120M	2017	0.45%
AI-Driven Income (AIIQ)	AIIQ	~$30M	2018	0.75%
QRAFT AI	QRFT / various	~$50M	2019	0.75%
WisdomTree AI	Various	~$200M	2023	0.45%
Global X AI	Various	~$1B	2016	0.68%

D.4.2 The honest verdict

Per most published reviews (2025–26):

- **AIEQ has underperformed the S&P 500 over 5+ years.** IBM Watson is not a magic alpha source.
- **AIIQ has had mixed performance**; income-focused limits the AI leverage.
- **The thematic AI ETFs** (AI-as-sector, not AI-as-alpha) have done well — but that's a sector bet, not an alpha engine.

The lesson: "AI-driven" ETFs are mostly marketing. The retail buyer pays an extra 0.3–0.5% for a wrapper around a mediocre model. The institutional AI funds (Renaissance, etc.) don't sell retail products — and the public funds that claim AI tend to underperform.

D.5 — The retail vs. institutional asymmetry

A useful frame: **in 2026, a solo retail quant with $500/mo budget has access to ~70% of what an institutional quant with $5M/year budget has access to.**

Capability	Retail 2026	Institutional 2015	Asymmetry
Compute	$50/mo cloud	$500K/year cluster	Mostly closed
Data	Free + $200/mo	$500K/year data budget	Mostly closed
Backtesting	Lean open-source	Kdb+/internal	Mostly closed
Execution	IBKR	Co-lo + smart router	Wide gap
ML tooling	Open source	Same open source	Closed
LLM	$20/mo API	$50K/year internal	Mostly closed
Latency	100ms	1µs	Wide gap
Capital	$100K	$10B	Wide gap
Compliance	DIY	$500K/year	Wide gap

The takeaway: if your edge depends on compute, data, or latency, retail loses. If your edge depends on insight, research discipline, or creativity, retail competes.

D.6 — The solo-quant playbook

If you're a solo retail quant:

1. **Pick a niche the institutions don't cover.** Microcaps, regional equities, niche crypto.
2. **Run your own gate stack.** Adopt the G1–G28 + G29–G31 specification verbatim; the canonical reference is `strategies/_templates/reference/gate_stack_reference_G1–G28.md` in any Macion-Capital-style workspace.
3. **Use a multi-agent research platform of the kind Macion Capital operates.** The orchestrator-worker + read-only-adversary pattern is replicable on any Linux box with a Python venv; the methodology is the open-source part, the agent prompts are the IP.
4. **Use Lean + Vectorbt + FinRL for backtesting.** All free.
5. **Use QuantConnect for paper trading.** Free tier is enough to start.
6. **Use IBKR for live trading.** $0 commission tier available.
7. **Budget $200–500/mo for AI tools.** Claude Code, AlphaSense (or FinChat), Databento (free credit).
8. **Document everything.** The audit trail matters even for solo quants — your future self will thank you.

See `10_cost_model_phased_roadmap.md` for the cost scenarios for a small fund.

D.6.1 Sample prompt skeletons (illustrative, not operational)

A Macion-Capital-style research platform uses 15 specialised agents (orchestrator + 13 workers + 1 read-only adversary), each with a charter Markdown file at `.claude/agents/`. The prompts below are **illustrative starting points** for a small team that wants to replicate the orchestrator-worker pattern. They are **not** battle-tested; they are sketches to adapt, not deploy.

```
# Role 1: Data Collector (excerpt)
"""
You are the data-collector agent. Your single mission: assemble a cl
time-aligned panel for the requested universe and date range.

Constraints:
- Every price/yield must have a timestamp, source, and lineage fielc
- Reject any source that fails integrity_check().
- Never invent values; refuse rather than fabricate.

Output: {csv_path, audit_trail_path, integrity_score}
"""

# Role 2: Factor Miner (excerpt)
"""
You are the factor-miner agent. Your mission: propose candidate alph
that could explain a market anomaly, given a clean panel and a hypot

Constraints:
- Every factor must be FALSIFIABLE — specify the exact entry/exit ru
- No 'the model finds a pattern' claims. Name the variables and the
- Bias log: tag every candidate with bias_class ∈ {lookahead, surviv
  selection, publication, data-snooping, regime-overfitting, none}.

Output: {factor_id, hypothesis, mechanism, formula, bias_tags, gate_
"""

# Role 3: Gate Judge (excerpt)
"""
You are the gate-judge agent. Your mission: apply a specific gate (e
DSR, G9 PBO, G12 min-BTL) to a candidate factor and return a PASS/FA
verdict with a numeric anchor.

Constraints:
- ALWAYS compute the gate numerically. Never use qualitative judgmer
- If the underlying test is impossible (e.g., not enough trials for
  FAIL — do not pass on insufficient data.
- Cite the formula and the inputs; no magic numbers.

Output: {gate_id, numeric_value, threshold, verdict, evidence_path}
"""

# Role 4: Report Writer (excerpt)
"""
You are the report-writer agent. Your mission: assemble a research m
from a typed handoff packet (factor + gates + verdict + evidence).
```

```
Constraints:
- Mirror the research-template.md structure verbatim.
- Embed charts as PNGs; do not reference 'see attached' without a path.
- The verdict and the gates must be reproducible from the script + data
  cited; if a step is not reproducible, tag [VERIFY] in the memo.

Output: {memo_path, charts_dir, evidence_paths, verdict, kill_patter
"""
```

Discipline note: every agent above outputs a **typed packet**, not prose. The strategy-scribe consumes the packet, validates the schema, and writes the registry. **No prose handoffs; no Slack-style "agent A told agent B".** This is the contract discipline that keeps multi-agent systems debuggable. The 31-gate eval harness is implemented **scipy-free in numpy**, owned by the principal — the gate stack is the audit trail, not the agent's call.

Part E. Build-vs-Buy, Cost Model & Roadmap

What this part answers. The operational playbook — what to buy, what to build, what it costs, and the 4-phase roadmap for Macion Capital. Implemented as a denser table-heavy format rather than the workbook's eight-part engine — this is operational reference material.

Chapter E1. Build-vs-Buy Matrix

BOTTOM LINE Buy commodity infrastructure; build the differentiating layers — the alpha logic, the gate stack, the eval harness. Everything else is negotiable on cost.

The decision matrix. For every layer of the AI-in-quant stack, what's the 2026 default — buy, build, or hybridize — and what's the realistic cost?

E.1 — The matrix (2026)

E.1.0 The visual decision matrix (one-page)

The 10 components of the AI-in-quant stack, with a traffic-light row per layer. **Default = the option a small fund should take on day 1.**

Component	Default	Build hours	Buy cost ($/yr)	Hybrid?	Risk
Market data	**Buy**	—	$500–$250K	Optional	Low
NLP extraction	**Buy (Y0); Build (Y2+)**	500	$30K–$100K	Yes	Medium
Backtest engine	**Build (Lean free)**	5,000	Free	Yes	Low
ML platform	**Buy**	—	$5K–$50K	Yes	Low
Vector DB	**Buy**	—	$300–$24K	Yes	Medium
Code assistant	**Buy**	—	$1.2K–$24K	No	Low
Compliance	**Buy**	—	$5K–$50K	No	Low
LP reporting	**Build**	1,200	$50K	Optional	Medium
OMS / execution	**Buy**	—	$10K–$100K	No	High
Risk engine	**Buy**	—	$25K–$200K	No	High

Reading the matrix: - **Default = Buy** when vendor competition is healthy and the cost of building > the cost of buying. Most ops layers (data, ML platform, vector DB) fall here. - **Default = Build** when the layer is **differentiating** (backtest engine: every quant's edge is in the alpha logic; LP reporting: every fund's brand is in the letter). The build hours reflect the **first-time** investment, not the steady-state maintenance. - **Risk = High** when the layer has a small vendor pool and is operationally critical (OMS, risk engine). The default is still to buy, but the **vendor-lock-in** is a real cost. - **Hybrid** when the vendor provides 80% of the value and the differentiating 20% must be built. NLP is the canonical hybrid: buy AlphaSense Enterprise for breadth, build in-house for IP-sensitive extraction.

The two-stage architecture (canonical, used in B1, F.5, closing):

```
AI agents (read-only)        ← propose / factor-mine / write code
      │
      ▼
Gate stack (G1–G28 + G29–G31) ← numeric anchors: DSR, PBO, min-BTL
      │
      ▼
Decision (KILL / ITERATE / PROMOTE)
```

This is the **principle** behind every layer decision in the matrix below: the gate stack is the judge; the agents are read-only; the vendor is the cheapest path to the right tool.

E.1.1 Market data layer

Vendor	Pricing (2026)	Coverage	AI features	Build alternative
Databento	$125 free credit; ~$0.0001–$0.01 per record	US equities, futures, options, FX, crypto; 19 PB normalized, 3M+ symbols, 4 ns PTP timestamps	None direct; MCP server community preview	Self-host via IBKR API + CCXT
Polygon.io (rebrand → Massive.com in 2025/26)	$0 free; $30/mo Starter; $99/mo Developer	US stocks, options, forex, crypto	None	Self-host via IBKR
Nasdaq Data Link (Quandl)	Free tier; $50+/mo Premium	Wide alt-data marketplace	NLP for some datasets	Self ven
Alpha Vantage	Free 25/day; $50/mo Pro	US stocks, FX, crypto	None	Self IBK
Tiingo	Free 500 symbols; $30/mo	108,700+ global securities; US/CN stocks; ETFs/MFs; news	None direct	Self
Intrinio	$150/mo Individual (AI-Ready); $333→$999/mo Startup; $1,250+/mo Enterprise	US equities/options/fundamentals	**GenAI integration with Claude and ChatGPT; MCP Server**	Self
EODHD (EOD Historical Data)	Free; $19.99/mo EOD; $29.99/mo EOD+Intraday; $59.99/mo Fundamentals; €79.99+/mo commercial	Global EOD + intraday + fundamentals	**MCP Server with 72 tools; OpenAPI 3.1 spec with 74 endpoints; 28 pre-built AI agent skills; ChatGPT Assistant**	Self
Refinitiv / LSEG	$20K+/yr Enterprise; LSEG Workspace quote-based	Global, institutional-grade; 40,000+ customers / 400,000+ end users	**LSEG Workspace AI "Deep Research" agent (2025–26); AI Explainability docs; Microsoft strategic AI/data-cloud partnership**	Not buil reta
Bloomberg	$24K–$28K+/yr/terminal (CONFIRMED industry-standard estimate; Bloomberg doesn't publish)	Global, full suite; ~350,000 Terminal subs	**BloombergGPT** (50B params, 363B Bloomberg + 345B general tokens, arXiv:2303.17564); Terminal AI Insights / AI News / AI Chart / **BQuant Agent**	Not buil reta

Default for Macion Capital Y0–Y1: Databento ($125 free + pay-as-you-go) for crypto + futures; IBKR API for live execution; FRED for macro. Intrinio AI-Ready tier ($150/mo) is the smartest retail upgrade path when MCP/Claude integration matters.

E.1.2 NLP / document extraction

Vendor	Pricing (2026)	Capabilities	Build alternative
AlphaSense	Quote-based Enterprise (~$10K–$25K+/seat/yr historical reference); Market Intelligence + Enterprise Intelligence tiers	500M+ premium documents; Generative Search (sentence-level citations, multi-agent); Deep Research (frontier reasoning); Smart Synonyms; Sentiment Indices. **7,000+ enterprise customers; 85%+ of S&P 100; 70% of top 50 hedge funds (CONFIRMED**	FinGPT + Llama-3 fine-tune

Vendor	Pricing (2026)	Capabilities	Build alternative
		2026-07-07 via alpha-sense.com)	
Hebbia	Six-figure min	Matrix UI for LLM on doc sets	Custom RAG pipeline
RavenPack	$20K+/yr (quote-based)	80+ NLP fields incl. 20+ sentiment indicators; **Bigdata.com = "AI agents for finance" platform with FT content integration (CONFIRMED 2026-07-07)**. Scale: 12M entities / 7.1M companies / 4.5M individuals / 60K+ products / 13 languages / 7K+ event categories / 20+ yr archive	FinBERT legacy
FinChat / ChatIQ	$30/mo consumer; $500/mo pro	Finance LLM with citations	Custom RAG
In-house (FinGPT + Llama-3)	$200/mo API + 1 FTE	Domain-specific	—

Default for Macion Capital: in-house FinGPT for general extraction; FinChat for finance-specific queries. Skip AlphaSense until AUM > $50M.

E.1.3 Backtesting

Vendor	Pricing (2026)	Engine	AI features
QuantConnect	Free–$100/mo	Lean (C#)	AI code completion
SigTech	$20K+/yr	Proprietary event-driven	ML signals
WorldQuant BRAIN	Free + paid comps	Expression language	AI alpha suggestions
OpenBB Platform	Free; $50/mo Pro	Python wrapper	AI co-pilot (2026)
Lean (self-hosted)	Free	Same as QuantConnect	Same

Default for Macion Capital: Lean self-hosted (free) + Vectorbt PRO ($200/mo) + custom Python.

E.1.4 ML platforms

Vendor	Pricing (2026)	Use case
Weights & Biases	Free; $50/mo/seat Pro	Experiment tracking
MLflow	Free self-hosted	Experiment tracking + model registry
DVC	Free + cloud	Data versioning
Feast	Free self-hosted	Feature store
Tecton	Enterprise pricing	Managed feature store
BentoML	Free self-hosted	Model serving
Ray Serve	Free (Anyscale $50/mo Pro)	Distributed serving
Modal	Free tier; $30/mo Pro	Serverless GPU
Lightning AI	Free tier; $30/mo Pro	Research-to-prod

Default for Macion Capital: MLflow self-hosted + DVC + BentoML + Modal (for GPU bursts).

E.1.5 Vector databases

Vendor	Pricing (2026)	Use case
Chroma	Free self-hosted	Local dev
Qdrant	Free self-hosted; $25/mo Cloud	Production
Pinecone	Free tier; $70/mo Standard	Managed
Weaviate	Free self-hosted; $25/mo Cloud	Hybrid search
Milvus	Free self-hosted; Zilliz $65/mo	Billion-scale

Default for Macion Capital: Chroma (local) + Qdrant (production).

E.1.6 Code / agent assistants

Vendor	Pricing (2026)	Build alternative
Claude Code	$20–$200/mo	—
Cursor	$20/mo	Continue (free, open source)
GitHub Copilot	$10/mo	Codeium (free)
Cody	$9/mo	—
Aider	Free	—

Default for Macion Capital: Claude Code ($100/mo) + Cursor ($20/mo).

E.1.7 Compliance + risk

Vendor	Pricing (2026)	Use case
ComplyAdvantage	$5K+/yr	AML/KYC
Lucinity	$10K+/yr	AML/KYC
Solidus Labs	Enterprise	Crypto-native AML
Two Sigma Venn	Enterprise	Risk analytics
MSCI Barra	$25K+/yr	Risk analytics
Aladdin (BlackRock)	$50K+/yr	Full ODD

Default for Macion Capital Y0–Y1: DIY compliance with LLM help; upgrade at AUM > $100M.

E.1.8 LP-facing reporting

Vendor	Pricing (2026)	Use case
Addepar	$50K+/yr	Wealth reporting
Orion	$25K+/yr	Wealth reporting
In-house + Claude	$200/mo API + 1 FTE	Custom

Default for Macion Capital: in-house + Claude API + python-pptx for PDF reports.

E.2 — The build-vs-buy decision rule

For every component, ask three questions:

1. **What's the vendor moat?** High = buy; low = build.
2. **What's the regulatory exposure?** High = buy (a vendor's compliance is their product); low = build.
3. **What's the IP sensitivity?** Low = buy; high = build (or buy infrastructure only).

Apply this rule to every line item in the matrix.

E.3 — The "build" cost estimates (rough, 2026)

For components where build makes sense, here's a rough engineering-hour estimate:

Component	Engineering hours	FTE-years	Notes
Custom backtest engine	~5,000	2.5	Lean / Zipline already exist — only build if you need something custom
Alt-data extraction pipeline	~500	0.25	Per data source
LLM-based research agent	~200	0.1	Mostly prompts + retrieval
Vector DB + RAG pipeline	~300	0.15	Chroma + LangChain
MLOps platform	~1,000	0.5	MLflow + DVC + BentoML
Compliance automation	~800	0.4	Mostly workflow + audit logs
LP reporting dashboard	~1,200	0.6	Custom visualization
Order management integration	~1,500	0.75	Per broker
Risk engine	~2,000	1.0	Don't DIY; buy MSCI Barra or DIY with open source
Eval harness (G1–G31)	~500	0.25	Already implemented in Macion Capital's research platform (workspace-local)

At $150K/year for a mid-level engineer, the **build cost** for the full stack is roughly **$1.5M in engineering** — assuming you can hire 2 FTEs for 2 years. That's higher than the buy cost (~$300K/year), but you own the IP.

The hybrid is optimal: buy commodity infrastructure, build the differentiating layers (eval harness, alpha logic, ops workflow).

See `10_cost_model_phased_roadmap.md` for the cost scenarios.

Chapter E2. Cost Model & 4-Phase Roadmap for Macion Capital

BOTTOM LINE Talent is ~67% of cost at every stage — the AI tooling is leverage, not cost, and a 4-phase roadmap delivers Macion Capital's research platform first (gate stack + 15 agents + 31 skills + tool layer), then the agentic layer, then the data layer, then governance.

The operational playbook. What does it cost to build an AI-augmented quant fund at three stages (seed, Series-A, mid-tier), and what's the 4-phase implementation roadmap tailored to Macion Capital?

E.1 — Three cost scenarios (2026)

The line-item annual cost for an AI-augmented quant fund at three stages. All numbers in USD.

E.1.1 Scenario A: Seed stage ($5–25M AUM, 1–2 FTE)

Line item	Cost ($/yr)	Notes
Data		
Databento (pay-as-you-go)	500	$125 free credit + ~$500 top-up
Polygon.io Starter	360	$30/mo
Alpha Vantage Pro	600	$50/mo
FRED	0	Free
Subtotal data	1,460	
Compute / infra		
AWS / GCP / CoreWeave	6,000	1× reserved 4-vCPU + 16GB RAM + 1 GPU
Datadog (or Grafana Cloud)	1,200	$100/mo for observability
Storage (S3)	600	~50 GB research data
Subtotal infra	7,800	
Talent		
Principal (sweat equity)	0	—
1 junior quant / engineer	80,000	PH-based, ~$80K all-in
Subtotal talent	80,000	
Vendors / SaaS		
QuantConnect Pro	240	$20/mo
Claude Code / Cursor	1,200	$100/mo combined
Notion / Obsidian	120	$10/mo
GitHub Team	240	$20/mo
Vector DB (Qdrant Cloud)	300	$25/mo
Subtotal vendors	2,100	
Ops / compliance / legal		
PH securities counsel (advisory)	5,000	Quarterly retainer
Auditor (pre-launch audit)	3,000	—
Admin (none — too small)	0	—
Subtotal ops	8,000	
TOTAL	**99,360**	~$100K/year

The honest framing: at seed stage, **talent + ops dominates** (~$88K of $100K). The AI tooling is the cheap part. The leverage is high because the principal's time is the binding constraint.

E.1.2 Scenario B: Series-A stage ($25–100M AUM, 3–6 FTE)

Line item	Cost ($/yr)	Notes
Data		
Databento Plus/Standard contract	50,000	Institutional-grade contract
Polygon.io Enterprise	36,000	$3K/mo
AlphaSense (1 seat)	12,000	~$1K/mo
FRED + Nasdaq Premium	2,000	—
Subtotal data	100,000	
Compute / infra		
AWS / GCP production cluster	60,000	Multi-GPU for ML training
Datadog / Grafana Cloud	12,000	—
Storage (S3 + glacier)	6,000	Larger research corpus
MLflow + BentoML (managed)	6,000	—
Subtotal infra	84,000	
Talent		
Principal	0	—
1 senior PM	250,000	—
2 mid quants / engineers	300,000	$150K each
1 ops / compliance	120,000	—
Subtotal talent	670,000	
Vendors / SaaS		
QuantConnect Team	1,200	—
Claude Code + Cursor + AlphaSense	5,000	—
Pinecone / Qdrant Cloud	3,600	—
Weights & Biases Teams	2,400	—
Subtotal vendors	12,200	
Ops / compliance / legal		
Counsel (engaged counsel-of-record)	25,000	Retainer
Auditor (Big-4 PH affiliate)	30,000	—
Admin (outsourced fund admin)	30,000	—
Custody (Tier 1 PH trust bank)	20,000	—
Insurance (D&O, E&O, cyber)	15,000	—
Subtotal ops	120,000	
TOTAL	**986,200**	**~$1M/year**

The framing: at Series-A, **talent is 68%** of total cost. AI infra is small but matters for the leverage it provides.

E.1.3 Scenario C: Mid-tier ($100M–1B AUM, 10–25 FTE)

Line item	Cost ($/yr)	Notes
Data		
Databento Unlimited contract	250,000	Full history + L3
Polygon.io Enterprise + L2	100,000	—
AlphaSense Enterprise (5 seats)	75,000	—
Bloomberg (limited seats)	60,000	3 seats at $20K each
RavenPack / similar	40,000	—
Subtotal data	525,000	
Compute / infra		
AWS / GCP production cluster	600,000	Multi-region, GPU-heavy
Datadog + Sentry	60,000	—
Storage (S3 + lakehouse)	80,000	—
Managed MLflow / BentoML	30,000	—
Subtotal infra	770,000	
Talent		
Principal	0	—
1 CIO	500,000	—
3 senior PMs	1,200,000	$400K each
6 mid quants	1,200,000	$200K each
4 engineers	800,000	$200K each
2 ops / compliance	300,000	$150K each
2 research associates	240,000	$120K each
Subtotal talent	4,240,000	
Vendors / SaaS		
Enterprise stack (Sigma, Bloomberg, etc.)	100,000	—
Claude Code Enterprise	24,000	$2K/mo
Pinecone Enterprise	12,000	—
Vector DB managed	24,000	—
Subtotal vendors	160,000	
Ops / compliance / legal		
Counsel (Big-4 affiliated)	100,000	—
Auditor (Big-4)	75,000	—
Admin (outsourced)	75,000	—
Custody (tier 1)	100,000	—
Insurance (full suite)	100,000	—
ODD / risk (in-house + vendor)	200,000	—
Subtotal ops	650,000	
TOTAL	**6,345,000**	**~$6.3M/year**

The framing: at mid-tier, talent is **67%** of cost. The AI/ML infra is **12%**. The leverage from good AI infra is the 10× productivity multiplier that makes the talent count feasible.

See `07_cost_curve.png`.

E.2 — The 4-phase roadmap for Macion Capital

A 4-phase implementation plan, tailored to Macion Capital's pre-launch state. The starting substrate is the principal's own research platform — 15 specialised agents, 31 skills, the G1–G28 + G29–G31 gate stack, the five-tool cron layer, and the banked `data/` cache — and each phase adds a layer on top. The platform is documented workspace-side in `CLAUDE.md` §2 + §9; this roadmap is the operational follow-through.

See `08_phased_roadmap.png`.

Phase 1: Activate the Macion Capital research platform (Months 0–3, Y0)

Goal: Stand up the principal's research platform on the Macion Capital data layer, with the gate stack + tool layer already in production at the workspace root. The "platform is the substrate; the data is the differentiator" pattern means Y0 work is mostly about wiring, observability, and daily-routine hygiene, not new engineering.

Activities:

1. **Verify the platform end-to-end.** The 15 agents + 31 skills at `.claude/agents/` + `.claude/skills/` are clean-room-forked to the Macion Capital workspace at `/Users/christianmacion/Macion Capital/` per `CLAUDE.md` §1. Activity 1 confirms each agent can complete its handshake (read CLAUDE.md → read its charter → read DATA_REGISTRY → record a no-op run) on the Macion Capital `data/` cache without external dependencies.
2. **Run the gate stack on a known strategy to validate.** Pick one archived strategy (e.g. the BTC 22:00 close-session candidate from `agent_outputs/research/`). Walk it through G1–G28 + G29–G31 end-to-end; confirm the gate verdicts reproduce; confirm the strategy-scribe records the verdict in `STRATEGY_REGISTRY.md`.
3. **Set up local observability.** Grafana + Loki for agent session logs is optional until a live venue is wired up; until then, the platform's own session-ledger files at `agent_outputs/` are the audit trail. Phase 1 keeps Grafana+Loki parked.
4. **Activate the cron layer.** Wire `sign_flip_monitor.py` (daily 22:00 UTC; emits `daily_reports/YYYY-MM-DD_sign_flip_monitor.md`); `daily_report_runner.py` (daily 06:30 UTC; populates `daily_reports/YYYY-MM-DD_daily_summary.md`); `registry_index.py` (Sunday 03:00 UTC; rebuilds the strategy-registry index). Sources of truth: `CLAUDE.md` §9.1 + the `scripts/` directory.
5. **Commit `scripts/` on every green-path** (per `CLAUDE.md` §9.4) so the audit trail reads "we ran X and got Y" without ambiguity. The principal-owned git init one-shot is `tools/init_git.sh --apply`.

Deliverables: - 15-agent + 31-skill research platform operational on the Macion Capital data layer (i.e. one clean daily run + one clean sign-flip pass + one clean Sunday registry-index rebuild) - 5 sanity-check research memos produced end-to-end (1 per value-chain stage: research → backtest → dev → production → reporting) - Sign-flip monitor emitting daily reports; CRITICAL flags acknowledged within 24h - Documentation: `CLAUDE.md` updated for own-fund posture; `OPERATING_MODEL.md` aligned to the 8-binding-rules + 5-cron + 5-tool contract - `daily_reports/` populated for every working day of Phase 1

Cost: ~$200/mo (compute + SaaS only; no new FTEs).

Phase 2: Add agentic AI layer (Months 3–9, Y0–Y1)

Goal: Add an LLM-copilot + agent harness on top of the inherited stack.

Activities:

1. Set up Claude Code (or Cursor + Claude) as the daily dev companion.
2. Build `quant-llm-copilot` skill — retrieval over kill log + memos + gate rulings.
3. Build `agent-system-auditor` skill — periodic self-audit of agent behavior vs. MAST failure taxonomy.
4. Add `quant-executor` agent design (paper-trading-ready; deferred until live venue).
5. Add a paper-trading harness (test against broker sandbox).

Deliverables: - LLM-copilot skill in production use (weekly) - Agent-system-auditor producing monthly reports - Paper-trading harness validated end-to-end - Documentation: new skills registered; SSOT updated

Cost: ~$500/mo (API + cloud).

Phase 3: Add data + alt-data (Months 9–18, Y1)

Goal: Build out the data layer beyond what's in the banked Macion Capital `data/` cache.

Activities:

1. Pull Databento data for target universes (CME futures, ES, MBT, MET).
2. Add AlphaSense or in-house FinGPT for filings/transcripts.
3. Build a vector-DB-backed research corpus (Qdrant + Claude/GPT-4 embeddings).
4. Add 1 junior FTE (PH-based, ~$80K/year all-in) for data engineering.
5. Begin live paper-trading on 1–2 strategies.

Deliverables: - Databento data live for 3+ universes - AlphaSense or in-house NLP pipeline operational - 1–2 strategies in continuous paper-trading - Documentation: DATA_REGISTRY.md updated

Cost: ~$100K/year (data + 1 FTE + vendors).

Phase 4: LLMOps + governance (Months 18–36, Y1–Y2)

Goal: Production-grade LLMOps, formal MRM, LP-facing reporting.

Activities:

1. Implement the full MLFinLab / MLflow + DVC stack for model versioning.
2. Build the model-risk-management documentation suite (model inventory, validation reports, monitoring).
3. Build the LP-facing reporting agent (monthly letter generator).
4. Engage counsel-of-record + auditor (if AUM > $25M).
5. Add 1 mid-level engineer for platform work.
6. Begin live trading on 1+ validated strategy.

Deliverables: - MLflow + DVC operational - MRM documentation complete for all live models - Monthly LP letter generator in production - Counsel-of-record + auditor engaged - Live trading on 1+ strategy

Cost: ~$300K/year (1 FTE + counsel + auditor + data).

E.3 — Cost-stance discipline

For Macion Capital, the binding constraints are: - **Talent is the dominant cost** (67% in seed stage, similar in all stages) - **AI tooling is leverage, not cost** — a $1K/mo Claude bill that produces 100× productivity is good ROI - **Vendor lock-in is the hidden cost** — every contract should have an exit clause - **Build vs. buy favors hybrid** — buy infrastructure, build differentiating layers

E.3.1 The "AI is leverage, not cost" frame

A junior engineer with Claude Code produces ~3× the output of one without. For a 2-FTE team, that's the equivalent of 6 FTEs — at the cost of 2 FTEs + $1K/mo of AI tooling. **The AI tooling pays for itself within the first week.**

E.3.2 The vendor lock-in audit

Every year, audit the vendor stack: - Are we paying for capability we don't use? - Is there an open-source alternative that's production-ready? - Is the vendor's pricing increasing faster than our usage? - Do we have an exit clause / data export path?

A 2024 example: many funds were locked into expensive OMS contracts; after 2024, several open-source OMS (e.g., LEAN OMS, FlexTrade alternative) became production-ready. The exit was free; the savings were 6 figures.

E.3.3 The "build only if it's your edge" rule

- **Build**: alpha logic, gate stack, eval harness, ops workflow — these are your edge
- **Buy**: data, compliance, risk, OMS, code assistant — these are commodity
- **Hybrid**: backtest engine, ML platform, vector DB — infra can be open source; the workflow is yours

See `09_build_vs_buy_matrix.md` for the full decision matrix.

Part F. Risk, Governance & Frontier

What this part answers. What can go wrong, what regulators are saying, and where the field is going in 2026–2028. The risk layer is where most AI-in-finance write-ups are weakest — this is the blow-up catalog.

Chapter F1. Risk, Governance & Frontier

BOTTOM LINE The ML risk heatmap has 10 named risks with known defenses — model collapse, reward hacking, herding, and prompt injection are the load-bearing ones; the regulatory timeline (EU AI Act 2025, SEC final 2026 H1) sets the binding constraints.

> **What can go wrong, what regulators are saying, and where the field is going.** The risk layer is where most AI-in-finance write-ups are weakest — they're all "look at this cool thing" and zero "look at how this blows up." This part is the blow-up catalog.

The two-stage architecture (canonical, used in B1, E.2, closing):

```
AI agents (read-only)        ← propose / factor-mine / write code
       │
       ▼
Gate stack (G1–G28 + G29–G31) ← numeric anchors: DSR, PBO, min-BTL
       │
       ▼
Decision (KILL / ITERATE / PROMOTE)
```

The LLM is **never the alpha**. The gate stack is the judge. This is the standing rule.

F.1 — The ML risk heatmap

The single most important risk-management view for an AI-augmented quant fund. See `09_risk_heatmap.png`.

F.1.1 The 10 risks, by frequency × severity

Risk	Frequency	Severity	Notes
Model collapse / regime shift	Medium	High	The most common "AI worked in backtest, failed in live" failure
Reward hacking (RL agent)	Low	High	When reward function is misspecified
LLM hallucination in research	High	Medium	Mitigated by numeric gate stack
Data poisoning	Very low	Very high	Adversarial input attacks
Herding / AI–AI feedback	Medium	High	If many funds use same vendor
Prompt injection in trading	Low	Medium	LLM agent action injection
Overfit to vendor signals	High	Medium	The FINSABER null
Concentration on one vendor	Medium	Medium	Vendor outage / pivot
Compliance / marketing claim	Medium	High	"AI-driven" claims attract SEC
Cost blowout (token spend)	High	Low	Surprises more than kills

F.1.2 The defenses

For each risk, there's a known defense:

Risk	Defense
Model collapse	Walk-forward + regime decomposition + multiple-testing defenses
Reward hacking	Multi-objective reward + pre-registered risk gates
LLM hallucination	Numeric gate stack + read-only adversary
Data poisoning	Adversarial validation + outlier detection
Herding	Vendor diversification + anti-correlation checks
Prompt injection	Input validation + structured-output contracts
Overfit	Holdout discipline + de-selection re-tests
Vendor concentration	Multi-vendor strategy + open-source fallbacks
Compliance	No "AI-driven" claims; let the gate stack speak
Cost blowout	Per-task token budgets + model tiering

F.2 — The regulatory landscape (2025–2026)

F.2.1 SEC (US)

- **Marketing Rule (17 CFR 275.206(4)-1)** — applies to AI-driven performance claims; backtested-only funds face stricter pre-conditions. Captures AI-generated marketing as adviser advertising (chatbots/virtual agents, AI-targeted social media, AI-extracted performance figures must meet net-of-fees and fair-representation obligations). **Effective 2021-05-04**; final adopted 2019-12-19 (Release IA-5653). **[CONFIRMED 2026-07-07]** — `https://www.sec.gov/rules/final/2020/ia-5653.pdf`
- **SEC Predictive AI / Conflicts-of-Interest Proposal (Release Nos. 33-11210; 34-97989; 34-97990; IA-6176; File No. S7-12-23)** — published 2023-07-31; would require broker-dealers and investment advisers to identify and eliminate/neutralize conflicts of interest from "covered technology" (predictive data analytics, AI, ML, algorithms, optimization engines) used in investor interactions
- **Disposition as of 2026-07-07: STILL PROPOSED — not finalized.** Under Chair Atkins (since April 2025) the proposal has remained unenacted; no re-proposal, no withdrawal publication; effectively dormant. SEC has shifted to lighter-touch enforcement using existing fiduciary, Reg BI, Marketing Rule, and books-and-records authority. **[CONFIRMED 2026-07-07]** — `https://www.sec.gov/rules/proposed/2023/ia-6176.pdf` and `https://www.sec.gov/rules/proposed/2023/34-97990.pdf`
- **SEC Staff Statement on "AI Washing" (2024)** — IA Risk Alert addressing firms making misleading AI claims; signals the SEC's de facto posture: use existing anti-fraud authority, don't wait for new prescriptive rules
- **Reg S-P / Reg S-ID** — privacy + identity-theft implications for AI-driven KYC/AML

F.2.2 CFTC (US)

- **No standalone AI-specific rule from the CFTC as of 2026-07-07.** Existing authorities under CEA §4b (manipulation) and CFTC Reg 180.1 apply to AI/algorithmic conduct. **[CONFIRMED 2026-07-07]**
- **Reg AT** (proposed) — algorithmic trading recordkeeping; AI-specific implications deferred; no finalized action
- **CFTC Division of Enforcement advisories** (2023–24) address insider-trading/AI-driven information advantage and market-manipulation risk from algorithmic/automated trading
- **CFTC Innovation Hub** (`cftc.gov/About/Innovation`) — industry engagement without binding rulemaking

F.2.3 FINRA

- **Regulatory Notice 24-09** (June 27, 2024) — the load-bearing FINRA AI document for 2024–26; explicitly reminds members that FINRA rules are **technology-neutral** and apply equally to AI/LLM-using firms. Covers both proprietary and third-party AI tools; encourages pre-deployment evaluation; cites Member Obligation obligations
- **The four named obligation areas** (per Notice 24-09, primary text CONFIRMED 2026-07-07): 1. **Technology governance** — establish a framework for AI/ML use, including approval, change management, and version control 2. **Model risk management** — supervised validation, performance monitoring, and ongoing review analogous to SR 11-7 for traditional models 3. **Data privacy and integrity** — ensure training and inference data are accurate, complete, and properly sourced 4. **Reliability and accuracy** — implement controls to validate outputs before they reach customers or the market
- **Anchor rules cited in Notice 24-09:**

- **FINRA Rule 2210 (Communications with the Public)** — applies to AI-generated or AI-assisted client communications
- **FINRA Rule 3110 (Supervision)** — applies to AI systems used in supervisory or operational roles
- **FINRA Rule 4511 (General Books and Records)** — applies to AI-generated records
- **Prior FINRA AI publication:** June 2020 report *"Artificial Intelligence (AI) in the Securities Industry"* — `https://www.finra.org/sites/default/files/2020-06/ai-in-the-securities-industry.pdf` **[CONFIRMED 2026-07-07]**
- **FINRA AI topic index:** `https://www.finra.org/rules-guidance/notices/by-topic?term=167031`

Sources (CONFIRMED): `https://www.finra.org/rules-guidance/notices/24-09` (HTML); PDF at `https://www.finra.org/sites/default/files/2024-06/regulatory-notice-24-09.pdf` (PDF metadata: created 2024-06-27; references Phil Shaikun / Meredith Cordisco; cites NIST.AI.100-1), retrieved 2026-07-07.

F.2.4 EU AI Act (Regulation EU 2024/1689)

- **Primary source (CONFIRMED 2026-07-07):** `https://eur-lex.europa.eu/eli/reg/2024/1689/oj` (OJ L 2024/1689, 12 July 2024)
- **Risk tiers:** prohibited practices (Article 5); high-risk systems (Annex III + Annex I product-safety); limited-risk transparency obligations (Article 50 — chatbot/GenAI/deepfake disclosure); GPAI (general-purpose AI) models with systemic-risk obligations (Articles 51–55)
- **Phased effective dates (CONFIRMED 2026-07-07 against `digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai`):**
- **Entered into force:** 2024-08-01
- **Prohibitions + AI-literacy obligations:** 2025-02-02
- **GPAI obligations:** 2025-08-02
- **Most high-risk Annex III obligations: 2027-12-02** *(post-Digital Omnibus political agreement 2026-05-07; was 2026-08-02 before the omnibus)*
- **High-risk systems embedded in regulated products (Annex I): 2028-08-02** *(post-Digital Omnibus; was 2027-08-02)*
- **Digital Omnibus (2026-05-07):** EU political agreement to simplify the AI Act — pushed Annex III + Annex I dates by ~16 months; extended prohibitions to non-consensual sexually explicit / intimate content + CSAM; expanded regulatory sandboxes (incl. EU-level); simplified requirements for small mid-caps. Primary sources: `digital-strategy.ec.europa.eu/en/news/eu-agrees-simplify-ai-rules-boost-innovation-and-ban-nudification-apps-protect-citizens`.
- **Implications for hedge funds (CRITICAL):**
- **Credit-scoring/creditworthiness AI** is explicitly classified as **high-risk** (Annex III §5(b)) — applies to any fund / fund-servicing entity extending consumer credit, **but NOT** to pure investment-management trading algorithms
- **Pure securities-trading AI is NOT classified as high-risk** under the Act; covered by existing market manipulation rules + MiFID II
- **LLMs** → limited risk under Article 50; transparency obligations (chatbot/GenAI disclosure)
- **Compliance/AML/credit-decision systems** within fund operations may trigger Annex III high-risk classification
- **High-risk obligations** (when triggered): data governance, technical documentation, human oversight, accuracy/robustness/cybersecurity (Articles 8–15), transparency (Article 13)
- **EU AI Office:** `https://digital-strategy.ec.europa.eu/en/policies/ai-office`

F.2.5 UK FCA / PRA

- **FCA FS24/2 — "AI Update" (April 2024)** — joint FCA/PRA feedback statement on AI/ML deployment in UK financial services; three FCA priorities: (1) safe AI adoption, (2) supporting market growth, (3) cross-sector coordination. **[CONFIRMED 2026-07-07]** — `https://www.fca.org.uk/publications/feedback-statements/fs24-2`
- **Note:** the previously-cited "PRA SS1/23" was a mislabel — SS1/23 is a PRA-only model-risk-management paper unrelated to AI. The canonical FCA/PRA AI publication is **FS24/2** above
- **FCA AI Lab** (launched 2024, expanded 2025–26) — industry sandboxes (AI Spotlight), services-related tools, regtech guidance; 2025–26 AI Innovation Showcase for AI-based regulatory compliance tooling
- **FCA supervisory posture:** no binding AI rule as of 2026-07-07; relies on existing SMCR (Senior Managers & Certification Regime), SYSC (operational risk), and COCON (Conduct Rules) to regulate AI use

F.2.6 Singapore MAS

- **FEAT principles (Fairness, Ethics, Accountability, Transparency)** — voluntary but expected
- **MAS AI Verify** — testing framework for AI fairness

Source: mas.gov.sg/regulation, retrieved 2026-07-07.

F.2.7 Hong Kong SFC

- **AI guidance (2024)** — risk-based, similar to FCA
- **Circular on AI trading** — 2024, focuses on governance

F.2.8 US Banking — the April 2026 MRM refresh (FIRST AI-explicit update since 2011)

- **OCC Bulletin 2026-13 + Federal Reserve SR 26-02** (April 17, 2026) — the interagency revised Model Risk Management guidance; **the first MRM refresh explicitly addressing AI/ML risks since 2011**. Rescinds OCC 2011-12. **[CONFIRMED 2026-07-07]**
- OCC Bulletin 2026-13: `https://www.occ.gov/news-issuances/bulletins/2026/bulletin-2026-13.html`
- Fed SR 26-02: `https://www.federalreserve.gov/supervisionreg/srletters/SR2602.`
- **Why this matters for Macion Capital:** even non-bank funds should adopt the 2026 MRM refresh provisions — they will be the binding floor for any future bank-dealer counterparty relationship, ODD vendor questionnaire, and EU-LP readiness assessment

F.2.9 Basel BCBS

- **AI guidance forthcoming (2026)** — expected to align with EU AI Act and the OCC/Fed MRM refresh

F.2.10 The Philippines (for Macion Capital)

- **SEC MC 7-2007** — investment adviser registration
- **SEC MC 33** — securities lending
- **RA 9160 (Anti-Money Laundering)** — applies to AI-driven KYC
- **BIR** — tax implications of AI vendor spend
- **NPC (National Privacy Commission)** — Data Privacy Act 2012 implications for alt-data

Source: sec.gov.ph, retrieved 2026-07-07.

See `10_compliance_timeline.png`.

F.3 — Notable AI-in-finance incidents (case studies)

F.3.1 Knight Capital (2012) — the pre-AI cautionary tale

- **What happened**: Knight Capital deployed a faulty algo in production; lost $440M in 45 minutes
- **Cause**: code deployment error; lack of kill switch
- **AI connection**: not AI-specific, but the lesson applies — always have a kill switch, always test in staging

F.3.2 Flash Crash 2010

- **What happened**: ~$1T erased from US equities in 36 minutes
- **Cause**: HFT + sell algorithms + cross-market feedback loops
- **AI connection**: anticipatory — modern AI agents could amplify this further

F.3.3 LTCM 1998

- **What happened**: Nobel-laureate hedge fund blew up on convergence trades
- **Cause**: model overconfidence + leverage + liquidity assumption failure
- **AI connection**: modern AI could replicate this with a model that confidently predicts an unsustainable spread

F.3.4 Archegos 2021

- **What happened**: Bill Hwang's family office imploded; $20B+ losses
- **Cause**: concentrated positions + hidden leverage + no risk management
- **AI connection**: AI-driven risk management would have flagged the concentration

F.3.5 Aidyia 2018 — AI fund blowup

- **What happened**: Hong Kong-based AI hedge fund closed after ~2 years
- **Cause**: unclear publicly; the AI didn't find enough alpha to cover costs
- **AI connection**: the cautionary tale for AI-first funds — AI doesn't guarantee alpha

F.3.6 FINSABER null result (2024)

- **What happened**: Stanford benchmarked LLM-based trading agents on long-run outperformance vs. buy-and-hold; **no significant alpha (p > 0.34; *N* = 6 LLMs across 2020–2023 historical windows)** — the canonical "do these agents actually beat the index?" benchmark
- **Cause**: LLM-based traders exhibit ~37% / ~19% memorization
- **AI connection**: the empirical proof that LLMs-as-traders are placebo-bound

F.3.7 FTX 2022 — not AI but instructive

- **What happened**: Sam Bankman-Fried's exchange imploded; ~$8B customer funds missing
- **Cause**: fraud + hidden leverage + no audit
- **AI connection**: an AI-driven compliance agent could have flagged the red flags; an AI-driven risk manager would have refused the leverage

---

F.4 — The compliance playbook for Macion Capital

A practical compliance posture for an AI-augmented fund at each stage.

F.4.1 Pre-launch (Y0)

- No public AI claims. Marketing uses "quantitative" + "systematic" — accurate.
- Internal AI use is documented; agent charters are versioned.
- Data Privacy Act compliance for any PH-resident data.
- Counsel-of-record engaged at the right time (per the existing legal dossier).

F.4.2 Seed stage (Y1, AUM < $25M)

- No public performance claims beyond GIPS-compliant composite.
- AI use documented in DDQ appendix.
- Basic KYC/AML with manual review + LLM-assisted.
- Quarterly counsel review.

F.4.3 Series-A stage (Y1–Y2, AUM $25–100M)

- Full GIPS composite presentation.
- AI use documented; specific techniques disclosed in DDQ.
- Formal MRM documentation (model inventory + validation reports).
- Annual counsel review.

F.4.4 Mid-tier (Y2+, AUM > $100M)

- Full MRM framework with model-risk-management committee.
- AI use disclosed in marketing materials with specific framework references.
- External auditor reviews MRM annually.
- Quarterly counsel review.

F.5 — The frontier (2026–2028)

F.5.1 Where the genuine frontier is

1. **Causal discovery** — moving from correlation to causation; double-ML (DML, arXiv:1707.01847), causal forests (arXiv:1510.04342), instrumental variables, synthetic controls; Python package DoubleML at `docs.doubleml.org`
2. **World models + synthetic data** — for stress testing, NOT for alpha training; DreamerV3 (arXiv:2301.04104), FinRL-Meta (arXiv:2211.03107)
3. **Multi-agent research systems** — the principal's 15-agent research platform is one of the early production instances; the academic literature is more mature. See FinCon (arXiv:2407.06567), FinMem (arXiv:2311.13743), MAST failure taxonomy (arXiv:2503.13657; NeurIPS 2025; **41–86.7% failure rate across 7 popular agentic frameworks and 1500+ test cases** — the highest-fidelity skeptic)
4. **AI safety applied to capital markets** — herding detection, anti-feedback-loop
5. **Federated learning across funds** — IP-preserving collaborative learning (theoretically possible; regulatory hard)
6. **Quantum-ML for portfolio optimization** — still research-only; no production win yet
7. **Constitutional AI for finance** — domain-specific constitutions for trading agents
8. **Conformal prediction for trading** — distribution-free prediction intervals for risk-managed position sizing; Gibbs/Candès (arXiv:2208.08401, JASA 2024); Kato CPPS (arXiv:2410.16333)
9. **Neural-SDEs + signature methods** — calibration-free option pricing, deep hedging, rough volatility; Cohen et al. (arXiv:2105.11053); SigFormer (arXiv:2310.13369, ICAIF 2023)
10. **Time-series foundation models** — zero-shot forecasting; PatchTST (arXiv:2211.14730, ICLR 2023), iTransformer (arXiv:2310.06625, ICLR 2024), Chronos (arXiv:2403.07815), TimesFM (arXiv:2310.10688), TimeGPT (arXiv:2310.03589), MOMENT (arXiv:2402.03885), Moirai-MoE (arXiv:2410.10469) — **Moirai-MoE in particular uses sparse Mixture-of-Experts routing inside a time-series Transformer, beating Moirai/TimesFM on 39 datasets** [STATED, arxiv.org/abs/2410.10469]
11. **State-space models (Mamba/S4) for finance** — long-context efficiency for high-frequency streams; 2024-2026 paper stream continues; **[NEEDS VERIFY]** for any canonical Mamba-finance paper
12. **Diffusion models for portfolio generation** — denoise-return-path sampling as a novel mechanism for portfolio weights vs mean-variance; 2025-2026 paper stream; **[NEEDS VERIFY]** for the canonical arXiv ID
13. **Process Reward Models (PRMs) for strategy evaluation** — step-level reward scoring of strategy reasoning chains; analog to SR 11-7 validation for LLM chains-of-thought; 2025-2026 paper stream; **[NEEDS VERIFY]** for the canonical arXiv ID
14. **Neuro-symbolic AI for finance** — IBM WatsonX.ai hybrid (neural + symbolic); compliance-traceability argument directly relevant for MRM and audit trail; no production quant-fund win yet [INFERRED]
15. **Self-play RL for market-making** — execution alpha (crossing the bid-ask), not directional alpha; 2025-2026 paper stream from Optiver / JPMorgan / Hudson River Trading [INFERRED; **[NEEDS VERIFY]** for canonical arXiv ID]
16. **FPGA real-time ML inference** — sub-microsecond inference for ultra-low-latency strategies; Optiver / Exablaze / AMD-Xilinx papers [INFERRED]; marginal relevance to small fund
17. **LLM-as-judge for MRM (the biggest 2026 compliance gap)** — using reasoning models to score model-validation evidence under SR 11-7 / OCC 2026-13 / Fed SR 26-02. The April 17 2026 OCC Bulletin 2026-13 refresh (the FIRST AI-explicit MRM guidance since SR 11-7 2011) explicitly contemplates AI-assisted validation. **This is the single largest fertile frontier item.** See F.2.8 for the OCC bulletin.

F.5.2 What's hype, what's real

Hype	Real
"AGI traders will replace quants"	LLMs are bad traders (FINSABER)
"AI funds will dominate"	AIEQ underperformed S&P 500 by 2–4%/yr over 8.5 years
"Foundation models will predict markets"	They won't; markets are non-stationary
"Reinforcement learning finds alpha"	RL finds execution alpha, not directional alpha
"Auto-features will replace researchers"	Auto-features surface candidates; humans still validate
"Multi-agent systems are dangerous"	They are, but the discipline (gates + adversary) makes them safe
"Vector DBs are overhyped"	Actually underhyped for proprietary corpus retrieval
"AI compliance will replace lawyers"	AI compliance copilots help; humans sign
"End-to-end LLM trading benchmarks are reliable"	**NO** — the **Alpha Illusion critique** (arXiv:2605.16895) flags temporal contamination, narrative fitting, and short-window Sharpe uncertainty in self-reported alpha. Apply P1–P6 reporting protocols before citing any LLM-trading Sharpe

F.5.3 The 2026–2028 predictions

1. **LLM-as-judge becomes standard in agentic systems.** With numeric anchors, not as a replacement.
2. **EU AI Act Annex III (high-risk) takes effect 2027-12-02** *(post-Digital Omnibus political agreement 2026-05-07; was 2026-08-02 in the original Regulation EU 2024/1689 schedule)* — drives formal MRM adoption globally. Even US funds will adopt to be EU-LP-ready. The OCC Bulletin 2026-13 + Fed SR 26-02 (April 17, 2026) refresh is the US banking-side parallel — non-bank funds should still adopt it.
3. **AI-driven ETFs continue to underperform.** The retail wrapper costs are too high.
4. **More "AI fund blowups" (cf. Aidyia 2018).** The hype cycle produces casualties.
5. **Vector DB + RAG becomes standard for proprietary research corpus.** The "RAG over your own work" pattern.
6. **Multi-agent systems get a formal failure taxonomy.** MAST (arXiv:2503.13657) is the start; production frameworks will adopt the discipline.
7. **AI safety applied to capital markets emerges as a discipline.** Herding, AI–AI feedback, vendor concentration risks get formal study.
8. **Conformal prediction overlays become a default risk layer** for AI-driven sizing — distribution-free coverage guarantees are a much cheaper alternative to full SR 11-7 validation for the retail tier.

9. **The Alpha Illusion reporting protocol (P1–P6) becomes table-stakes.** Backtests that don't meet the protocol are increasingly dismissed in 2027+ literature.

F.5.4 The evidence-quality ladder for AI-in-finance claims

A practical filter for the Macion Capital reader. Five rungs, ordered from weakest to strongest evidence. Every AI-in-finance claim should be assigned to a rung before being trusted.

Rung	Description	Example	Trust level
0 — Vendor marketing	"We use AI for X" with no metric, no comparison, no audit trail	AIEQ marketing copy ("Powered by IBM Watson"); hedge fund website copy ("We use cutting-edge AI")	**Don't trust** as evidence of alpha
1 — Careers signal	Job listing implies AI/ML work; no public artifact	PDT Partners "Senior SWE AI Tech" role; Balyasny 12-person internal AI team; Millennium 65 AI/ML-tagged roles	**Weak** — implies direction, not magnitude
2 — Public engineering blog	Technical write-up with code, architecture, or quantified results	Optiver "Engineering the Agentic SDLC" (75% time-to-market reduction, 85% engineering effort reduction); HRTWorker + HeraclesQL blog; Databento feed-handler architecture	**Medium** — peer-readable but not peer-reviewed
3 — Academic paper (peer-reviewed)	Paper with code, methodology, and replication-friendly design	AQR "Virtue of Complexity" (JoF 2024); Cohen "SigFormer" (ICAIF 2023); Gibbs & Candès conformal prediction (JASA 2024)	**Strong** — peer-reviewed but academic-vs-deployment gap remains
4 — Backtested live result	Publicly auditable net-of-cost performance, with walk-forward, MC, and full gate stack	None of the 17 firms in C1 publicly disclose rung-4 results [NEEDS VERIFY]	**Strongest** — but the rarity is itself the signal

The critical gap. Most institutional AI claims live on rungs 0–2. The C1 source-pack review found that **no major fund publicly discloses a rung-4 result** (backtested live net-of-cost with the full gate stack). Bridgewater's AIA Forecaster comes closest: "manages billions, generating alpha" (Bloomberg Jul 2024, `[reported, not a firm financial filing]`) — but without a numeric anchor, this is rung 1 (careers/marketing signal), not rung 4. Man Group's **AlphaGPT** ("proposes signals, writes code, runs backtests before any human sees the output") is rung 2 (public engineering blog) without a public performance benchmark.

The Macion Capital discipline. Apply this ladder to every claim you read: - **Rung 0–1:** Cite as a *signal of direction*; do not cite as evidence of alpha. - **Rung 2:** Cite as engineering context; reproduce locally before adopting. - **Rung 3:** Cite as method; check the assumptions against your own use case. - **Rung 4:** Cite as precedent; the rarity is the lesson.

The Macion Capital G1–G28 + G29–G31 gate stack is the **internal implementation** of rung 4: every memo + every backtest must clear the numeric anchors, with reproducibility + walk-forward + MC, before the strategy-scribe records it in registry. **A fund that only publishes rung 0–2 claims should be asked why.**

F.5.5 Reasoning-models in finance (2026 frontier)

The single biggest 2026 shift not yet integrated into v1.4. **Reasoning models** (Claude Sonnet 4.5 with extended thinking, OpenAI o3/o4, Gemini 2.5 Pro thinking, DeepSeek R1, Grok 3 reasoning) are the new primitive — they produce a chain-of-thought before answering, with a tunable budget for thinking tokens.

Adoption signals (confirmed)

User	Model	Use case	Source
NBIM (Norges Bank IM, ~$1.7–2.0T AUM)	**Claude Sonnet 4.5 + extended thinking**	Research synthesis, ESG screening across ~9,000 portfolio companies, commodity arbitrage research, risk analysis, compliance documentation, Claude Code for analyst/quant tooling (~20% weekly time savings; ~600 employees onboarded within 2 months of Nov 2024 launch)	[STATED, claude.com/customers/nbim, 2025–2026]
Man Group ($227.6B)	**Claude (Anthropic partnership Feb 2026)**	Alpha generation, large-dataset processing, risk-model analysis, application development	[STATED, man.com news-centre; Pensions & Investments; HedgeWeek]
Bridgewater / AIA Labs	**Fine-tuned Qwen3-235B** via Thinking Machines Tinker (2026-06-30)	Reached **84.7% average accuracy** on 6 financial document-triage tasks, beating frontier GPT/Claude/Gemini (50–78%) at fraction of inference cost	[STATED via secondary summary; primary URL `[NEEDS VERIFY]`]
Anthropic for Financial Services launch (2025-07-15)	Claude 4 + Claude Code + MCP connectors to FactSet, S&P Global, Morningstar, PitchBook, Palantir, Snowflake, Databricks, Daloopa, Box	Vendor-reported: Claude Opus 4 passed **5/7 levels Financial Modeling World Cup**; **83% on Vals AI Finance Agent benchmark**	[STATED, anthropic.com/news/claude-for-financial-services]

Cautionary literature

Source	Finding	URL
FINSABER (KDD 2026 D&B Oral; Edinburgh / Sungkyunkwan / UCLA / Oxford)	20-year, 100+ symbol unbiased backtests: "LLM strategies generally underperform traditional methods and simple baselines." Evaluated reasoning models: GPT-o1, DeepSeek-R1, **Fin-o1** (their finance-domain reasoning fine-tune). v6 2026-06-26.	[STATED, arxiv.org/abs/2505.07078]
The Alpha Illusion	Already in v1.4 — keep as load-bearing skeptic.	[STATED, arxiv.org/abs/2605.16895]
LLM Agents Do Not Replicate Human Market Traders	Mono-agent + heterogeneous "Battle Royale" LLM markets show systematically different dynamics from human traders.	[STATED, arxiv.org/abs/2502.15800]

The latency-cost tax (the production caveat)

Model	Reasoning latency	Inference cost multiplier
GPT-4o (no reasoning)	~2–5s	1×
Claude 3.7 Sonnet full-reasoning	~15–45s	5–10×
o3 high-reasoning	~30–120s	10–30×
Gemini 2.5 Pro thinking	~20–60s	8–20×

[INFERRED — multiple vendor dashboards; primary latency benchmarks `[NEEDS VERIFY]`]

Recommended placement

Reasoning models are a research-surface primitive, not a live-alpha primitive. The 2026 consensus: - **Use reasoning models** for: research synthesis (NBIM use case), memo drafting, ESG screening, document triage (Bridgewater use case), compliance documentation, anomaly explanation in research, walk-forward design. - **Do NOT use reasoning models** for: live order routing, sub-second execution, real-time risk decisions, anything where the 30–120s latency budget can't be paid. Reasoning-model accuracy does NOT scale linearly past 4–8 thinking tokens — there's a knee of diminishing returns.

The FINSABER null applies to reasoning models too: the most-fine-tuned reasoning model (Fin-o1) still doesn't beat simple baselines on 20-year backtests. **The reasoning primitive is a productivity layer, not an alpha source.** This is consistent with the standing rule (LLM is the orchestrator; gate stack is the judge).

F.6 — Practical recommendations

1. **Default to "no public AI claims".** Use "quantitative" + "systematic" — accurate.
2. **Build the agent-system-auditor early.** Catch MAST failures before they ship.
3. **Maintain a vendor concentration register.** Diversify.
4. **Document AI use internally.** Even for pre-launch funds.
5. **Engage counsel-of-record at AUM > $25M.** Before that, use an advisory retainer.

6. **Read the MAST failure taxonomy.** The 14 modes are the spec for what to defend against.
7. **Plan for the regulatory timeline.** EU AI Act is in force; SEC final rule expected 2026 H1.

See `12_references_bibliography.md` for citations.

Chapter F7. EU AI Act Day-1 compliance runbook (Annex III = 2027-12-02 per Digital Omnibus)

BOTTOM LINE EU AI Act compliance is a technical documentation discipline, not a checkbox — every high-risk AI system needs a 12-section Annex IV technical file, Article 12 logging from day 1, and Article 14 human-oversight with 2-of-3 multi-sig minimum; cheapest path for Macion Capital is to defer EU-resident LP acceptance until LP base forces it (~Q4 2027).

> **Updated for the 2026-05-07 Digital Omnibus political agreement.** The Annex III high-risk effective date moved from **2026-08-02 → 2027-12-02**; the Annex I embedded-AI effective date moved from **2027-08-02 → 2028-08-02**. Prohibitions (2025-02-02) and GPAI obligations (2025-08-02) remain unchanged. This runbook covers a Macion Capital–scale own-fund with EU-resident LPs or EU-domiciled ops; it is **not legal advice**.

The standing rule. EU AI Act compliance is a **technical documentation discipline**, not a checkbox. Every high-risk AI system (Article 6 + Annex III) needs an **Annex IV technical file** before deployment. The 12 sections of Annex IV (Recital 95) define what "compliant" means.

F.7.1 — Step 0: Inventory all AI systems (Article 6 + Annex III determination)

For each AI system in the fund, fill this table:

System	Purpose	Vendor / build	High-risk? (Article 6)	Annex III category	Day-1 deadline
`mc-research-agent`	Idea discovery	Custom + Claude API	NO (research only)	—	n/a
`mc-factor-miner`	Alpha generation	Custom + Databento	NO (read-only)	—	n/a
`mc-gate-judge-G1`	DSR computation	Custom numpy	NO (deterministic)	—	n/a
`mc-ncaa-judge`	LLM-as-judge	Claude API	**YES** (Annex III §5 — creditworthiness / financial risk)	**2027-12-02**	First live use
`mc-wallet-executor`	Trade execution	Turnkey + CDP	**YES** (Annex III §5)	**2027-12-02**	First live trade
`mc-portfolio-optimizer`	Mean-variance / risk-parity	Custom + cvxpy	NO (deterministic)	—	n/a
`mc-narrative-reader`	Filing/transcript extraction	Claude API	**YES** (Annex III §5)	**2027-12-02**	First live use
`mc-regime-detector`	HMM / change-point	Custom + hmmlearn	NO (read-only)	—	n/a

Rule of thumb for Macion Capital. Any system whose output **informs a trading decision that affects EU-resident investors** is high-risk per Annex III §5 (access to essential private services + financial services). LLM-as-judge + LLM-extracted narrative are high-risk; deterministic numerical code is not.

F.7.2 — Step 1: Annex IV technical file (12 sections per Recital 95)

For each high-risk system, build a `docs/ai_compliance/<system_id>_annex_iv.md` with the following structure:

```
1.  General description of the AI system (name, version, intended purpose)
2.  Provider identity (Macion Capital + relevant contact person)
3.  Scope of intended use (markets, AUM range, position size limits)
4.  Data sources + data governance (lineage, integrity checks, retention)
5.  Methodology (model architecture, training, validation, holdout)
6.  Performance metrics (gate-stack verdict, PBO, DSR, era stability)
7.  Human oversight design (Article 14)
8.  Accuracy + robustness + cybersecurity
9.  Transparency to deployers (LP-facing disclosure)
10. Logging + post-market monitoring (Article 12)
11. CE marking + EU Declaration of Conformity
12. Quality management system (per Article 17)
```

Concrete per-section requirements:

Section	What to write	Source
1	Name, version, intent	system spec doc
2	Macion Capital + EU agent for service of process	corporate doc
3	Markets (e.g. "US equities + EU govt bonds"), AUM band, size cap	mandate doc
4	Data sources, integrity gate, lineage	DATA_REGISTRY.md
5	Architecture diagram, training pipeline	`build_workbook.py` + agent templates
6	Gate-stack verdict (G1-G28), MAST defense map (G.5)	REGISTRY + KILL_LOG
7	Human-in-the-loop design (multi-sig, halt conditions)	Part G.6.4
8	Adversarial robustness, determinism score, rate limits	G.4 metrics
9	LP-facing language on AI usage (e.g. "AI-augmented research, human-approved trades")	LP memo
10	Decision log (Article 12) + audit retention policy	COMPLIANCE_LOG.csv
11	CE marking per Article 48 + Declaration of Conformity	legal
12	QMS per Article 17 (change management, incident response)	OPERATING_MODEL.md

F.7.3 — Step 2: Article 14 — Human oversight

Mandatory for every high-risk system. The technical implementation:

1. **Multi-sig approval on every wallet operation.** 2-of-3 human co-sign minimum; recommend 3-of-5 for >$10M AUM.
2. **Halt conditions.** The judge-agents emit a "HALT" signal if any gate fails. The wallet-executor agent MUST honor the halt unconditionally.
3. **Override log.** If a human overrides a HALT signal, log: timestamp, agent_id, gate_id, override_reason, human_id.
4. **Kill switch.** A documented, tested, drill-able kill switch that closes all positions in <60 seconds. Quarterly drill required.

Implementation pattern (Turnkey / Fireblocks).

```
# Pseudocode: 2-of-3 multi-sig policy
wallet_policy = {
    "version": "1.0",
    "default_rule": "deny",
    "rules": [
        {
            "condition": "tx.value_usd < 100_000",
            "approvers": ["human_pm", "human_compliance"],
            "threshold": 2,
            "signers_required": 2
        },
        {
            "condition": "tx.value_usd >= 100_000",
            "approvers": ["human_pm", "human_compliance", "human_cou
            "threshold": 3,
            "signers_required": 3
        }
    ],
    "halt_signals": ["G1_FAIL", "G4_FAIL", "G22_KILL"],
    "halt_action": "block_all_pending"
}
```

F.7.4 — Step 3: Article 12 — Logging

Mandatory technical + operational logging.

Log field	Required	Source
`timestamp_utc`	YES	All agent calls
`agent_id`	YES	All agent calls
`system_id`	YES	All agent calls
`prompt_hash`	YES	LLM-as-judge + LLM-narrative
`model_id`	YES	LLM calls
`input_summary` (truncated)	YES	All calls
`output_summary` (truncated)	YES	All calls
`gate_anchors`	YES	Gate-judge calls (DSR, PBO, etc.)
`verdict`	YES	All judge calls
`human_oversight_id`	YES	Wallet-executor calls
`override_reason`	conditional	Override events
`incident_id`	conditional	HALT events

Retention: Minimum 6 months (Article 12(1)); recommend 5 years for fund-of-fund ops.

Format: Append-only CSV or Parquet. Daily signature hash for tamper-evidence.

```
# Sample log entry
{
    "timestamp_utc": "2026-07-07T14:23:11.123Z",
    "agent_id": "mc-ncaa-judge-v1.2",
    "system_id": "mc-ncaa-judge",
    "prompt_hash": "sha256:abc123...",
    "model_id": "claude-sonnet-4-6",
    "input_summary": "strategy: bt_us_close_session, len: 2487 chars",
    "output_summary": "verdict: ITERATE, ncaa_score: 6.5, reason: ...",
    "gate_anchors": {"dsr_p": 0.18, "pbo": 0.62, "min_btl": 0.41, "era_pass": 0.55},
    "verdict": "ITERATE",
    "human_oversight_id": null,
    "override_reason": null,
    "incident_id": null
}
```

F.7.5 — Step 4: EU AI database registration

Per Article 49 + Article 71, register each high-risk system in the **EU AI database** before deployment. Registration requires: - Provider name + contact - System name + version - Intended purpose - Conformity assessment reference

Database URL: `digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai` (registration portal TBD by EC; expected 2027 H1).

F.7.6 — Step 5: Conformity assessment (Article 43)

For Macion Capital–scale (own-fund with EU LPs but no biometric / critical-infrastructure use case), the **self-assessment (Annex VI)** route is the default. The 8-step procedure:

1. Identify the applicable requirements (Articles 8-17)
2. Identify whether the system is subject to third-party assessment (not for our scale)
3. Apply harmonized standards (where available) or state-of-the-art
4. Complete the Annex IV technical file
5. Implement post-market monitoring (Article 72)
6. Issue the EU Declaration of Conformity
7. Apply the CE marking (Article 48)
8. Register in the EU AI database

F.7.7 — Step 6: Article 17 — Quality management system

The QMS binds all of the above. The minimal viable QMS for a small fund:

Element	Document	Owner
Risk management	Risk register + Article 9 risk assessment	PM
Data governance	DATA_REGISTRY.md	Data lead
Technical documentation	Annex IV files (per F.7.2)	Eng lead
Record-keeping	Article 12 logs (per F.7.4)	Compliance
Transparency	LP-facing disclosure	PM + counsel
Human oversight	Multi-sig policy (per F.7.3)	Eng + compliance
Accuracy + robustness	G.4 eval harness	Eng
Post-market monitoring	Live forward-OOS monitor	PM + research
Incident response	Kill-switch drill log	Compliance

F.7.8 — The realistic timeline for Macion Capital

Assuming the Day-1 deadline is **2027-12-02** for high-risk Annex III:

Quarter	Action
Q3-Q4 2026	Inventory AI systems; build Annex IV template; deploy Article 12 logging
Q1-Q2 2027	Fill Annex IV files; implement multi-sig wallet policy; build QMS docs
Q3 2027	Conformity assessment + EU Declaration of Conformity; test kill switch
Q4 2027	Register in EU AI database; CE mark; final compliance review with counsel
2027-12-02	Day-1: high-risk Annex III obligations take effect
2028-08-02	Day-1 for Annex I (embedded AI in regulated products) — not applicable to Macion Capital

Caveat. The Digital Omnibus political agreement of 2026-05-07 is not yet final law as of 2026-07-07. Council + Parliament formal adoption expected Q3 2026. **Verify with counsel before relying on the shifted dates.** The 2025-02-02 prohibitions and 2025-08-02 GPAI obligations are not affected by the omnibus and apply as scheduled.

F.7.9 — What's cheap vs expensive

Item	Cost (USD)	Note
Annex IV template + internal filing	$0 (in-house)	4-6 hours per system
Article 12 logging infra (logging + WORM storage)	$500-$2,000/mo	Self-hostable; Turnkey ships native logs
Multi-sig wallet policy (Turnkey / Fireblocks)	$0-$500/mo	Native to vendor
CE marking + EU Declaration of Conformity	$0 (in-house) + $5K-$25K (counsel review)	Counsel engagement required for high-risk
EU AI database registration	$0	Free
Counsel engagement (one-time, 2027 H1)	$25K-$100K	One-time setup
Ongoing QMS audits (annual)	$10K-$50K/yr	External auditor recommended

Total Year-1 compliance budget for Macion Capital scale: ~$50K-$200K one-time + $15K-$75K/yr ongoing. This is **separate from** the engineering time to fill the Annex IV files (a real cost: ~1 FTE × 2-3 months).

F.7.10 — The bottom line for the principal

EU AI Act compliance is **doable for a small fund**, but not free. The cheapest path is: 1. **Limit high-risk systems to the minimum** (i.e. don't deploy LLM-as-judge in the live trade loop; keep it in research). 2. **Pick a wallet vendor with audit-grade logs** (Turnkey, Fireblocks-via-Dynamic) — see C4.2. 3. **Build the Annex IV files in 2026 H2** so the 2027-12-02 deadline is met with margin. 4. **Engage counsel in 2026 Q4** for the conformity assessment + EU Declaration of Conformity. Do not skip this.

The **no-compliance** alternative is to **not accept EU-resident LPs** and not domicile any ops in the EU. For Macion Capital's pre-launch stage, that is the right answer — the LPs are not yet identified, and the cleanest path is to defer EU AI Act compliance until the LP base forces it.

See F.2.4 for the regulatory landscape; see F.5.5 for the LLM-as-judge use case; see C4 for the wallet-vendor layer; see Part G for the operational build.

Chapter G. Macion Capital operational layer — MCP menu + agents + gates + MAST defense + 4-phase roadmap

BOTTOM LINE The bridge from research to runtime: MCP server menu (Tier 1/2/3), agent template taxonomy (research/judge/reporter/executor), gate-stack numpy code patterns (DSR, PBO, min-BTL, slippage, NCAA), multi-agent eval metrics, MAST 14-mode defense map, and 4-phase operational roadmap.

The principal's own-fund track, made operational. Translates the workbook's vendor + framework + frontier research into a concrete, executable build plan for Macion Capital. Every recommendation is gated by the same G1–G28 gate stack documented in F.1–F.3 — agents *propose*, the gate stack *judges*, the wallet *executes*.

This section is the bridge from research to runtime. It consolidates: (G.1) the MCP server menu, (G.2) the agent template taxonomy, (G.3) gate-stack numpy code patterns, (G.4) multi-agent evaluation metrics, (G.5) the MAST 14-mode → defense map, and (G.6) the four-phase operational roadmap.

G.1 — MCP Server Menu (the agent-tool layer)

Model Context Protocol (MCP) is the 2026 standard for AI-tool connectivity. Each MCP server exposes a typed set of tools that any MCP-compatible agent (Claude Code, Cursor, LangGraph, ElizaOS) can call. Macion Capital's vendor selection across C1 + C4 collapses to the following MCP surface.

G.1.1 — Tier-1 servers (Y0, free or already-paid)

Server	What it exposes	Source	Cost	Tag
EODHD MCP	72 tools (equities/forex/crypto fundamentals + OHLCV + macro indicators + OpenAPI 3.1 + 28 AI agent skills)	`eodhd.com/mcp`	$20/mo (free tier)	**[STATED]**
Intrinio MCP	AI-Ready data feeds, GenAI integration, MCP server	`intrinio.com`	$150/mo (AI-Ready tier)	**[STATED]**
Databento MCP	Tick / 1m / daily historical, free $125 credit	`databento.com`	Free $125 credit → ~$200/mo	**[INFERRED]**
Polygon.io / Massive.com MCP	US equities + options + crypto real-time	`polygon.io`	$29-$99/mo	**[INFERRED]**
Alpha Vantage MCP	Fundamental + forex + crypto	`alphavantage.co`	Free tier / $50/mo premium	**[INFERRED]**
Tiingo MCP	EOD + news + fundamentals	`tiingo.com`	$10-$50/mo	**[INFERRED]**

G.1.2 — Tier-2 servers (Y1, mid-tier spend)

Server	What it exposes	Source	Cost	Tag
AlphaSense MCP	500M+ premium documents, 7,000+ enterprises, Tegus transcripts	`alpha-sense.com`	$20K-$100K+/yr	**[STATED]**
RavenPack Bigdata.com MCP	12M entities, 7.1M companies, 7K+ event categories	`ravenpack.com`	$25K-$100K+/yr	**[STATED]**
Kensho LLM API (S&P Global)	LLM-ready market + entity data, Anthropic/OpenAI/Google ecosystem	`kensho.com`	$50K-$500K/yr	**[STATED]**
KDB.AI + MCP	Time-series + vector hybrid, NVIDIA cuVS/CAGRA on KDB-X	`kx.com/kdb-ai`	$50K-$500K/yr	**[STATED]**

G.1.3 — Tier-3 servers (Y2+, compliance-grade)

Server	What it exposes	Source	Cost	Tag
Bloomberg BQuant MCP	Full Terminal data + BQuant Agent, ArcticDB integration	`bloomberg.com`	$24K-$28K/seat/yr	**[STATED]**
FactSet MCP	Fundamentals + ownership + estimates + transcripts	`factset.com`	$15K-$25K/seat/yr	**[STATED]**
S&P Capital IQ Pro MCP	Fundamentals + estimates + ownership + public filings	`capitaliq.com`	$15K-$25K/seat/yr	**[STATED]**
Anthropic for Financial Services MCP	FactSet + S&P Global + Morningstar + PitchBook + Palantir + Snowflake + Databricks connectors	`claude.com/financial-services`	Bundled with Claude API	**[STATED, 2025-07-15 launch]**
Coinbase AgentKit / CDP MCP	Onchain wallet + x402 + session keys	`docs.cdp.coinbase.com`	Usage-based	**[STATED]**
Crossmint MCP	Custodial-as-server MPC-AA wallets + Agentic Commerce	`crossmint.com`	Usage-based	**[STATED]**
Turnkey MCP	Enclave-secured WaaS, sub-100ms signing	`turnkey.com`	Usage-based	**[STATED]**
Nevermined MCP	x402 facilitator + agent payment credentials	`nevermined.ai`	Usage-based	**[STATED]**

G.1.4 — Macion Capital MCP selection

Phase	Default MCP server set	Cost / month	Notes
Y0 (pre-launch, $0 AUM)	EODHD + Alpha Vantage + Tiingo + Polygon Starter	~$70	Hobbyist tier, full data coverage for backtests
Y1 (first AUM, $1-10M)	+ Databento $200 + Intrinio $150 + KDB.AI dev $500	~$1,000	Add tick + L3 data; add vector DB for alt-data
Y2 (scale, $10-100M)	+ AlphaSense $1,500 + RavenPack $1,500 + Anthropic for FS $1,000	~$5,000	Add premium NLP + agent framework
Y2+ (compliance-grade, $100M+)	+ Bloomberg 2 seats $5K + FactSet 1 seat $1.5K + Turnkey $1K + Turnkey/Fireblocks wallet layer	~$15K	Full audit-grade MRM stack

Note on cost. All amounts USD/month. Multi-seat / annual contracts discount 20-40%. The Y0 tier is the cheapest viable stack; the Y2+ tier is the best-in-class stack (per D.3.2 vs D.3.3 in `09_build_vs_buy_matrix.md`).

G.2 — Agent Template Taxonomy

The Macion Capital agent zoo follows the **tool-use → propose → judge → act** pattern. Each agent template has a single mission, a typed input/output contract, and an MCP-server dependency list.

G.2.1 — Research agents (read-only, propose factors / alphas / theses)

Agent	Mission	Inputs	Outputs	MCP deps
data-collector	Assemble clean, time-aligned panels for a universe	Universe + date range + source list	`{csv_path, audit_trail_path, integrity_score}`	EODHD, Databento, Tiingo, Polygon
factor-miner	Mine candidate factors from fundamentals + alt-data	Universe + date range + signal type	`{alpha_id, factor_vector, ic, era_pass}`	Alpha Vantage, RavenPack, AlphaSense
narrative-reader	Extract thesis fragments from filings, transcripts, news	Document set + prompt	`{thesis_id, claims, evidence_chain, confidence}`	AlphaSense, Intrinio, RavenPack
regime-detector	Identify regime-shift candidates (vol, correlation, factor-regime)	Market data + macro series	`{regime_id, prob, breakpoints, recall}`	EODHD macro, FRED (free)

Agent	Mission	Inputs	Outputs	MCP deps
scientific-discovery	LLM-orchestrated literature search + cross-domain idea transfer	Topic + corpus hint	`{idea_id, prior_art, hypothesis, falsifier}`	OpenAlex (free), ArXiv MCP, Anthropic for FS

G.2.2 — Judge agents (run gate-stack, return verdicts)

Agent	Mission	Inputs	Outputs	MCP deps
gate-judge-G1-G7	Statistical rigor gate (DSR, PBO, min-BTL, era stability, drawdown, turnover, IC decay)	Strategy + backtest output	`{verdict, numeric_anchors, failure_flags}`	(none — internal)
gate-judge-G8-G14	Execution reality gate (slippage, fill rate, capacity, latency, regime-conditional, robustness, dependency)	Strategy + tick-data	`{verdict, slippage_estimate, capacity_estimate}`	Databento L3 (for slippage calibration)
gate-judge-G15-G21	Compliance + governance gate (Marketing Rule, EU AI Act, MRM, model documentation, audit log, change mgmt, segregation)	Strategy + agent template	`{verdict, article_references, log_completeness}`	(none — internal)
gate-judge-G22-G28	Frontier + reflection gate (LLM-as-judge for narrative alpha, conformal coverage, PRM, neuro-symbolic review, explainability, red-team)	Strategy + narrative output	`{verdict, ncaa_score, conformal_coverage, red_team_pass}`	(none — internal)
gate-judge-G29-G31	Crypto + DeFi + Privacy (venue lock, DeFi liquidation, ZK/TEE proof)	Strategy + venue + wallet ops	`{verdict, venue_pass, liquidation_pass, privacy_pass}`	Coinbase AgentKit (for venue ops), Nevermined (for x402)

G.2.3 — Reporter agents (write deliverables, no decision authority)

Agent	Mission	Inputs	Outputs	MCP deps
strategy-scribe	Convert research output → PM proposal + LP-facing memo	Strategy + backtest + verdict	`{pm_proposal.md, lp_memo.md, slide_deck.md}`	(none — internal)
decision-log-writer	Append verdict + reasoning to KILL_LOG + REGISTRY	Verdict + gate_anchors	`{registry_row, kill_log_row}`	(none — internal)
compliance-briefer	Convert agent-wallet ops + LLM-judge calls → EU AI Act / FINRA / OCC bundle	Agent logs + article refs	`{article_12_log.csv, occ_2026_13_log.csv}`	(none — internal)

G.2.4 — Executor agents (only when promoted)

Agent	Mission	Constraints	MCP deps
wallet-executor	Sign + broadcast a strategy-blessed tx	Mandate: tx pre-approved by gate-stack + 2-of-3 multi-sig + size cap	Coinbase AgentKit, Turnkey, Fireblocks
rebalancer	Maintain collateralization / liquidation buffer	Mandate: bounded by health-factor floor	(none — internal)

The standing rule. Only the **wallet-executor** and **rebalancer** are allowed to move capital. All other agents are read-only proposers or judges. **No agent shall issue a trade without gate-stack verdict AND human multi-sig co-sign** until AUM exceeds the human-bandwidth threshold (TBD; recommended $25M AUM).

G.3 — Gate-Stack Numpy Code Patterns

The G1–G28 gate stack is the load-bearing safety layer. Below are **numpy / vectorbt-pro / MLFinLab code patterns** for the highest-stakes gates. These are illustrative starting points — adapt to your data + backtest engine.

G.3.1 — G1: Deflated Sharpe Ratio (DSR)

```
import numpy as np
from scipy import stats

def deflated_sharpe(returns: np.ndarray, n_trials: int, risk_free: f
    """
    Bailey & López de Prado, "The Deflated Sharpe Ratio" (2014).
    Adjusts Sharpe for the number of trials (multiple-testing bias).

    returns : array of period returns (e.g. daily)
    n_trials: number of backtest variants tried (for deflated adjust
    """
    excess = returns - risk_free
    sharpe = excess.mean() / excess.std(ddof=1) * np.sqrt(252)
    # Skewness/kurtosis of return distribution
    skew = stats.skew(excess)
    kurt = stats.kurtosis(excess, fisher=False)  # excess=False → re

    # Expected max Sharpe under null (N trials, T observations)
    e_max_z = (1 - np.euler_gamma) * stats.norm.ppf(1 - 1/n_trials)
              + np.euler_gamma * stats.norm.ppf(1 - 1/(n_trials * np
    e_max_sharpe = e_max_z + (e_max_z**2 - 1) / (4 * (len(returns))*
    # DSR p-value: P(actual Sharpe > observed | null of max(e_max_sh
    z = (sharpe - e_max_sharpe) * np.sqrt(len(returns) - 1) \
        / np.sqrt(1 - skew * sharpe + (kurt - 1)/4 * sharpe**2)
    p_value = 1 - stats.norm.cdf(z)

    return {"sharpe": sharpe, "dsr_p_value": p_value, "e_max_sharpe"

# Use: kill if DSR p_value > 0.05 (i.e. cannot reject null of max-tr
```

G.3.2 — G2: Probability of Backtest Overfitting (PBO)

```
def probability_of_backtest_overfit(
    returns_matrix: np.ndarray,  # shape: (T, N) where N = number of
    n_blocks: int = 16,          # number of CSCV blocks
    seed: int = 42,
) -> dict:
    """
    Bailey, Borwein, López de Prado, "The Probability of Backtest Ov
    Combinatorially Symmetric Cross-Validation (CSCV).

    returns_matrix : (T, N) — T observations, N variants
    """
    rng = np.random.default_rng(seed)
    T, N = returns_matrix.shape
    omega = rng.permutation(T)
    blocks = np.array_split(omega, n_blocks)

    # Compute logit of OOS performance per CSCV split
    logit_perf = []
    for s in range(n_blocks):
        test_idx = blocks[s]
        train_idx = np.concatenate([blocks[i] for i in range(n_block
        # Rank variants by in-sample performance
        is_perf = returns_matrix[train_idx].mean(axis=0)
        os_perf  = returns_matrix[test_idx].mean(axis=0)
        best_is  = np.argmax(is_perf)
        # OOS performance of the IS-best variant
        logit_perf.append(os_perf[best_is] / returns_matrix[test_idx

    logit_perf = np.array(logit_perf)
    # PBO = fraction of splits where the OOS best variant is below t
    pbo = np.mean(logit_perf < np.median(logit_perf))
    return {"pbo": pbo, "logit_perf": logit_perf}

# Use: kill if PBO > 0.50
```

G.3.3 — G3: minimum Backtest Length (min-BTL)

```
def min_backtest_length(returns: np.ndarray, target_sharpe: float, alpha: float = 0.05) -> int:
    """
    López de Prado, "Advances in Financial ML" §11.5 / Bailey & López de Prado (2012).
    Minimum number of observations needed to reject Sharpe = 0 at confidence alpha.

    returns : daily return series of the strategy
    target_sharpe : strategy's observed annualized Sharpe
    """
    skew = stats.skew(returns)
    kurt = stats.kurtosis(returns, fisher=False)
    z_alpha = stats.norm.ppf(1 - alpha)
    n_min = 1 + (1 - skew * target_sharpe
                 + ((kurt - 1) / 4) * target_sharpe**2) * (z_alpha / target_sharpe)**2
    return int(np.ceil(n_min))

# Use: if len(returns) < min_btl × 1.5 → kill ("insufficient evidence")
```

G.3.4 — G4: Era stability (regime-conditional)

```
def era_stability(
    returns: np.ndarray,
    era_boundaries: np.ndarray,  # e.g. [2018-01-01, 2020-01-01, 2022-01-01, 2024-01-01]
    min_sharpe_per_era: float = 0.5,
) -> dict:
    """
    K1.4 / G4: strategy must clear a minimum Sharpe in M-of-N eras.

    era_boundaries : array of timestamps marking era starts.
    """
    eras = []
    for i in range(len(era_boundaries) - 1):
        mask = (returns.index >= era_boundaries[i]) & (returns.index < era_boundaries[i+1])
        r_era = returns[mask].values
        if len(r_era) < 60:
            continue
        sharpe = r_era.mean() / r_era.std(ddof=1) * np.sqrt(252)
        eras.append({"start": era_boundaries[i], "end": era_boundaries[i+1], "sharpe": sharpe})

    n_pass = sum(1 for e in eras if e["sharpe"] >= min_sharpe_per_era)
    pass_rate = n_pass / len(eras)
    return {"eras": eras, "n_pass": n_pass, "pass_rate": pass_rate}

# Use: kill if pass_rate < 0.6 (i.e. fails in >40% of eras)
```

G.3.5 — G9: Slippage-aware re-evaluation

```
def slippage_aware_sharpe(
    backtest_returns: np.ndarray,
    signal_changes: np.ndarray,  # boolean series, True when position changes
    bid_ask_half_spread: float,  # e.g. 0.0005 for 5bps
    impact_coefficient: float = 1e-6,  # Almgren-Chriss linear coefficient
    adv: float,  # average daily volume (in same units as notional)
) -> dict:
    """
    G9: subtract estimated slippage from each turnover event.
    Returns net-of-cost Sharpe vs naive Sharpe.
    """
    naive_sharpe = backtest_returns.mean() / backtest_returns.std(ddof=1) * np.sqrt(252)
    turnover_per_event = bid_ask_half_spread + impact_coefficient * (1.0 / adv)
    total_slippage = signal_changes.sum() * turnover_per_event
    # Distribute slippage linearly across days (approx)
    daily_slippage = total_slippage / len(backtest_returns)
    net_returns = backtest_returns - daily_slippage
    net_sharpe = net_returns.mean() / net_returns.std(ddof=1) * np.sqrt(252)
    return {"naive_sharpe": naive_sharpe, "net_sharpe": net_sharpe,
            "slippage_bps_per_trade": turnover_per_event * 1e4}

# Use: kill if net_sharpe < 0.5 (the friction-cost threshold)
```

G.3.6 — G22: LLM-as-judge for narrative alpha

```
import anthropic

def ncaa_judge(
    client: anthropic.Anthropic,
    strategy_doc: str,           # Markdown research memo
    rubric: str = "default",     # see below
    model: str = "claude-sonnet-4-6",
) -> dict:
    """
    Narrative-Coherence Audit (NCAA) — LLM-as-judge gate.
    Returns {verdict, narrative_score, coherence_score, novel_premis
    """
    rubric_text = {
        "default": """You are auditing a quant strategy research mem
evidence quality. Score 0-10 each for:
- NARRATIVE: Does the thesis have a clear, non-tautological causal m
- COHERENCE: Are all claims internally consistent, no contradictions
- NOVEL_PREMISE: Does the edge require a fact not already priced by
- EVIDENCE_QUALITY: Are backtest claims supported by regime-conditic
Return JSON: {"narrative": N, "coherence": N, "novel_premise": N, "e
PASS criteria: all four scores >= 7 AND verdict = "PROMOTE".""",
    }[rubric]

    response = client.messages.create(
        model=model,
        max_tokens=2048,
        messages=[{"role": "user", "content": f"{rubric_text}\n\n===
    )
    import json
    return json.loads(response.content[0].text)

# Use: kill if verdict != "PROMOTE" or any score < 7
```

G.4 — Multi-Agent Evaluation Metrics

When multiple agents contribute to a single decision, you need metrics to evaluate the **decision system**, not the agents individually.

G.4.1 — Decision-quality metrics

Metric	Formula	Use
Decision precision (DP)	`TP_decisions / (TP_decisions + FP_decisions)`	Of "promote" calls, how many produced live alpha
Decision recall (DR)	`TP_decisions / (TP_decisions + FN_decisions)`	Of all real alpha candidates, how many did the system catch
Decision F1	`2 * DP * DR / (DP + DR)`	Single-number summary of system accuracy
Gate violation rate (GVR)	`n_violations / n_decisions`	How often a strategy bypassed a gate
Time-to-decision (TTD)	median time from research-completion to verdict	Operational latency
Audit-trail completeness (ATC)	`n_logged_articles / n_required_articles` (EU AI Act Annex IV)	Compliance coverage

G.4.2 — Agent-isolation metrics

Metric	Description
Decision determinism	For fixed input + seed, how often does the multi-agent system produce the same verdict? (target: ≥0.95)
Adversarial robustness	With a perturbed input (small noise on features), how often does the verdict change? (target: ≤0.05)
Conformal coverage	For conformal prediction layers, empirical coverage should be ≥ target coverage (e.g. 0.90)
LLM-judge agreement	For G22/G26/G29, inter-judge Cohen's κ on 100-sample audit (target: ≥0.6)

G.4.3 — Sample eval harness (skeleton)

```
def eval_agent_system(
    agent_pipeline: callable,
    held_out_memos: list,        # 100+ annotated strategy memos
    ground_truth_verdicts: list, # ["PROMOTE", "KILL", ...]
    perturbation_fn: callable = None,
) -> dict:
    """Run a held-out evaluation of the multi-agent decision system.
    preds = [agent_pipeline(m)["verdict"] for m in held_out_memos]

    # Decision precision/recall/F1
    from sklearn.metrics import precision_recall_fscore_support
```

```
labels = ["PROMOTE", "ITERATE", "KILL"]
p, r, f, _ = precision_recall_fscore_support(ground_truth_verdicts, preds,
                                             labels=labels, zero_division=0)

# Decision determinism
det_scores = []
if perturbation_fn:
    for m in held_out_memos[:20]:
        v_orig = agent_pipeline(m)["verdict"]
        v_pert = agent_pipeline(perturbation_fn(m))["verdict"]
        det_scores.append(int(v_orig == v_pert))

return {
    "precision": dict(zip(labels, p)),
    "recall":    dict(zip(labels, r)),
    "f1":        dict(zip(labels, f)),
    "determinism": np.mean(det_scores) if det_scores else None,
    "n_eval":    len(held_out_memos),
}
```

G.5 — MAST 14-Mode Defense Map

The MAST taxonomy (NeurIPS 2025, arXiv:2503.13657) names 14 distinct failure modes for ML in finance. Each maps to one or more of the G1–G28 gates + a concrete defense pattern.

MAST mode	Description	Gate-stack defense	Engineering pattern
1. Data leakage	Train/test split that allows future info to leak	G1 (DSR), G3 (min-BTL), G4 (era stability)	Strict temporal CV; embargo gap; future-mask on features
2. Non-stationarity	Distribution shift over time	G4 (era stability)	Walk-forward; rolling re-fit; regime detector (G.2.1)
3. Overfitting	Excessive in-sample tuning	G1 (DSR), G2 (PBO)	Cross-validation; complexity penalty; CSCV
4. Backtest overfitting	Multiple-testing bias across N variants	G1, G2	Familywise error correction; deflated Sharpe
5. Survivorship bias	Dropped delisted instruments	G22 (NCAA) + data audit	Include delisted tickers; point-in-time universe
6. Look-ahead bias	Using information not yet available	G3 (min-BTL) + code review	Strict date-of-knowledge audit; reproducible build
7. Selection bias	Cherry-picked universe	G22 (NCAA)	Pre-register universe + signals before backtest
8. Redundant features	Collinear inputs inflate IC artificially	G8 (execution reality)	Variance-inflation-factor pruning; PCA + whitening
9. Improper target	y-leakage or target construction artifact	G22 (NCAA) + G29 (crypto)	Construct target from raw prices; null target test
10. Spurious correlations	Numeric coincidence	G1, G2, G22	Bonferroni; hold-out period; out-of-sample era
11. Causation confusion	Confusing correlation with mechanism	G22 (NCAA)	Causal discovery (DML, do-calculus); placebo tests
12. Capacity / market impact	Strategy breaks at scale	G9 (slippage)	Almgren-Chriss; capacity curve; ADV-limit
13. Latency mismatch	Backtest assumes faster execution than real	G8 (execution)	Realistic fill model; queue-position penalty
14. Reporting bias	Highlight winners, hide losers	G22 (NCAA) + G23 (audit log)	Pre-registered reporting; full KILL_LOG; agent-judge for selective reporting

The standing rule. Every research memo MUST cite which MAST modes were checked + which defenses were applied. The G22 (NCAA) judge + the KILL_LOG + the audit trail are the load-bearing enforcement.

G.6 — Four-Phase Operational Roadmap

Translating the workbook into a sequenced build.

G.6.1 — Phase 1 (Weeks 1-4): Agent infrastructure + MCP plumbing

Build: 1. MCP server menu (G.1.1 — Y0 tier: EODHD + Alpha Vantage + Tiingo + Polygon Starter) 2. Agent template skeleton (Claude Code + Cursor + LangGraph scaffolding) 3. Gate-stack code: G1 (DSR), G2 (PBO), G3 (min-BTL), G4 (era stability), G9 (slippage) 4. Decision-log writer → KILL_LOG + REGISTRY

Milestone: First end-to-end pass through `data-collector → factor-miner → gate-judge → decision-log`. Decision determinism ≥0.80 on 20-sample audit.

G.6.2 — Phase 2 (Weeks 5-12): Research agents + narrative-judge

Build: 1. Research agents: data-collector, factor-miner, narrative-reader, regime-detector 2. G22 NCAA judge (LLM-as-judge for narrative + evidence quality) 3. MAST 14-mode defense map integration (G.5) 4. Decision-quality metrics (G.4) + held-out eval harness 5. Compliance log scaffolding (EU AI Act Annex IV template)

Milestone: First end-to-end backtest with full gate-stack verdict. Audit-trail completeness ≥0.85.

G.6.3 — Phase 3 (Weeks 13-24): Causal discovery + alt-data + scaling

Build: 1. Scientific-discovery agent (literature + cross-domain idea transfer) 2. Alt-data vendor tier (AlphaSense / RavenPack) with MCP integration 3. Causal discovery stack (DML, causal forests, IV) for thesis-validation 4. Conformal prediction layer for uncertainty quantification 5. Wallet-vendor POC (Turnkey or Coinbase AgentKit) — **observe-only**

Milestone: First promoted strategy with full audit-trail + LLM-judge verdict + net-of-cost backtest. Decision precision ≥0.60 on a 50-strategy historical audit.

G.6.4 — Phase 4 (Weeks 25+): Live trading + EU AI Act compliance

Build: 1. Wallet-executor + rebalancer agents (with human multi-sig co-sign until $25M AUM) 2. EU AI Act Annex IV technical-file generator 3. FINRA Rule 2210 / 3110 + OCC 2026-13 / Fed SR 26-02 compliance bundle 4. 2-of-3 multi-sig human-in-the-loop on every wallet operation 5. LP-facing reporting pipeline (pm_proposal + lp_memo generators)

Milestone: First live trade, fully audited, with EU AI Act + FINRA + OCC compliance package attached. Decision recall ≥0.40 on live forward-OOS (i.e. catching 40%+ of real alpha is the floor; >60% is the goal).

G.7 — The standing rules (recap)

The workbook's standing rules, now operationalized:

1. **Agents are read-only proposers.** Only wallet-executor + rebalancer may move capital.
2. **The gate stack is the judge.** No strategy is promoted without ≥G1-G31 verdict.
3. **The wallet is the actuator.** Every tx is multi-sig, audit-logged, EU-AI-Act-compliant.
4. **The alpha source is NOT the LLM.** The alpha source, if any, is in the strategy the agent orchestrates.
5. **Honesty over impressiveness.** Every claim reduces to a measurable mechanism + a primary URL.

See F.5.4 for the evidence-quality ladder (apply to every vendor / framework / agent claim). See F.5.5 for the reasoning-model latency-cost tax (apply before deploying any LLM in the decision loop). See Part C3 for the cross-cutting infra this operational layer rides on.

This section is the bridge. Research ends here; runtime begins.

Chapter H. Production prompt engineering for quant agents

BOTTOM LINE 13 production techniques (CoT, ReAct, Self-Consistency, CAI, ToT, Reflexion, PRM, RAG, function-calling, chaining, PAL, RLAIF, LLM-as-judge) and 6 worked prompt patterns (data-collector, factor-miner, NCAA-judge, narrative-reader, compliance-briefer, strategy-scribe); the financial domain picks its techniques by what they PREVENT, not by what they enable.

The discipline of telling a financial-ML agent what to do, precisely, safely, and reproducibly. This section translates the workbook's agent templates (G.2) into concrete prompt-programming techniques + worked patterns. It complements the gate-stack (G.1–G.28) by giving the **judge-agents** their inputs.

Tag legend: [STATED] = direct quote from primary source; **[INFERRED]** = derived from public signals; **[PEER-SUPP]** = content from a peer research agent (treated as informational, not as user approval).

H.1 — The 13 production techniques

Each technique has a one-line definition + arXiv / blog primary source. These are the techniques that survived from the 2024-2026 wave of prompt-engineering research.

#	Technique	Definition	Source
1	**Chain-of-Thought (CoT)**	"Think step by step" → forces intermediate reasoning	Wei et al. 2022, arXiv:2201.11903
2	**ReAct**	Reason + Act in interleaved loop (tool use + reflection)	Yao et al. 2022, arXiv:2210.03629
3	**Self-Consistency**	Sample N chains → majority vote	Wang et al. 2022, arXiv:2203.11171
4	**Constitutional AI (CAI)**	Critique-revise against principles	Anthropic 2022, arXiv:2212.08073
5	**Tree-of-Thoughts (ToT)**	Branching search with backtracking	Yao et al. 2023, arXiv:2305.10601
6	**Reflexion**	Verbal self-reflection between attempts	Shinn et al. 2023, arXiv:2303.11381
7	**Process Reward Models (PRM)**	Step-level scoring of reasoning chains	Lightman et al. 2023, arXiv:2305.20050; arXiv:2412.11006
8	**Retrieval-Augmented Generation (RAG)**	Ground in retrieved documents	Lewis et al. 2020, arXiv:2005.11401
9	**Function-calling / Tool-use**	Structured JSON output for tool invocation	OpenAI 2023, Anthropic 2024
10	**Prompt Chaining**	Decompose → sequential LLM calls	LangChain 2022; Anthropic cookbook
11	**Program-Aided Language (PAL)**	Offload computation to Python interpreter	Gao et al. 2022, arXiv:2211.10435
12	**Constitutional + Self-Critique (RLAIF)**	Generate → critique against rules → revise	Anthropic 2023; arXiv:2309.00267
13	**LLM-as-Judge**	One LLM scores another's output	Zheng et al. 2023, arXiv:2306.05685

The standing rule. The financial domain picks its techniques by what they **prevent**, not by what they enable: - **RAG** prevents hallucination of facts (filings, transcripts, prices). - **PAL** prevents arithmetic errors (Sharpe, PBO, slippage math). - **Constitutional AI** prevents policy violations (EU AI Act, FINRA). - **LLM-as-Judge** prevents single-judge bias. - **Self-Consistency** prevents temperature-induced noise. - **Reflexion** prevents repeated mistakes across turns.

H.2 — The 6 worked prompt patterns

Each pattern is a starting point — adapt to your data + agent stack.

H.2.1 — Pattern A: Data-Collector (RAG + PAL)

```
# Pattern A: RAG-grounded data collection + arithmetic-grounded integrity check
SYSTEM_PROMPT = """
You are the data-collector agent. Your mission: assemble a clean, time-aligned
panel for the requested universe and date range.

Constraints:
- Every price/yield must have a timestamp, source, and lineage field.
- Reject any source that fails integrity_check() — never silently fill.
- For every numerical claim, write Python and verify via the Python interpreter.
- Never invent values; refuse rather than fabricate.
- Cite the source URL for every record.

Workflow:
1. Parse the request → {universe, start_date, end_date, fields}
2. For each field, call retrieve_docs(field) for the documentation.
3. Build a Python script that pulls + validates + dedupes + stores.
4. Run integrity_check() and gate the result.
5. Output JSON: {csv_path, audit_trail_path, integrity_score}
"""
```

H.2.2 — Pattern B: Factor-Miner (CoT + Self-Consistency)

```
# Pattern B: Generate N factor candidates, self-consistency vote
SYSTEM_PROMPT = """
You are the factor-miner agent. Your mission: propose 10 candidate factors
for the requested universe, with explicit causal mechanisms.

Constraints:
- Every factor MUST have a non-tautological causal mechanism.
- Every factor MUST be expressed as a vectorized Python function.
- Reject any factor that is a function of only the target (look-ahea
- For each factor, write a step-by-step causal chain (Chain-of-Thoug

Workflow:
1. Read the request (universe, target variable, era).
2. For each of 10 seeds, generate a factor proposal with a causal ch
3. Self-consistency: sample 3 reformulations; majority-vote the mech
4. Score each factor by IC, IC decay, era stability.
5. Output JSON: [{alpha_id, mechanism, code, ic, decay, era_pass}, .
"""
```

H.2.3 — Pattern C: Gate-Judge (LLM-as-Judge + Constitutional)

```
# Pattern C: G22 NCAA judge — LLM-as-judge against a constitutional
SYSTEM_PROMPT = """
You are the NCAA (Narrative Coherence Audit) judge. Your mission: au
quant strategy research memo for narrative coherence + evidence qual

Constitution (you MUST enforce all 7 principles):
1. NON-TAUTOLOGY: The thesis must NOT be "stocks with high X go up b
   they have high X." It must cite a non-trivial causal mechanism.
2. REGIME CONDITIONALITY: The thesis must specify the regime(s) wher
   works and the regime(s) where it fails.
3. NET-OF-COST: All backtest claims must be net of realistic frictic
4. HOLD-OUT: The thesis must report holdout-period performance.
5. KILL-CRITERIA: The thesis must pre-register the conditions under
   it would be killed.
6. CAPACITY: The thesis must specify its capacity limit.
7. EVIDENCE-QUALITY: Every claim must cite a primary source URL or a
   reproducible data path.

Workflow:
1. Read the strategy doc.
2. Score each principle 0-10.
3. For each principle, cite the exact sentence in the doc that you s
4. If any principle < 7, return KILL; else PROMOTE or ITERATE.
5. Output JSON: {verdict, scores: {...}, evidence: {...}, reason}
"""
```

H.2.4 — Pattern D: Narrative-Reader (RAG + CAI)

```
# Pattern D: Filing/transcript extraction — RAG-grounded, CAI-enforc
SYSTEM_PROMPT = """
You are the narrative-reader agent. Your mission: extract thesis fra
from filings, earnings transcripts, and news.

Constraints:
- Every claim MUST cite the document + page/section.
- Every claim MUST include a verbatim quote.
- Reject any claim where the quote is paraphrased rather than verbat
- Reject any claim that contains speculative language ("may", "coulc
  "expects to") — these are forward-looking statements, not facts.

Constitutional principles:
- FACTUALITY: Only extract what is directly stated, not what is impl
- COMPLETENESS: Note "not mentioned" for absent context, never siler
- TIMESTAMP: Every claim includes the document date.

Output JSON: [{claim_id, claim_text, quote, source_doc, page, date},
"""
```

H.2.5 — Pattern E: Compliance-Briefer (RAG + PAL + CAI)

```
# Pattern E: Convert agent-wallet ops into EU AI Act / FINRA / OCC b
SYSTEM_PROMPT = """
You are the compliance-briefer agent. Your mission: convert the ager
operation log into a compliance bundle for EU AI Act Article 12 + FI
2210/3110 + OCC Bulletin 2026-13 + Fed SR 26-02.

Constraints:
- Every field MUST be machine-parseable JSON; no prose.
- Every override event MUST be flagged with reason + human_id.
- For every halt event, compute the affected positions and the time-
- Cite the article / rule for every field.

Constitutional principles:
```

```
- COMPLETENESS: If a field is missing, output "MISSING" — never guess.
- DETERMINISM: The same input MUST produce the same output.
- CHAIN-OF-CUSTODY: Include signature hashes for all source logs.

Output JSON: {ai_act_article_12: [...], finra_2210: [...], occ_2026_13: [...]}
"""
```

H.2.6 — Pattern F: Strategy-Scribe (Reflexion + Self-Consistency)

```
# Pattern F: Convert research output into a PM proposal + LP memo
SYSTEM_PROMPT = """
You are the strategy-scribe agent. Your mission: convert a research memo
+ backtest + verdict into a PM proposal + LP-facing memo.

Workflow:
1. Read the research memo (markdown).
2. Read the backtest output (CSV/Parquet).
3. Read the gate-stack verdict (JSON).
4. Generate a PM proposal (4-8 sections, 1-2 pages).
5. Generate an LP memo (2-4 sections, 1 page, jargon-free).
6. REFLECT: Did I introduce any claim not in the source materials?
   If yes, REWRITE the offending section.
7. Self-consistency: Re-generate the LP memo; flag inconsistencies.
8. Output: {pm_proposal.md, lp_memo.md, reflection_notes.md}
"""
```

H.3 — Common failure modes in production

Failure	Cause	Defense
Hallucinated fact	LLM invents a number / source	RAG + citation requirement + Constitutional
Arithmetic error	LLM mis-computes Sharpe / PBO	PAL (offload to Python interpreter)
Constitutional drift	LLM ignores a principle under long context	Re-state constitution in last user turn
Verbose output	LLM over-explains	Output schema constraint (JSON / function-calling)
Stubborn hallucination	LLM persists even after a critique	Reflexion: explicit "if contradicted, update" loop
Inconsistent verdicts	Same input → different verdicts across runs	Temperature=0 + Self-Consistency voting
Look-ahead bias	LLM uses target to inform features	Pre-register features; let gate-stack catch it

H.4 — The model-routing principle

Not all LLM calls deserve the most expensive model. Match model to task:

Task	Recommended model	Why
Research ideation	Claude Sonnet 4.6 (ext. thinking)	Long-context, nuanced, high-quality ideation
Data extraction	Claude Haiku 4.5 or GPT-4o-mini	Fast, cheap, structured output
Backtest code generation	Claude Sonnet 4.6 (ext. thinking)	Reasoning + code quality both matter
LLM-as-judge (G22 NCAA)	Claude Sonnet 4.6 (ext. thinking)	Audit-grade consistency
Compliance-briefer	Claude Sonnet 4.6 (no thinking)	Determinism preferred over nuance
LP memo draft	Claude Opus 4.8	Highest-quality prose, but only for the final draft

The cost rule. Most LLM calls in a research pipeline are NOT the most expensive model. A typical research session: 200 calls total, of which 5 are Opus, 50 are Sonnet-extended-thinking, 145 are Haiku. This is the **3-tier** model routing pattern that Man Group and Bridgewater have confirmed.

The latency rule. Extended-thinking models add **30-120s per call** and **10-30× the cost** of non-thinking models. Use them only for the highest-stakes decisions (verdict gates, code generation, ideation). Data extraction and structured output do not benefit from extended thinking.

H.5 — Connection to the gate stack

The **gate-stack judge-agents (G22, G23, G24, G26, G29) are LLM-as-judge calls**. The prompt patterns in H.2.3 + H.2.5 are the canonical templates. Every judge-agent MUST: - Cite the article / gate-id it is enforcing - Score 0-10 on each principle - Output machine-parseable JSON (not prose) - Log to the Article 12 audit log (per F.7.4) - Co-sign with a deterministic gate (G1, G2, G3, G4, G9) when possible

The LLM-as-judge is a **safety net**, not the primary judge. **The numeric gates (G1-G9) are the load-bearing defense.** LLM judges catch what numeric gates miss (narrative coherence, constitutional compliance, novel-premise, evidence quality). Numeric gates catch what LLM judges miss (overfitting, multiple-testing bias, slippage, regime shift).

The standing rule: every strategy verdict requires BOTH a numeric gate stack verdict AND an LLM-judge verdict. Mismatch → human review.

See G.4 for multi-agent evaluation metrics. See F.5.5 for reasoning-model latency-cost tradeoffs. See F.7 for the EU AI Act Article 12 logging requirements.

Chapter I. Reference: entity index + decision tree + architecture diagrams

BOTTOM LINE The quick-lookup material — 8 entity sub-indices (wallet vendors, agent frameworks, data vendors, institutional firms, retail platforms, cloud/infra, regulators, academic canon), 2 decision trees (build-vs-buy + AI-claim evidence-quality), 4 ASCII architecture diagrams, gate-stack verdict thresholds, and cost quick-reference (Y0 $200/mo → Y2+ $50K/mo → institutional $1-5M/yr).

Quick-reference material for the principal. This section consolidates the workbook's long-form research into lookup tables + decision trees + ASCII diagrams. It is **not new content** — it is a re-organization of the existing material into a one-page-per-decision form.

I.1 — Entity index (vendor / framework / firm / regulator)

Sortable by category, with the section reference for each entity.

I.1.1 — Wallet + MPC vendors (see C4.2)

Vendor	Section	Tag	Cost band	Best for
Coinbase AgentKit / CDP	C4.2.1	**[STATED]**	Usage-based	Base-native, Y1 first AUM
Crossmint	C4.2.2	**[STATED]**	Usage-based	Multi-agent + fiat ramp
Skyfire	C4.2.3	**[STATED]**	Usage-based	KYA tokens, web checkout
Nevermined	C4.2.4	**[STATED]**	Usage-based	x402 + Stripe/PayPal
Privy	C4.2.5	**[STATED]**	Free tier	Y0 learn + POC
Dynamic (acq. Fireblocks 10/2025)	C4.2.6	**[STATED]**	Usage-based	Multi-vendor / resilience
Turnkey	C4.2.7	**[STATED]**	Usage-based	Y2 compliance-grade

I.1.2 — Agent frameworks (see C4.3)

Framework	Section	Tag	Best for
ElizaOS (ai16z)	C4.3.1	**[STATED]**	Full agent OS
Truth Terminal	C4.3.2	**[STATED]**	Memetic launch (not alpha)
Virtuals Protocol (Luna, aixbt)	C4.3.3	**[INFERRED]**	Social-token meta
Wayfinder (Parallel)	C4.3.4	**[INFERRED]**	Base-specific routing
Brian (fetch.ai)	C4.3.5	**[INFERRED]**	Cross-chain DeFi
Colony / Bounty-on-Base	C4.3.6	**[INFERRED]**	Agent-to-agent escrow

I.1.3 — Data vendors (see D.3, C3.6)

Vendor	Section	Tag	Cost band	Best for
EODHD	G.1.1	**[STATED]**	$20/mo (free tier)	Y0 hobbyist
Alpha Vantage	G.1.1	**[INFERRED]**	Free / $50/mo	Y0 hobbyist
Tiingo	G.1.1	**[INFERRED]**	$10-$50/mo	Y0 hobbyist

Vendor	Section	Tag	Cost band	Best for
Polygon.io / Massive.com	G.1.1	**[INFERRED]**	$29-$99/mo	US real-time
Databento	G.1.1	**[INFERRED]**	Free $125 → $200/mo	Tick + 1m
Intrinio AI-Ready	G.1.1	**[STATED]**	$150/mo	MCP server
AlphaSense	G.1.2 + C3.6	**[STATED]**	$20K-$100K+/yr	Premium NLP
RavenPack Bigdata.com	G.1.2 + C3.6	**[STATED]**	$25K-$100K+/yr	12M entities
Kensho (S&P Global)	G.1.2 + C3.6	**[STATED]**	$50K-$500K/yr	LLM-ready API
KDB.AI	G.1.2 + C3.2	**[STATED]**	$50K-$500K/yr	Vector + time-series
Bloomberg Terminal	G.1.3	**[STATED]**	$24K-$28K/seat/yr	Best-in-class data
FactSet	G.1.3	**[STATED]**	$15K-$25K/seat/yr	Fundamentals
S&P Capital IQ Pro	G.1.3	**[STATED]**	$15K-$25K/seat/yr	Fundamentals + filings

I.1.4 — Institutional firms (see C1)

Firm	Section	Tag	Notable
Renaissance Technologies	C1.1	**[INFERRED]**	150 researchers, closed
Two Sigma	C1.2	**[STATED]**	Venn → Insight Partners (Jan 2026)
D. E. Shaw	C1.3	**[STATED]**	PyTorch/JAX/Triton ML JD $275-350K
Citadel	C1.4	**[STATED]**	$67B AUM (Jan 2026), AI Assistant
Point72 / Cubist	C1.5	**[STATED]**	$45.7B AUM, Turion $1.5B
Bridgewater	C1.6	**[STATED]**	$2B ML-fund, AIA Forecaster
Balyasny	C1.7	**[STATED]**	Charlie Flanagan CAIO (2026)
Millennium	C1.8	**[STATED]**	37 AI + 28 ML tagged roles
AQR	C1.9	**[STATED]**	Kelly/Xiu/Malamud paper series
Man Group	C1.10	**[STATED]**	Anthropic partnership Feb 2026, AlphaGPT
Winton	C1.11	**[INFERRED]**	"100+ academics, 0 AI/ML on homepage"
HRT	C1.12	**[INFERRED]**	$1B/yr AI spend (LOW CONFIDENCE)
PDT / Schonfeld	C1.13	**[STATED]**	4 explicit AI-titled roles
Squarepoint	C1.14	**[STATED]**	$226.9B AUM, 0 public AI mention
WorldQuant	C1.15	**[STATED]**	4M alphas in Alpha Factory
BlackRock Aladdin	C1.16	**[STATED]**	LLM Suite
JPM COIN	C1.17	**[STATED]**	LLM Suite + IndexGPT

I.1.5 — Retail / indie platforms (see C2)

Platform	Section	Tag	Best for
QuantConnect	C2.1	**[STATED]**	515K quants, Lean engine
WorldQuant BRAIN	C2.2	**[STATED]**	550K+ users, 4M alphas
Numerai	C2.3	**[STATED]**	Erasure + NMR + meta-model
Composer	C2.4	**[STATED]**	$32/mo, SoFi-owned
Trade Ideas	C2.5	**[STATED]**	$178/mo Premium
AIEQ (EquBot)	C2.6	**[STATED]**	8.5yr track: +10.02% ann. (underperforms SPY)
Zipline / Lean / Backtrader	C2.7	**[STATED]**	OSS backtesters
Capitalise.ai (acq. Kraken)	C2.8	**[STATED]**	Plain-English automation

I.1.6 — Cloud + infra (see C3)

Vendor	Section	Tag	Best for
CoreWeave	C3.1	**[STATED]**	~250K H100/H200/Blackwell, Jane Street customer
Optiver (Slurm + GraphRunner)	C3.1	**[STATED]**	Rob Hahn, Dec 2025
Man/OMI (Graphcore IPU)	C3.1	**[INFERRED]**	Academic partnership
Confluent (Kafka 4.3)	C3.3	**[STATED]**	"Compliant AI Agents" Jun 2026
AMD-Xilinx FPGA	C3.4	**[STATED]**	Optiver/HRT/Exablaze low-latency
Databento (Napatech NT200A02)	C3.4	**[INFERRED]**	100G NIC for tick capture
Modal	G.1.2	**[INFERRED]**	Serverless GPU
W&B	G.1.2	**[STATED]**	ML platform (Free → Enterprise)

I.1.7 — Regulators + frameworks (see F.2)

Body	Section	Tag	Latest
SEC Marketing Rule (IA-5653)	F.2.1	**[STATED]**	Effective 2021-05-04
SEC Predictive AI (IA-6176)	F.2.2	**[STATED]**	Still PROPOSED (dormant under Atkins)
FINRA Notice 24-09	F.2.3	**[STATED]**	AI/ML compliance
EU AI Act (Reg 2024/1689)	F.2.4 + F.7	**[STATED]**	Annex III 2027-12-02 (post-Digital Omnibus)
OCC Bulletin 2026-13	F.2.5	**[STATED]**	2026-04-17, first MRM refresh since 2011
Fed SR 26-02	F.2.5	**[STATED]**	2026-04-17, joint with OCC
FCA FS24/2	F.2.6	**[STATED]**	April 2024
CFTC (no AI rule)	F.2.7	**[STATED]**	Existing CEA §4b + Reg 180.1
FINRA June 2020 AI report	F.2.8	**[STATED]**	Baseline reference
Digital Omnibus 2026-05-07	F.2.4 + F.7	**[STATED]**	Political agreement, Annex III shift

I.1.8 — Academic + skeptic canon (see F.5)

Paper	Section	Tag	Role
FINSABER (KDD 2026 D&B Oral)	F.5.5	**[STATED]**	arXiv:2505.07078 — LLM traders don't beat baselines
Alpha Illusion	F.5.0	**[STATED]**	arXiv:2605.16895 — P1-P6 reporting protocol
MAST	F.5.0 + G.5	**[STATED]**	arXiv:2503.13657 — 14 failure modes
FINSABER companion (LLM agents)	F.5.5	**[STATED]**	arXiv:2502.15800 — "Do Not Replicate Human Market Traders"
Process Reward Models	F.5.1	**[STATED]**	arXiv:2412.11006
Conformal prediction (Gibbs/Candès)	F.5.1	**[STATED]**	JASA 2024
Neural-SDEs (Cohen)	F.5.1	**[STATED]**	arXiv:2105.11053
Time-series foundation models	F.5.1	**[STATED]**	PatchTST, iTransformer, Chronos, TimesFM, Moirai-MoE

I.2 — Decision tree: "Should I buy or build X?"

```
START
  │
  ├─ Question 1: Is X a commodity (data, infra, compliance) or a moa
  │    │
  │    ├─ Commodity → Question 2
  │    └─ Moat      → Build (you cannot buy alpha; you can only buy i
  │
  ├─ Question 2: Is X on a 12-month price-decline curve (LLM APIs, v
  │    │
  │    ├─ Yes → Buy the cheapest viable, refactor every 6 months
  │    └─ No  → Question 3
  │
  ├─ Question 3: Is X a regulatory surface (wallet ops, audit log, m
  │    │
  │    ├─ Yes → Buy from a compliance-grade vendor (Turnkey, Fireblo
  │    └─ No  → Question 4
  │
  ├─ Question 4: Is X core to your thesis (factor-mining, gate-stack
  │    │
  │    ├─ Yes → Build (with maximum canonicalization)
  │    └─ No  → Question 5
  │
  └─ Question 5: Is there an OSS implementation with >1K GitHub star
       │
       ├─ Yes → Use OSS; contribute back
       └─ No  → Buy (you will not beat a commercial team in 6 months)
```

The 100× principle. The cheapest stack covers 70% of the value; the best-in-class stack covers 99%. The 30× cost difference covers 30% more value, not 30× more. **Choose the cheapest stack that you can sleep on.** See D.3.2 + D.3.3 in `09_build_vs_buy_matrix.md`.

I.3 — Decision tree: "Is this AI claim marketing or real?"

```
START
  │
  ├─ Rung 0: "We use AI" + no metric  → MARKETING (kill)
  │   (Example: AIEQ homepage copy without performance disclosure)
  │
  ├─ Rung 1: Job listing implies AI/ML work  → HIRING SIGNAL
  │   (Example: PDT "Applied ML Scientist" role)
  │
  ├─ Rung 2: Public engineering write-up  → TECHNICAL SIGNAL
  │   (Example: Optiver "Engineering the Agentic SDLC")
  │
  ├─ Rung 3: Peer-reviewed paper + code  → ACADEMIC SIGNAL
  │   (Example: AQR "Virtue of Complexity" JoF 2024)
  │
  └─ Rung 4: Backtested live result, net-of-cost  → DEPLOYABLE
      (Example: ??? — no major fund publicly discloses rung-4 results as of 2026-07-07)
```

The 4-rung ladder is the **evidence-quality filter** in F.5.4. Apply it to every claim before adopting it.

I.4 — Architecture diagrams

I.4.1 — Two-stage architecture (research → gate → decision)

```
AI agents (read-only)
      │
      │  propose / factor-mine / write code
      ▼
Gate stack (G1–G28 + G29–G31)
      │
      │  numeric anchors: DSR, PBO, min-BTL
      ▼
Decision (KILL / ITERATE / PROMOTE)
```

The canonical 4-line ASCII figure. **Same diagram appears in B1, F.5, E.1, and the closing block.** This is the load-bearing abstraction of the workbook.

I.4.2 — Multi-agent topology (Macion Capital)

```
                ┌──────────────────┐
                │ LP / PM / Counsel │   ←── human approval
                └──────────────────┘
                          │ mandate
                          ▼
┌─────────┐  ┌─────────┐  ┌─────────┐
│ data-   │  │ factor- │  │narrative│   ←── research agents (propose)
│collector│  │ miner   │  │ reader  │
└─────────┘  └─────────┘  └─────────┘
     │            │            │
     └────────────┼────────────┘
                  │ proposal
                  ▼
        ┌───────────────┐
        │  gate-stack   │   ←── judge agents (numeric + LLM)
        │ G1–G28 + G29  │
        └───────────────┘
                  │ verdict
                  ▼
        ┌───────────────┐
        │strategy-scribe│   ←── reporter (PM proposal + LP memo)
        └───────────────┘
                  │ decision log
                  ▼
        ┌───────────────┐
        │ wallet-       │   ←── only agent allowed to move capital
        │ executor      │
        └───────────────┘
                  │ signed tx
                  ▼
        ┌───────────────┐
        │ Article 12    │   ←── tamper-evident, 6mo+ retention
        │ audit log     │
        └───────────────┘
```

I.4.3 — AI-wallet-agent stack (C4)

See C4.7 for the agent → wallet → compliance stack.

I.4.4 — Cross-cutting infra (C3)

```
┌──────────────────┐
│ Vendor (Coinbase)│   ←── Application
│ / Open-source    │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ Vector + Time    │   ←── Data layer (KDB.AI, Pinecone, Chroma)
│ Series DB        │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ GPU + Kafka +    │   ←── Infra layer (CoreWeave, Confluent)
│ Flink + FPGA     │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ AWS / Azure /    │   ←── Cloud layer
│ GCP / On-prem    │
└──────────────────┘
```

I.5 — Quick reference: gate-stack verdict thresholds

Gate	Threshold	Kill if	Reference
G1 DSR	p-value	> 0.05	Bailey & López de Prado 2014
G2 PBO	probability	> 0.50	Bailey, Borwein, López de Prado 2015
G3 min-BTL	observations	< 1.5× min_btl	Bailey & López de Prado 2012
G4 era stability	M-of-N eras pass	< 60%	Macion Capital gate stack
G9 net Sharpe	net-of-cost Sharpe	< 0.50	Macion Capital gate stack
G10 max DD	max drawdown	> 25%	Macion Capital gate stack
G15 Marketing Rule	advisor advertising	non-compliant	SEC IA-5653
G22 NCAA	LLM-as-judge score	any principle < 7	F.5.4 + H.2.3
G23 EU AI Act Art. 12	logging completeness	< 100%	F.7.4
G29 venue lock	Topstep-ON	fails on Topstep	Macion Capital standing rule (`CLAUDE.md` §6)

I.6 — Cost quick-reference

Tier	Monthly cost (USD)	What you get
Y0 hobbyist	$70-$200	EODHD + Tiingo + Alpha Vantage + W&B + Polygon Starter + free OSS backtester
Y1 first AUM	$1,000-$3,000	+ Databento $200 + Intrinio $150 + KDB.AI dev $500 + Anthropic Sonnet $1,000 + OSS QuantConnect Team
Y2 scale	$5,000-$10,000	+ AlphaSense $1,500 + RavenPack $1,500 + Anthropic for FS $1,000 + MLflow $500
Y2+ compliance-grade	$15,000-$50,000	+ Bloomberg 2 seats $5K + FactSet 1 seat $1.5K + Turnkey $1K + Fireblocks $5K + W&B Enterprise $1K
Institutional (Renaissance tier)	$1M-$5M/yr	250K GPU cluster + KDB-X + Confluent + Bloomberg 50 seats + FactSet 30 seats + KDB.AI Enterprise + compliance team

Note. All amounts USD/month unless noted. Multi-seat / annual contracts discount 20-40%. The Y0 tier is the cheapest viable; the Y2+ tier is the best-in-class (per D.3.2 vs D.3.3). See `10_cost_model_phased_roadmap.md` for the full cost model.

I.7 — Where to look for what

If you want to know...	Go to
What AI in hedge funds actually looks like	C1 (institutional) + C2 (retail/indie)
What infra is shared across firms	C3 (cross-cutting infra)
What's wrong with AI vendor claims	F.5.4 (evidence-quality ladder)
Whether reasoning models are useful for finance	F.5.5 (reasoning models)
Whether AI agents can hold wallets	C4 (AI wallet agents)
What regulators care about	F.2 (regulatory landscape) + F.7 (EU AI Act runbook)
What to build for Macion Capital	G (Macion Capital operational layer)
How to write agent prompts	H (prompt engineering)
What the cheapest viable stack is	D.3.2 + I.6
What the best-in-class stack is	D.3.3 + I.6
What 14 failure modes to defend against	G.5 (MAST defense map)
What the F.5 frontier looks like	F.5 (frontier techniques)

I.8 — The standing rules (workbook-wide recap)

1. **Agents are read-only proposers.** Only wallet-executor + rebalancer may move capital.
2. **The gate stack is the judge.** No strategy is promoted without G1-G31 verdict.
3. **The wallet is the actuator.** Every tx is multi-sig, audit-logged, EU-AI-Act-compliant.
4. **The alpha source is NOT the LLM.** The alpha source, if any, is in the strategy the agent orchestrates.
5. **Honesty over impressiveness.** Every claim reduces to a measurable mechanism + a primary URL.
6. **Vendor claims need evidence-quality rungs.** Marketing (rung 0) ≠ hiring (rung 1) ≠ engineering (rung 2) ≠ academic (rung 3) ≠ backtested live (rung 4).
7. **Cost discipline is a feature.** Cheapest stack that you can sleep on. The 100× principle.
8. **Compliance is a technical discipline, not a checkbox.** Article 12 logs, Annex IV files, multi-sig policies — all from day 1.
9. **De-select before promoting.** First-pass positives get a non-overlapping holdout re-test.
10. **Net-of-cost edge.** Backtests must include realistic friction. Never report gross.

This index + decision tree + architecture reference is meant to be the first thing the principal looks at on a given day. The full material is the chapters; the index is the lookup.

Part Appendix. References & Bibliography

Every claim in this document should trace to a source here. Dates retrieved: 2026-07-07 unless otherwise noted. The bibliography is maintained in `12_references_bibliography.md` and updated on every revision.

Every claim in this document should trace to a source here. Dates retrieved: 2026-07-07 unless otherwise noted.

Academic papers

Paper	URL	Notes
Anthropic — Multi-agent research system	https://www.anthropic.com/research/multi-agent-research-system	The +90.2% / ~15× tokens benchmark
MAST: Multi-Agent System Failure Taxonomy	https://arxiv.org/abs/2503.13657	NeurIPS 2025; 41–86.7% failure rate
FINSABER	https://arxiv.org/abs/2411.02337	Stanford 2024; LLM-as-trader null result
Alpha Illusion critique	https://arxiv.org/abs/2605.16895	P1–P6 reporting protocol for LLM-trading benchmarks; flags temporal contamination, narrative fitting, short-window Sharpe
FinMem (LLM trading agent, layered memory)	https://arxiv.org/abs/2311.13743	Yu et al. 2023
FinCon (LLM multi-agent, conceptual verbal RL)	https://arxiv.org/abs/2407.06567	Yu/Yao/Li et al. 2024
FinRL-Meta (market environments + benchmarks)	https://arxiv.org/abs/2211.03107	NeurIPS 2022 Datasets & Benchmarks
FinGPT (open-source financial LLM)	https://arxiv.org/abs/2306.06031	Yang/Liu/Wang; rev. Nov 2025
PIXIU / FinMA	https://arxiv.org/abs/2306.05443	Xie et al. 2023
CFGPT (Chinese financial LLM)	https://arxiv.org/abs/2309.10654	Li et al. 2023
InvestLM	https://arxiv.org/abs/2309.13064	Yang/Tang/Tam 2023
Double/Debiased ML (Chernozhukov)	https://arxiv.org/abs/1707.01847	AER 2018; Python pkg at docs.doubleml.org
Causal Forest (Wager & Athey)	https://arxiv.org/abs/1510.04342	JASA 2018
Conformal prediction (Gibbs/Candès)	https://arxiv.org/abs/2208.08401	JASA 2024; applied to stock-vol
Conformalized Quantile Regression	https://arxiv.org/abs/1905.03222	Romano et al. 2019
Conformal Predictive Portfolio Selection	https://arxiv.org/abs/2410.16333	Kato 2024
Neural-SDE market models (arbitrage-free)	https://arxiv.org/abs/2105.11053	Cohen/Reisinger/Wang 2021
SigFormer (signature transformers for deep hedging)	https://arxiv.org/abs/2310.13369	ICAIF 2023
PatchTST (time-series transformer)	https://arxiv.org/abs/2211.14730	ICLR 2023
iTransformer (inverted transformers)	https://arxiv.org/abs/2310.06625	ICLR 2024
Chronos (time-series foundation)	https://arxiv.org/abs/2403.07815	Amazon AWS AI Labs
TimesFM (Google time-series foundation)	https://arxiv.org/abs/2310.10688	Das et al. 2023
TimeGPT-1 (Nixtla)	https://arxiv.org/abs/2310.03589	Garza et al. 2023
MOMENT (open time-series foundation)	https://arxiv.org/abs/2402.03885	ICML 2024
Moirai-MoE (time-series MoE)	https://arxiv.org/abs/2410.10469	Salesforce 2024
DeepLOB (LOB CNN)	https://arxiv.org/abs/1808.03668	IEEE TSP 2019
Concept Bottleneck Models (TCAV precursor)	https://arxiv.org/abs/2007.04612	ICML 2020
Counterfactual Fairness	https://arxiv.org/abs/1703.06856	Kusner et al. NeurIPS 20
Counterfactual Explanations (GDPR)	https://arxiv.org/abs/1711.00399	Wachter/Mittelstadt/Rus 2017
Process Reward Model (entropy-regularized)	https://arxiv.org/abs/2412.11006	2024
Anthropic system cards	https://www.anthropic.com/system-cards	Claude 3 (Mar 2024) through Sonnet 5 (Jun 2026)
MAST: Multi-Agent System Failure Taxonomy	https://arxiv.org/abs/2503.13657	NeurIPS 2025; 41–86.7% failure rate
DSPy: Compiling Declarative Language Model Calls	https://arxiv.org/abs/2310.03714	Khattab et al. 2024
Reflexion: Language Agents with Verbal Reinforcement Learning	https://arxiv.org/abs/2303.11381	Shinn et al. NeurIPS 202
ReAct: Synergizing Reasoning and Acting in Language Models	https://arxiv.org/abs/2210.03629	Yao et al. 2022
Tree of Thoughts	https://arxiv.org/abs/2305.10601	Yao et al. 2023
Constitutional AI	https://arxiv.org/abs/2212.08073	Anthropic 2022
Self-RAG	https://arxiv.org/abs/2310.11511	Asai et al. 2023

Paper	URL	Notes
CRAG: Corrective Retrieval Augmented Generation	https://arxiv.org/abs/2401.15884	Yan et al. 2024
Advances in Financial Machine Learning	Marcos López de Prado, 2018	CPCV, triple-barrier, meta-labeling
The Deflated Sharpe Ratio	Bailey & López de Prado, JPM 2014	DSR
Probability of Backtest Overfitting	Bailey, Borwein, López de Prado, Zhu, JPM 2015	CSCV-based PBO
TimeGAN	Yoon et al. NeurIPS 2019	Time-series synthesis
QuantGAN	Wiese et al. 2020	Financial GAN
BloombergGPT	https://arxiv.org/abs/2303.17564	50B params, 363B Bloomberg tokens + 345B general; Wu et al. 2023
FinGPT	https://github.com/AI4Finance-Foundation/FinGPT	Open-source financial LLM
Generative Agents: Interactive Simulacra of Human Behavior	Park et al. 2023	Stanford

Vendor / product documentation

Vendor	URL	Notes
QuantConnect	https://www.quantconnect.com/pricing	Lean engine; cloud quant; 515K quants (2026-07-07); Ask Mia + Agentic AI Assistants
WorldQuant BRAIN	https://www.worldquantbrain.com	Crowdsourced alpha; 550K+ users / 16K+ consultants / 400K+ data fields
Numerai	https://numer.ai	AI hedge fund tournament; NMR staking
Databento	https://databento.com/pricing	Market data; $125 free credit; L3/MBO historical at retail prices
Polygon.io / Massive.com	https://polygon.io → https://massive.com	US stocks, options, crypto; rebrand 2025/26
AlphaSense	https://www.alpha-sense.com	Enterprise search over filings; 7,000+ enterprise / 85%+ S&P 100 / 70% top 50 HFs (2026-07-07); Tegus 2024 ~$660M
Hebbia	https://hebbia.com	Matrix UI for LLM on documents
RavenPack / Bigdata.com	https://www.ravenpack.com	News NLP; Bigdata.com = "AI agents for finance"; FT content integration
Kensho (S&P Global)	https://www.kensho.com	LLM-ready API; Anthropic/OpenAI/Google/Snowflake/AWS/Cohere/Databricks/LangChain partners; $550M 2018 acquisition
LSEG Workspace AI	https://www.lseg.com/en/products/workspace	"Deep Research" agent; Microsoft strategic AI/data-cloud partnership
Bloomberg Terminal + BloombergGPT	https://www.bloomberg.com/professional	$24K–$28K/seat/yr; BloombergGPT = 50B params / 363B Bloomberg + 345B general tokens
SigTech	https://www.sigtech.com	Backtesting + signals
OpenBB	https://openbb.co	Open-source Bloomberg alternative; 70.2K stars (2026-07-07)
Lean (QuantConnect engine)	https://github.com/QuantConnect/Lean	Open-source backtester; 20.4K stars, 13,246 commits
Vectorbt	https://github.com/polakowo/vectorbt	Vectorized backtesting; 8.2K stars; v1.1.0 (2026-07-05)
FinRL	https://github.com/AI4Finance-Foundation/FinRL	RL for finance; 15.6K stars
Qlib	https://github.com/microsoft/qlib	AI-first quant platform
MLFinLab	https://github.com/hudson-and-thames/mlfinlab	ML tools for finance; 4.9K stars; paid license
Freqtrade	https://github.com/freqtrade/freqtrade	Crypto bot; 52.1K stars; FreqAI ML extension
Jesse	https://github.com/jesse-ai/jesse	Crypto event-driven; JesseGPT LLM-native authoring
Nautilus Trader	https://github.com/nautechsystems/nautilus_trader	Rust-native deterministic OMS; ~24.5K stars (2026 fastest-rising)
PyBroker	https://github.com/edtechre/pybroker	ML-driven strategies; walkforward + bootstrap
Intrinio	https://intrinio.com/pricing	AI-Ready $150/mo; GenAI + MCP integration
EODHD	https://eodhd.com/pricing	MCP Server with 72 tools; OpenAPI 3.1; AI agent skills
Hugging Face	https://huggingface.co	Model hub; FinBERT ProsusAI/finbert-tone

Vendor	URL	
Weights & Biases	https://wandb.ai	Experiment tracki
MLflow	https://mlflow.org	Experiment tracki
DVC	https://dvc.org	Data versioning
BentoML	https://github.com/bentoml/BentoML	Model serving
Feast	https://feast.dev	Feature store
Prefect	https://prefect.io	Pipeline orchestra
Dagster	https://dagster.io	Pipeline orchestra
Modal	https://modal.com	Serverless GPU
Pinecone	https://www.pinecone.io	Vector DB
Qdrant	https://qdrant.tech	Vector DB
Weaviate	https://weaviate.io	Vector DB
Chroma	https://www.trychroma.com	Vector DB
Composer.trade	https://www.composer.trade	No-code AI portfo
Trade Ideas	https://www.trade-ideas.com	Holly AI; Premiun
Kavout	https://kavout.com/pricing-plans	AI stock ranking; l
Benzinga Pro	https://benzinga.com/pro/pricing	Benzinga AI Essen
Capitalise.ai	https://www.capitalise.ai	NL-to-algo; acquir
FinChat	https://finchat.io	Finance LLM chat
AIEQ	https://stockanalysis.com/etf/aieq/	AI ETF (EquBot / SPY ~12–14%
IBKR	https://www.interactivebrokers.com	Retail broker; TWS
TradeStation	https://www.tradestation.com	Retail broker; "Ins
Binance / Bybit / Coinbase	—	Crypto exchanges;

Claude / Anthropic

Source	URL	Notes
Claude docs — multi-agent research	https://platform.claude.com/docs/en/docs/agents/multi-agent-research	Multi-agent research system overview
Claude Code	https://docs.claude.com/en/docs/claude-code	CLI agent
MCP	https://modelcontextprotocol.io	Model Context Protocol
Anthropic engineering blog	https://www.anthropic.com/engineering	Engineering posts
Anthropic research	https://www.anthropic.com/research	Research papers

Firm engineering blogs / careers pages

Firm	URL	Notes
Renaissance	(no public engineering blog)	Careers: renfund.com
Two Sigma	https://www.twosigma.com/articles	Engineering + research posts
D. E. Shaw Research	https://www.deshawresearch.com	Science publications
Citadel	https://www.citadel.com/careers	Job listings
Point72	https://www.point72.com/careers	Job listings
Bridgewater AIA Labs	https://www.bridgewater.com/ai-labs	Public blog
AQR	https://www.aqr.com/Research	Academic-style research
Man Group	https://www.man.com/insights	Research papers
Hudson River Trading	https://www.hudsonrivertrading.com/blog	Engineering posts
WorldQuant BRAIN	https://www.worldquantbrain.com	Platform
BlackRock	https://www.blackrock.com/corporate/insights	Industry insights
JPMorgan	https://www.jpmorgan.com/technology/ai	AI research blog

Regulatory primary text

Source	URL	Notes
SEC Marketing Rule (IA-5653)	https://www.sec.gov/rules/final/2020/ia-5653.pdf	17 CFR 275.206(4)-1; effective 2021-05-04; captures AI-generated marketing
SEC Predictive AI proposal (IA-6176, 33-11210)	https://www.sec.gov/rules/proposed/2023/ia-6176.pdf	2023-07-31; **STILL PROPOSED / dormant under Atkins** as of 2026-07-07
SEC Predictive AI proposal (Release 34-97990)	https://www.sec.gov/rules/proposed/2023/34-97990.pdf	Companion broker-dealer release
SEC "AI Washing" Risk Alert (2024)	https://www.sec.gov/news/press-release/2023-149	Staff statement on misleading AI claims
FINRA June 2020 AI report	https://www.finra.org/sites/default/files/2020-06/ai-in-the-securities-industry.pdf	Pre-Notice 24-09 baseline
FINRA AI topic index	https://www.finra.org/rules-guidance/notices/by-topic?term=167031	Index page for FINRA AI notices
FINRA Regulatory Notice 24-09	https://www.finra.org/rules-guidance/notices/24-09	2024-06-27; LLM/GenAI obligations
FINRA Notice 24-09 (PDF)	https://www.finra.org/sites/default/default/files/2024-06/regulatory-notice-24-09.pdf	Direct PDF; metadata confirms 2024-06-27
CFTC PressRoom / Speeches	https://www.cftc.gov/PressRoom/SpeechesTestimony	No standalone AI rule 2023–2026
CFTC Innovation Hub	https://www.cftc.gov/About/Innovation	Industry engagement without binding rulemaking
EU AI Act (Regulation EU 2024/1689)	https://eur-lex.europa.eu/eli/reg/2024/1689/oj	Primary source; OJ L 2024/1689, 12 July 2024
EU AI Act plain-English Article 13	https://artificialintelligenceact.eu/article/13/	Transparency obligations summary
EU AI Office	https://digital-strategy.ec.europa.eu/en/policies/ai-office	Implementing body
FCA FS24/2 (AI Update)	https://www.fca.org.uk/publications/feedback-statements/fs24-2	April 2024; correct citation (not "FS23/2")
FCA AI Lab	https://www.fca.org.uk/initiatives/ai-lab	Industry sandbox 2024–26
OCC Bulletin 2026-13 (MRM refresh)	https://www.occ.gov/news-issuances/bulletins/2026/bulletin-2026-13.html	April 17, 2026; first MRM refresh addressing AI/ML since 2011
Federal Reserve SR 26-02 (MRM refresh)	https://www.federalreserve.gov/supervisionreg/srletters/SR2602.htm	April 17, 2026; companion to OCC
Federal Reserve SR 11-7 (MRM 2011)	https://www.federalreserve.gov/boarddocs/srletters/2011/sr1107a1.pdf	Origi guida supe SR 2 AI/M
OCC Bulletin 2011-12 (rescinded)	https://www.occ.gov/static/rescinded-bulletins/bulletin-2011-12.pdf	Resc 2026
MAS FEAT principles	https://www.mas.gov.sg/publications/consultations/2018-feat-principles	Fairn Ethic Acco Tran
PH SEC	https://www.sec.gov.ph	PH s regul
BIR	https://www.bir.gov.ph	PH t
NPC (PH privacy)	https://www.privacy.gov.ph	Data Act

Industry reports

Source	URL	Notes
AIMA DDQ	https://aima.org	Institutional standards
ILPA DDQ	https://ilpa.org	LP-side standards
GIPS	https://www.gipsstandards.org	Performance standards
McKinsey AI survey	https://www.mckinsey.com/capabilities/quantumblack/our-insights	Annual state of AI
BCG AI report	https://www.bcg.com/publications/2023/ai-adoption	Annual
CFA Institute	https://www.cfainstitute.org	Industry standards

Books

- **Marcos López de Prado**, *Advances in Financial Machine Learning* (2018)
- **Marcos López de Prado**, *The 7 Reasons Most Machine Learning Investments Fail* (2022)
- **Marcos López de Prado**, *Machine Learning for Asset Managers* (2020)
- **Ernie Chan**, *Algorithmic Trading* (2013) + *Machine Trading* (2017)
- **Andreas Clenow**, *Following the Trend* (2013) + *Stocks on the Move* (2015)
- **Michael Lewis**, *The Man Who Solved the Market* (2019, on Renaissance)
- **Sebastian Mallaby**, *The More or Less Definite History of the World's Most Famous Hedge Fund* (on Long-Term Capital)
- **Gregory Zuckerman**, *The Man Who Solved the Market* (2019)

Internal Macion Capital assets cited

The following Macion-Capital-OWN documents are the canonical sources for every workspace-level claim in this workbook. Every path below resolves at the Macion Capital tree root (`/Users/christianmacion/Macion Capital/`) unless prefixed with ~ (workspace-skill location).

Asset	Path
Macion Capital CLAUDE.md	`/Users/christianmacion/Macion Capital/CLAUD`
Macion Capital MODEL_ROUTING.md	`/Users/christianmacion/Macion Capital/MODEL_`

Asset	Path	Notes
Macion Capital OPERATING_MODEL.md	`/Users/christianmacion/Macion Capital/OPERATING_MODEL.md`	Operating model: session hygiene, kill patterns, "show, don't tell" rule, chart-pipeline contract
Macion Capital DATA_REGISTRY.md	`/Users/christianmacion/Macion Capital/DATA_REGISTRY.md`	Data inventory with provenance + staleness checks per file
Macion Capital STRATEGY_REGISTRY.md	`/Users/christianmacion/Macion Capital/STRATEGY_REGISTRY.md`	Active strategies (the canonical registry the gate-stack verdicts write into)
Macion Capital KILL_LOG.csv	`/Users/christianmacion/Macion Capital/KILL_LOG.csv`	KILL verdicts with reason codes (one of the four named kill patterns)
Macion Capital gate stack reference	`/Users/christianmacion/Macion Capital/strategies/_templates/reference/gate_stack_reference_G1-G28.md`	G1–G28 + G29–G31 (canonical gate definitions, kill patterns, statistical tests)
Macion Capital statistical tests reference	`/Users/christianmacion/Macion Capital/strategies/_templates/reference/_statistical_tests_reference.md`	DSR / PBO / min-BTL / walk-forward / era-split / de-selection formulas
Macion Capital agent + skill index	`/Users/christianmacion/Macion Capital/.claude/agents/`, `.claude/skills/`	15 agents + 31 skills, each with a charter Markdown
Macion Capital chart pipeline	`/Users/christianmacion/Macion Capital/scripts/agent_chart_pipeline.py`	The 5-PNG contract (equity, drawdown, walkforward, era-split, gate heatmap) per backtest
Macion Capital institutional pitch audit	`/Users/christianmacion/Macion Capital/docs/startup_buildout/02_INSTITUTIONAL_QUANT_GOLD_STANDARD_AUDIT.md`	The 215-page gold-standard reference used by counsel + the principal
Macion Capital legal dossier for counsel	`/Users/christianmacion/Macion Capital/docs/startup_buildout/04_MACION_CAPITAL_LEGAL_DOSSIER_FOR_COUNSEL.pdf`	215-page counsel handoff bundle
Macion Capital counsel brief (short)	`/Users/christianmacion/Macion Capital/docs/startup_buildout/05_COUNSEL_BRIEF_SHORT.pdf`	15-page distil for the first counsel meeting
Workbook format spec	`~/.claude/skills/workbook/reference/format_spec.md`	The teaching format this PDF was built with

Verification notes

- **Pricing is dated 2026-07.** Verify before committing budget; vendor pricing moves fast.
- **Firm-level AI claims are INFERRED from public signals** (careers, blogs, talks) unless explicitly PUBLIC.
- **Regulatory text was retrieved from primary government URLs.** Always re-verify before quoting.
- **Statistical claims** (e.g., +90.2%, ~15× tokens, 41–86.7% failure rate) are sourced from the cited paper. Always cite.
- **WebSearch and WebFetch were partially unavailable** during this document's preparation (2026-07-07). Some URLs were retrieved manually. Future revisions should re-verify.

v1.4 enrichment — institutional source pack (C1 + C3)

All retrieved 2026-07-07 from publicly accessible primary URLs. The `[STATED]` / `[INFERRED]` / `[NEEDS VERIFY]` tags in C1 map to this bibliography.

Institutional firm pages (17 firms)

Firm	Source	
Renaissance Technologies	`en.wikipedia.org/wiki/Renaissance_Technologies`	[STAT PhD-h Welch
Two Sigma	`en.wikipedia.org/wiki/Two_Sigma`; `twosigma.com/articles/icml-2025-key-ideas-on-llms-human-ai-alignment-and-more/` (Firdaus Janoos, 2025-07-31)	[STAT by Jar 2017–
D. E. Shaw	`deshaw.com/careers/machine-learning-researcher-4954`; `deshaw.com/careers/applied-ai-engineer-5375`; `deshaw.com/careers/ai-product-engineer-private-equity-platform-5637`	[STAT Triton ML Re 350K; JD wit requir
Citadel / Citadel Securities	`linkedin.com/company/citadel-securities/`; `claude.com/solutions/financial-services` (Atte Lahtiranta customer page); `en.wikipedia.org/wiki/Citadel_LLC`; `en.wikipedia.org/wiki/Citadel_Securities`	[STAT Claud Assist NYC I
Point72 / Cubist	`point72.com/cubist/`; `en.wikipedia.org/wiki/Point72_Asset_Management`; `careers.point72.com`	[STAT (ex-W joined Point7 ~$1.5l Quant Analy
Bridgewater / AIA Labs	`bridgewater.com/aia-labs`; `bridgewater.com/in-the-news/bloomberg-on-bridgewaters-latest-innovations-in-ai-ml`; `job-boards.greenhouse.io/bwaltpostings/jobs/8463808002`; `en.wikipedia.org/wiki/Bridgewater_Associates`; `en.wikipedia.org/wiki/Jasjeet_Sekhon`; `en.wikipedia.org/wiki/David_Ferrucci`; `claude.com/customers/bridgewater`; `bridgewater.com/team`; `digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai`	[STAT Bloom Sonali a firm **ML-d** (corre v1.4, n source **from Goog** Wikip **Ferru** Wikip IBM 2 Cognit for Ad "de-op tabula [NEEI Foreca genera perf d VERII breako
Balyasny	`en.wikipedia.org/wiki/Balyasny_Asset_Management`; `bamfunds.com/`; `bamfunds.com/news-and-insights/balyasny-openai-feature`; `claude.com/solutions/financial-services` (Damian Miraglia customer quote)	[STAT Chief vendo "AI eq analys persor 2024)
Millennium	`mlp.com/people/technology/`; `career.mlp.com/careers`; `en.wikipedia.org/wiki/Millennium_Management`	[STAT tagged

Firm	Source	Notes
		2026; AI Lab; Vlad Torgovnik CIO
AQR	aqr.com/Insights/Research; multiple Kelly/Xiu/Malamud papers (see paper section below)	[STATED] Kelly/Xiu/Malamud ML paper series (2019–2024)
Man Group	man.com/news-centre/man-group-anthropic-partnership (Feb 2026); man.com/insights/catching-angels-before-the-fall (Jun 2026); man.com/technology; github.com/man-group/ArcticDB; oxford-man.ox.ac.uk/; job-boards.eu.greenhouse.io/mangroup/jobs/4882441101; en.wikipedia.org/wiki/Man_Group	[STATED] Man × Anthropic firmwide Claude deployment (Feb 2026); AlphaGPT "proposes signals, writes code, runs backtests before any human sees the output"; ArcticDB 2.4k GitHub stars; OMI/Graphcore IPU partnership; CPython 3.14 zlib-ng contribution (May 2026); pip perf (Jul 2025)
Winton	en.wikipedia.org/wiki/Winton_Group; winton.com; winton.com/news	[STATED] ~175 employees, 100+ academics; Hivemind spinoff 2018; homepage has ZERO "AI"/"ML"/"alt-data"
Hudson River Trading	en.wikipedia.org/wiki/Hudson_River_Trading; hudsonrivertrading.com/; hudsonrivertrading.com/hrtbeat	[STATED, LOW CONFIDENCE — single Wikipedia citation, no firm primary source] $1B/yr AI spend (2025-26); HRTWorker + HeraclesQL + SystemVerilog language server
PDT Partners / Schonfeld	schonfeld.com/careers/; pdtpartners.com/careers; en.wikipedia.org/wiki/PDT_Partners	[STATED] Senior SWE AI Tech, Full Stack Eng AI Tech, AI Data Eng, AI Strategy Analyst (parent Schonfeld); Applied ML Scientist (PDT)
Squarepoint	en.wikipedia.org/wiki/Squarepoint_Capital	[STATED] $226.9B AUM; École polytechnique founder pedigree; STG + AVT + Hull Partners/STG Securities acquisitions; **no AI/ML in Wikipedia** [NEEDS VERIFY]
WorldQuant	en.wikipedia.org/wiki/WorldQuant; worldquant.com/; worldquantbrain.com; platform.worldquantbrain.com	[STATED] 4M alphas in Alpha Factory; 550K+ BRAIN users; 80K IQC applicants in 2025 (vs 11K in 2018)
BlackRock / Aladdin	blackrock.com/corporate/insights	[STATED] Aladdin AI Labs founded 2024
JPMorgan (COIN, IndexGPT, LLM Suite)	jpmorgan.com/technology/ai; LLM Suite announcement (May 2024)	[STATED] ~200,000 employees with LLM Suite access; ~$2B annual AI spend

Cross-cutting infra sources (C3)

Topic	Source	Notes
CoreWeave GPU cloud	coreweave.com/blog (multiple posts)	[STATED] ~250K NVIDIA GPUs; Jane Street confirmed customer (Nov 2024); Mosaic AI Research Lab (Jun 2026); Slurm + GraphRunner; ~360 MW contracted power
Optiver agentic SDLC	optiver.com/engineering-blog (Rob Hahn, Dec 2025)	[STATED] Slurm + GraphRunner for AI workload scheduling in low-latency trading environment
KDB-AI + KDB-X	kx.com; kx.com/customer (B2C2, ADSS, Vanguard); kx.com/blog; github.com/KxSystems/kx-skills	[STATED, vendor-reported, not independently audited] 30× faster backtesting; 80% infra reduction on KDB-X
Apache Kafka 4.3	confluent.io/blog (Confluent "Compliant AI Agents with Stateful Stream Processing", Jun 15 2026)	[STATED] Kafka 4.3.0 (Jun 2026): KRaft, Tiered Storage, Share Groups; canonical 2026 AI-agent compliance pattern
AMD Optiver FPGA	optiver.com/engineering-blog (Kevin Sprague, May 2025)	[STATED] Optiver/AMD FPGA partnership; AMD PEAK pack
HRT engineering	hudsonrivertrading.com/hrtbeat	[STATED] HRTWorker + HeraclesQL + SystemVerilog language server
Databento low-latency	databento.com/blog	[STATED] Napatech NT200A02 100G FPGA NIC + NVMe flash + UDP/TCP + kernel bypass + C++23 + Rust + Zstandard + core/NUMA pinning + CBBO-1s schema
Open-source	github.com/man-group/ArcticDB; github.com/man-group/man-imperial-algothon-2026; kx.com (kx-skills plugins); python.org (CPython 3.14 zlib-ng)	[STATED] Man Group's CPython 3.14 zlib-ng contribution (May 2026); pip perf work (Jul 2025)
Alt-data vendors	alphasense.com; ravenpack.com; hebbia.com; facteus.com; mscience.com; planet.com; spire.com; rsmetrics.com; eaglealpha.com	Vendor pricing pages; **not independently audited**

Skeptical canon (added in v1.4, all arXiv pre-prints)

Paper	arXiv ID	Why it matters
MAST: Multi-Agent System Failure Taxonomy (NeurIPS 2025)	arXiv:2503.13657	41-86.7% failure rate across 7 popular agentic frameworks, 1500+ test cases (per Table 2)
FINSABER (Stanford 2024)	arXiv:2411.02337	LLM-as-trader: no significant alpha (p > 0.34; N = 6 LLMs across 2020-2023 historical windows)
BacktestBench (pre-print)	arXiv:2605.17937	Speculative 26xx numbering — preprint per source pack Appendix A
MarketSenseAI (pre-print)	arXiv:2604.17327	Speculative 26xx numbering — preprint per source pack Appendix A
LiveTradeBench (pre-print)	arXiv:2511.03628	Pre-print, not peer-reviewed

Dead domains (C3.7)

- safegpt.com (SafeGraph) — DEAD, lapsed after 2023
- thasos.ai — DEAD
- sentieo.com — ACQUIRED by AlphaSense 2021
- accern.com — DEAD
- balyasny.com — DEAD; firm is at bamfunds.com
- millenniummanagement.com — DEAD (GoDaddy parked); firm is at mlp.com

---

Maintained by the principal. Last updated 2026-07-07.

Where this leaves us

THE TWO-STAGE ARCHITECTURE (CANONICAL — USED IN B1, E.2, HERE)

```
  +============================+
   AI agents (read-only)         ← propose / factor-mine / write code
           |
           v
   Gate stack (G1–G28 + G29–G31) ← numeric anchors: DSR, PBO, min-BTL
           |
           v
   Decision (KILL / ITERATE / PROMOTE)

+============================+
```

The LLM is never the alpha. The gate stack is the judge.

THE VERDICT

AI is not optional for a 2026 hedge fund — but the AI that wins is the AI that respects the gate stack. A pre-launch fund like Macion Capital should adopt the inherited multi-agent research platform, add an LLM copilot + agent harness, then data + alt-data, then LLMOps + governance — **in that order**, with explicit numeric anchors at every step.

No first-mover advantage in the LLM. First-mover advantage in the gate stack that surrounds it.

S1 · THE STANDING RULE

Never let the LLM be the alpha. Make it the orchestrator and let the gate stack be the judge. The Macion Capital methodology enforces this rule at every stage of the lifecycle, and the principal's 15-agent research platform is the operational implementation.

The LLM's job is to `propose`. The gate stack's job is to `judge`. The wallet's job is to `execute`. No agent other than the executor ever moves capital, and the executor must co-sign with two humans on any trade above the size threshold.

S2 · THE 4-PHASE ROADMAP

PHASE-BY-PHASE BUILD · MACION CAPITAL

Phase	Months	Outcome
1	0-3	Activate Macion Capital's 15-agent research platform · MCP plumbing · gate-stack scaffold
2	3-9	Add agentic AI · NCAA judge · multi-agent eval harness
3	9-18	Data + alt-data layer · Causal discovery · Conformal pred.
4	18-36	Live trading · Annex IV · EU AI Act conformity · counsel handoff

TOTAL PAGES	SOURCE CHAPTERS	PRIMARY SOURCES	FIRMS PROFILED
60	**18**	**200+**	**17**
v1.6 rendered PDF	5 new in v1.6	incl. EU AI Act, FINRA 24-09	institutional + retail/indie

S3 · THE EVIDENCE BASE

200+ sources across academic papers (Anthropic multi-agent, MAST, FINSABER, AQR Kelly/Xiu/Malamud paper series, DSPy, López de Prado), vendor documentation (Databento, Polygon, AlphaSense, QuantConnect, WorldQuant, Numerai, KDB.AI, CoreWeave, Confluent, Optiver, HRTWorker), regulatory primary text (**FINRA Notice 24-09** confirmed at finra.org/rules-guidance/notices/24-09; **SEC Marketing Rule**; **SEC 2023 AI proposal** still dormant; **EU AI Act Regulation EU 2024/1689** via Digital Omnibus 2026-05-07; **OCC 2026-13 + Fed SR 26-02**; **FCA FS24/2**; **FINRA June 2020 AI report**; CFTC no-standalone-AI-rule), and firm engineering blogs (Two Sigma, Citadel, Bridgewater AIA Labs, AQR, Man × Anthropic partnership, Hudson River Trading `hrtbeat`, WorldQuant, BlackRock, JPMorgan, D. E. Shaw, Balyasny, Millennium, Man AHL, OMI/Graphcore, Optiver, Jane Street, Databento).

VERSION HISTORY · V1.0 → V1.6

v1.0	2026-06	Initial scaffold · Parts A-F + Appendix · 32 pages.
v1.1	2026-06	*+ FINRA 24-09 primary + firm URL corrections.*
v1.2	2026-06	*+ QuantConnect 515K, Lean 20.4K, AIEQ track record, AlphaSense, Kensho, MCP server trend.*
v1.3	2026-06	*+ SEC + EU AI Act + OCC 2026-13 + Fed SR 26-02 + FCA FS24/2 + F.5 frontier section.*
v1.4	2026-07	*+ P0 honesty fixes + full C1 rewrite (17 firms) + new Part C3 cross-cutting infra + F.5.4 evidence-quality ladder.*

v1.5	2026-07	*Bridgewater $1.6B → $2B + Sekhon/Ferrucci retraction + EU AI Act Annex III 2027-12-02 per Digital Omnibus + F.5.5 reasoning models.*
v1.6	2026-07	**+ C4 AI wallet agents · G Macion Capital ops · F.7 EU AI Act runbook · H prompt engineering · I reference index · institutional cover + TOC + closing bookend.**

AUTHOR & STEWARD

Christian T. Macion — Founder & Analyst, Macion Capital. Authored and self-attested. The Macion Capital gate stack (G1–G28 + G29–G31) and 15-agent research platform are the principal's own engineering, originated in earlier workspaces and clean-room forked to the Macion Capital track on 2026-06-30.

WORKSPACE

/Users/christianmacion/Macion Capital/AI_Research/
18 source markdown files · maintained on every revision.

DISTRIBUTION

Principal-only archive. No external circulation. No client data. No PII. No alpha cartography. Counsel-handoff copy (TBD) carries the same restrictions.

MACION CAPITAL

"The methodology outlives the strategy.
The gate stack outlives the model.
The discipline outlives the fund."

— MACION CAPITAL · WORKBOOK V1.6 · 2026-07-07